
Software Requirements Specification

For

Licensing of Certifying Authorities and Dashboard for DSCs

Version 1.9

Prepared by e-Governance Solutions Group

C-DAC Delhi

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Table of Contents

Revision History	3
1. Introduction	4
1.1 Purpose	4
1.2 Document Conventions	4
1.3 Intended Audience and Reading Suggestions	5
1.4 Product Scope	5
1.5 References	5
2. Overall Description	6
2.1 Product Perspective	6
2.2 Product Functions	6
2.3 User Classes and Characteristics	22
2.4 Use Case Scenario	27
2.5 Operating Environment	38
2.6 Design and Implementation Constraints	39
2.7 User Documentation	41
2.8 Assumptions and Dependencies	41
3. External Interface Requirements	44
3.1 User Interfaces	44
3.2 Hardware Interfaces	56
3.3 Software Interfaces	56
3.4 Communications Interfaces	57
4. System Features	58
4.1 Responsive Design	58
4.2 2-Factor Authentication	58
4.3 Role-based Authorization	58
4.4 Data Management	58
4.5 Data Security & Privacy	58
4.6 System Integrity	58
4.7 Bot Prevention	58
4.8 Integration	59
4.9 Online Payment	59
4.10 Instant Notification	59
4.11 Document Authentication	59
4.12 Scalability	59
4.13 Log Management	59
4.14 Visualization & Reporting	59
5. Other Nonfunctional Requirements	60
5.1 Performance Requirements	60
5.2 Safety Requirements	60
5.3 Security Requirements	60
5.4 Special Requirements	60
5.5 Software Quality Attributes	61
5.6 Business Rules	64
Appendix A: System Architecture	65
Appendix B: Application Architecture	66
Appendix C: Process Flow Diagram	67
Appendix D: Data Flow Diagram	82
Appendix E: Use Case Diagram	105

Revision History

Name	Date	Reason For Changes	Version
SRS-Licensing of CAs & Dashboard for DSCs	18/03/2024	-	1.0
SRS-Licensing of CAs & Dashboard for DSCs	05/04/2024	Modified	1.1
SRS-Licensing of CAs & Dashboard for DSCs	12/04/2024	Modified	1.2
SRS-Licensing of CAs & Dashboard for DSCs	17/04/2024	Modified	1.3
SRS-Licensing of CAs & Dashboard for DSCs	24/04/2024	Modified	1.4
SRS-Licensing of CAs & Dashboard for DSCs	03/05/2024	Modified	1.5
SRS-Licensing of CAs & Dashboard for DSCs	09/05/2024	Modified	1.6
SRS-Licensing of CAs & Dashboard for DSCs	20/06/2024	Modified	1.7
SRS-Licensing of CAs & Dashboard for DSCs	02/12/2024	Modification	1.8
SRS-Licensing of CAs & Dashboard for DSCs	12/12/2024	Modification	1.9

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the portal for Licensing of Certifying Authorities and Dashboard for DSCs. It will explain the functions, features and interfaces of the software, what the software will do, the constraints under which it must operate and how the software will react to external stimuli.

This document gives specifications for the software, which will be referred to as Licensing of Certifying Authorities and Dashboard for DSCs. The software can be used to submit an application form for obtaining a license from O/o Controller of Certifying Authorities (CCA) to operate as a Certifying Authority (CA). A Certifying Authority can issue Digital Signature Certificate (DSC) only after being duly licensed by the O/o CCA as per provisions of the IT Act.

This document contains a minimal set of features/ functionalities that the software will provide, and a set of functional and non-functional requirements that the software must meet.

1.2 Document Conventions

Headings: -

Text: - Bold

Font-Size: - 14

Highlighting: - Times

Sub Headings: -

Text: - Bold

Font-Size: - 12

Highlighting: - Times

Paragraph Content: -

Text: - Simple

Font-Size: - 11

Highlighting: - Times

Table Content: -

Text: - Simple

Font-Size: - 10

Highlighting: - Times

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, testers and managers for the purpose of understanding the design of the software in terms of different perspectives. Further, this document contains functionalities and characteristics of the software along with the working environment. It also includes other information related to software such as external interface requirements, features and other non-functional requirements.

1.4 Product Scope

Development of software to manage the applications submitted for grant of license to operate as a CA under the IT Act. The proposed software will cater to or take care of almost every aspect focusing on the ease of submitting and tracking fresh application for grant of license and also application for renewal of license.

Following lists the major scopes identified under the software being proposed: -

- **Intent for License**
- **Verification & Login**
- **License Application**
- **Applicant Audit**
- **License Issuance**
- **eSign Application**
- **Internal & Annual Audit**
- **Renewal Application**
- **Renewal Audit**
- **License Renewal**
- **Renewal of eSign Service**
- **Cessation of License**
- **Dashboard & Reports**

1.5 References

- [1] IEEE Standards Association (2018, Last Accessed June 2021) Systems and software engineering – Life cycle processes – Requirements engineering Standard. (IEEE 29148 - 2018) URL <https://standards.ieee.org/standard/29148-2018.html>
- [2] IEEE Standards Association (2011, Last Accessed June 2017) Systems and software engineering – Life cycle processes – Requirements engineering Standard. (IEEE 29148 - 2011) URL <https://standards.ieee.org/standard/29148-2011.html>

2. Overall Description

2.1 Product Perspective

At present, an individual/ organization has to submit the application along with required documents offline to the O/o CCA for grant of license to operate as a CA. A portal for Licensing of Certifying Authorities and Dashboard for DSCs is required to be designed & developed to facilitate Individual & Organization to efficiently submit application along with all the documents that are essential to substantiate the claim for award of license to operate as a CA.

Applicants will receive notifications via email, SMS and over dashboard regarding various events which will take place as per process.

Auditors will be able to submit the report of audits carried out by them. CCA will be able to issue the license based on the recommendations of report review committee.

Licensees will be able to get alerts regarding expiry of their licenses and accordingly, they can apply for renewal of the same before due date.

Moreover, O/o CCA will get detailed view of time-based and/or region-wise data related to certificate issuance by the licensed CAs. Software will also generate reports which can be downloaded in different formats.

2.2 Product Functions

Component-wise functionalities may be described as below: -

1. Intent for License

- a. Authorized Officer of CCA can generate a Unique Code for Intent Registration by the interested user (Individual/ Company/ Firm/ Government Agency) to become a licensed Certifying Authority (CA) upon assessment of ability & capacity of the user to become a licensed CA.
- b. Applicant can Submit/ Register her/ his Intent to become a CA using the Unique Code generated & shared by the O/o CCA.

2. Verification & Login

- a. Login credentials will be generated for the user and shared over the registered email address/ mobile.
- b. Authorized Officer of CCA can register the Empaneled Audit Agency by submitting the Audit Agency Registration Form.
- c. Login credentials will be generated for the registered Audit Agencies and shared over

their registered email address/ mobile.

- d. Only authorized users can login and access the software via 2-factor authentication process and mechanism.

3. License Application

- a. Logged-in applicant can submit the application based on the selected category & services to be offered by paying the applicable fee either online or offline, uploading payment proof and required necessary documents, including CPS, to support the application along with any third-party testing report. The O/o CCA will be notified via email/ SMS.
- b. Authorized Officer of CCA can verify the receipt of payment and upload the proof of payment received.
- c. Upon verification of payment, Authorized Officer of CCA can scrutinize the application and supporting documents including CPS. Any discrepancies found are notified to the applicant via email/ SMS.
- d. The applicant can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 3 (three) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- e. If the application is rejected, the applicant will be notified about application rejection via email/ SMS.
- f. Once all discrepancies are successfully resolved, an application number will be generated by the software. Acknowledgement will be sent to applicant upon completion of scrutiny and due approval from the CCA regarding acceptance of application over email/ SMS.
- g. Application Review Committee can review the application & supporting documents including CPS, notify the applicant about discrepancies in application, if any, via email/ SMS.
- h. The applicant can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 3 (three) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- i. If the application is rejected, the applicant will be notified about application rejection via email/ SMS.
- j. Upon successful review by the Committee, the applicant will be notified via email/ SMS to initiate the process of Audit at his/ her own end from one of the Empaneled Auditors,

after due approval from the CCA.

4. Applicant Audit

- a. The Applicant can appoint audit agency offline by entering the details of the appointed agency in the software for the said audit. The Audit Agency will be notified via email/ SMS.
- b. Audit Agency can nominate the names of minimum 2 (two) Auditors for the conduction of said audit out of which 1 (one) must be having a valid CISA certification while the other should be at least a valid ISO 27001 certified professional. The Agency must upload documents in this regard. The Applicant will be notified via email/ SMS.
- c. Audit Agency can submit undertakings, define the scope of audit and provide a schedule for the audit. The O/o CCA will be notified via email/ SMS.
- d. On receipt of approval from the CCA regarding appointment of the Audit Agency, Audit Team, Audit Scope & Audit Schedule, the Audit Team can commence the audit process.
- e. The Auditor can view the application & supporting documents, including CPS, for the audit.
- f. On completion of audit, the Audit Agency can submit the audit report along with the list of NCs, if any. The Applicant will be notified via email/ SMS.
- g. The Applicant can submit the action taken report along with supporting documents against the issued NCs, if any. The Audit Agency will be notified via email/ SMS.
- h. The Auditor can verify action taken report and supporting documents submitted by the applicant against the issued NCs. The Applicant will be notified via email/ SMS.
- i. The Applicant can endorse, via digital signature, the submitted audit report by the Auditor. The O/o CCA will be notified via email/ SMS.
- j. The Report Review Committee can review the uploaded audit report, NCs and action taken report; notify the applicant of any identified shortcomings in the report via email/ SMS.
- k. The Applicant can fulfill the shortcomings and submit documents in support, if any. The O/o CCA will be notified via email/ SMS. This correction process is limited to a maximum of 2 (two) attempts. Failure to address shortcomings within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- l. If the application is rejected, the Applicant will be notified about the application rejection via email/ SMS.
- m. Upon successful report review, the application will be recommended for Issuance of License by the Committee. The Applicant will be notified via email/ SMS.

5. License Issuance

- a. The CCA can issue “in-principle approval” for grant of license to the applicant upon the conformance of the audit report by the Committee. The Applicant will be notified via email/ SMS.
- b. The Applicant can submit the required Bank Guarantee, CSR, and Agreement, as applicable. The O/o CCA will be notified via email/ SMS.
- c. Authorized Officer of CCA first verifies the receipt of bank guarantee and uploads the proof of confirmation received.
- d. Upon verification of bank guarantee, CCA can issue CA License to the applicant on receipt of required CSR & Agreement. The Applicant will be notified via email/ SMS.

6. eSign Application

- a. After becoming a CA, Licensee can submit application for becoming eSign Service Provider by selecting their eKYC method(s) and uploading the necessary approvals for these methods, downloading a system-generated cover letter, digitally signing it and uploading the same. Additionally, the Licensee must indicate the API versions they intend to use and accordingly upload the corresponding audit report & documents, including CPS. The O/o CCA will be notified via email/ SMS.
- b. Authorized Officer of CCA can scrutinize and review the submitted application, along with supporting documents, including CPS, and applicable fee, if any. Any discrepancies found are notified to the licensee via email/ SMS.
- c. The license can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 2 (two) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- d. If the application is rejected, the licensee will be notified about application rejection via email/ SMS.
- e. The CCA can issue approval for providing eSign services to the Licensee on successful review of the application. The Licensee will be notified via email/ SMS.

7. Internal & Annual Audit

- a. Authorized Officer of CCA can prepare and publish a schedule for regular annual audits of the Licensees. The Licensee will be notified via email/ SMS.
- b. The Licensee can get performed the half yearly internal audits either by its own team or through an external agency and submit the reports. The O/o CCA will be notified via email/ SMS.

- c. The Licensee can initiate their annual audit process as per the schedule shared by the O/o CCA.
- d. The Licensee can appoint audit agency offline by entering the details of the appointed agency in the software for the said audit with the condition that same audit agency can't be engaged in consecutive years. The Audit Agency will be notified via email/ SMS.
- e. Audit Agency can nominate the names of minimum 2 (two) Auditors for the conduction of said audit out of which 1 (one) must be having a valid CISA certification while the other should be at least a valid ISO 27001 certified professional with the condition that same auditors can't be engaged in consecutive years. The Agency must upload documents in this regard. The Licensee will be notified via email/ SMS.
- f. Audit Agency can submit the undertakings, define the scope of audit and provide a schedule for the audit. The O/o CCA will be notified via email/ SMS.
- g. On receipt of approval from the CCA regarding appointment of the Audit Agency, Audit Team, Audit Scope & Audit Schedule, the Audit Team can commence the audit process.
- h. The Auditor can view previous audit reports of the Licensee and perform the assigned audit.
- i. On completion of audit, the audit agency can submit the audit report along with the list of NCs, if any. The Licensee will be notified via email/ SMS.
- j. The Licensee can submit the action taken report along with the supporting documents against the issued NCs, if any. The Audit Agency will be notified via email/ SMS.
- k. The Auditor can verify action taken report and supporting documents submitted by the licensee against the issued NCs. The Licensee will be notified via email/ SMS.
- l. The Licensee can endorse via digital signature about the submitted audit report by the Auditor. The O/o CCA will be notified via email/ SMS.
- m. The Report Review Committee can review the uploaded audit report, NCs and action taken report; notify the licensee of any identified shortcomings in the report via email/ SMS.
- n. The Licensee can fulfill the shortcomings and submit supporting documents, if any. The O/o CCA will be notified via email/ SMS. This corrective process is limited to a maximum of 2 (two) attempts. Failure to address shortcomings within these attempts may lead to any stringent action like show-cause notice followed by suspension of license from the O/o CCA or more attempts may be given depending upon the Approval of CCA.
- o. The Licensee can submit reply to the show-cause notice issued by the O/o CCA. The O/o CCA will be notified via email/ SMS.
- p. Upon successful report review and/ or acceptance of reply to the show-cause notice from the Licensee, audit process for the year will be marked complete on approval of CCA.

The Licensee will be notified via email/ SMS.

- q. CCA can issue notice for suspension of license to the licensee upon non-acceptance of reply submitted by the Licensee. The Licensee will be notified via email/ SMS.

8. Renewal Application

- a. The Licensee can pay the applicable fee and submit the application for renewal of the CA License by uploading the necessary required documents, including CPS & any third-party testing report, to support the application in between 45 to 90 days before the validity expiry of the existing license. The O/o CCA will be notified via email/ SMS.
- b. Authorized Officer of CCA first verifies the receipt of payment and uploads the proof of payment received.
- c. Upon verification of payment, Authorized Officer of CCA can scrutinize the submitted application, supporting documents, including CPS, and applicable fee. Any discrepancies found will be notified to the licensee via email/ SMS.
- d. The Licensee can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 2 (two) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the licensee depending upon the Approval of CCA.
- e. If the application is rejected, the licensee will be notified about application rejection via email/ SMS.
- f. Once all discrepancies are successfully resolved, an application number will be generated by the software. Acknowledgement will be sent to licensee upon completion of scrutiny and due approval from the CCA regarding acceptance of application via email/ SMS.
- g. Application Review Committee can review the application & supporting documents including CPS, notify the Licensee about discrepancies in application, if any, via email/ SMS.
- h. The Licensee can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 2 (two) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the licensee depending upon the Approval of CCA.
- i. If the application is rejected, the licensee will be notified about application rejection via email/ SMS.
- j. Upon successful review by the Committee, the Licensee will be notified via email/ SMS to initiate the process of Audit at his/ her own end from one of the Empaneled Auditors,

after due approval from the CCA.

9. Renewal Audit

- a. The Licensee can appoint audit agency offline by entering the details of the appointed agency in the software for the said audit. The Audit Agency will be notified via email/ SMS.
- b. Audit Agency can nominate the names of minimum 2 (two) Auditors for the conduction of said audit out of which 1 (one) must be having a valid CISA certification while the other should be at least a valid ISO 27001 certified professional. The Agency must upload documents in this regard. The Licensee will be notified via email/ SMS.
- c. Audit Agency can submit the undertakings, define the scope of audit and provide a schedule for the audit. The O/o CCA will be notified via email/ SMS.
- d. Upon due approval from the CCA regarding the appointment of the Audit Agency, Audit Team, Audit Scope & Audit Schedule, the Audit Team can commence the Audit process.
- e. The Auditor can view the application and supporting documents for renewal audit.
- f. On completion of audit, the audit agency can submit the audit report along with the list of NCs, if any. The Licensee will be notified via email/ SMS.
- g. The Licensee can submit the action taken report along with the supporting documents against the issued NCs, if any. The Audit Agency will be notified via email/ SMS.
- h. The Auditor can verify action taken report and supporting documents submitted by the licensee against the issued NCs. The Licensee will be notified via email/ SMS.
- i. The Licensee can endorse via digital signature about the submitted audit report by the Auditor. The O/o CCA will be notified via email/ SMS.
- j. The Report Review Committee can review the uploaded audit report, NCs and action taken report; notify the licensee of any identified shortcomings in the report via email/ SMS.
- k. The Licensee can fulfill the shortcomings and submit documents in support, if any. The O/o CCA will be notified via email/ SMS. This correction process is limited to a maximum of 2 (two) attempts. Failure to address shortcomings within these attempts may lead to rejection of application or more attempts may be given to the licensee depending upon the Approval of CCA.
- l. If the application is rejected, the licensee will be notified about the application rejection via email/ SMS.
- m. Upon successful report review, the application will be recommended for Renewal of License by the Committee. The Licensee will be notified via email/ SMS.

10. License Renewal

- a. The CCA can issue an “in-principle approval” for the grant of license renewal to the licensee upon conformance of the Audit report by the Committee. The Licensee will be notified via email/ SMS.
- b. The Licensee can submit the required Bank Guarantee, Agreement, CSR etc., as applicable. The O/o CCA will be notified via email/ SMS.
- c. Authorized Officer of CCA first verifies the receipt of bank guarantee and uploads the proof of confirmation received.
- d. Upon verification of bank guarantee, CCA can issue Renewed License to the Licensee on receipt of required CSR & Agreement. The Licensee will be notified via email/ SMS.

11. Renewal of eSign Service

- a. After the renewal of CA License, Licensee can submit application renewal of eSign Service by selecting their eKYC method(s) and uploading the necessary approvals for these methods, downloading a system-generated cover letter, digitally signing it and uploading the same. Additionally, the Licensee must indicate the API versions they intend to use and accordingly upload the corresponding audit report & documents, including CPS. The O/o CCA will be notified via email/ SMS.
- b. Authorized Officer of CCA can scrutinize and review the submitted application, along with supporting documents, including CPS, and applicable fee, if any. Any discrepancies found are notified to the licensee via email/ SMS.
- c. The license can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 2 (two) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- d. If the application is rejected, the licensee will be notified about application rejection via email/ SMS.
- e. The CCA can issue approval for renewal of eSign services being provided by the Licensee on successful review of the application. The Licensee will be notified via email/ SMS.

12. Cessation of License

- a. The Licensee can submit the application for cessation of license, issued by CCA, by uploading the necessary required documents (as per Section 21 of the IT Act), to support the application in between 60 to 90 days before the validity expiry of the license. The O/o CCA will be notified via email/ SMS.

- b. Authorized Officer of CCA can scrutinize & review the submitted application and supporting documents. Any discrepancies found will be notified to the licensee via email/ SMS.
- c. The Licensee can rectify the discrepancies notified and attach supporting documents, if any. The O/o CCA will be notified via email/ SMS. This process will continue for maximum of 2 (two) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the licensee depending upon the Approval of CCA.
- d. If the application is rejected, the licensee will be notified about application rejection via email/ SMS.
- e. Once all discrepancies are successfully resolved, an application number will be generated by the software. Acknowledgement will be sent to licensee via email/ SMS and the license will stand ceased upon expiry after due approval from CCA.

13. Dashboard & Reports

a. Monthly Data Submission:

- i. Licensees can submit their monthly data regarding Digital Signature Certificates (DSCs) and eSign certificates issued, as well as Application Service Providers (ASP) on-boarded by them. The O/o CCA will be notified via email/ SMS.

DSC/ eSign Issued (Monthly)

Sr. No.	Column Name	Data Type	Validation
1	State/ Union Territory	VARCHAR(40)	Required
2	Month	VARCHAR(2)	Required
3	Year	VARCHAR(4)	Required
4	DSC Issued	VARCHAR(5)	Required, Number
5	eSign Issued	VARCHAR(5)	Required, Number

ASP Details

Sr. No.	Column Name	Data Type	Validation
1	ASP Name	VARCHAR(75)	Required
2	ASP Category	VARCHAR (25)	Required
3	State/ UT	VARCHAR (40)	Required
4	Date of Onboarding	DATE	Required

b. Graphical Representation of Data using Bar Graph and Pie Chart:

i. Access by Authorized Officers of CCA:

1. Authorized Officers of CCA can view the number of eSign and DSCs issued by CAs on a monthly, yearly, and cumulative basis.
2. Authorized Officers of CCA can view state-wise data of eSign and DSCs issued by CAs on a monthly, yearly, and cumulative basis.

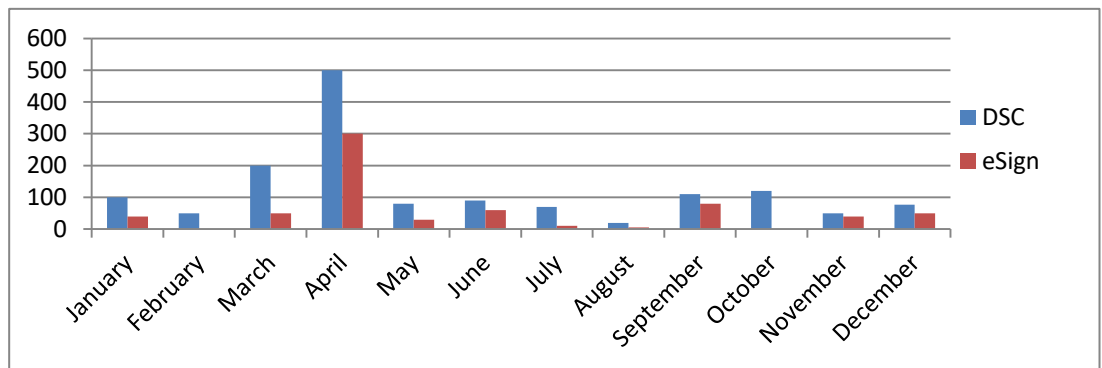
3. Authorized Officers of CCA can view individual CAs' data of issued eSign and DSCs on a state-wise, monthly, yearly, and cumulative basis.
4. Authorized Officers of CCA can view the number of issued eSign and DSCs for a specific period.

Monthly/Yearly data of eSign/DSC Issued by Individual/All CAs

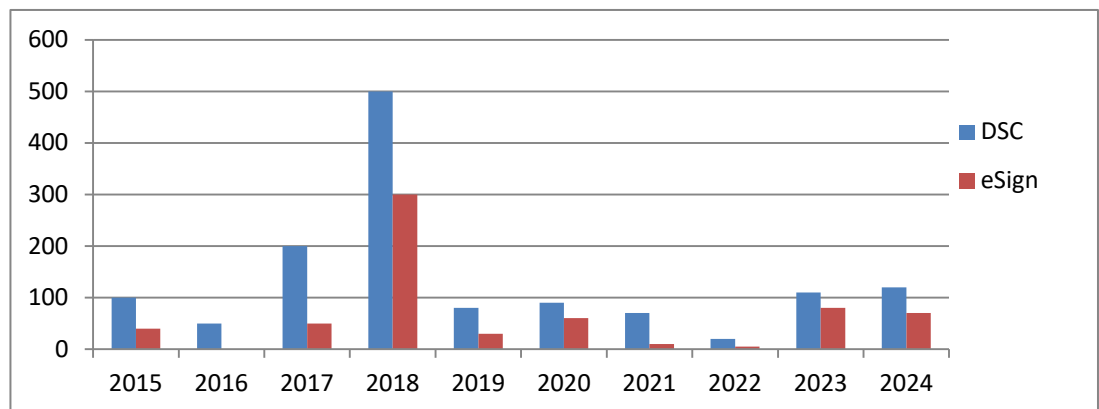
Filter Type:

- a. Year: <Particular / Multiple/All>
- b. Month: <Particular /Multiple/All>
- c. CA: <Particular/Multiple/All>
- d. State/UT: <Particular/Multiple/All>

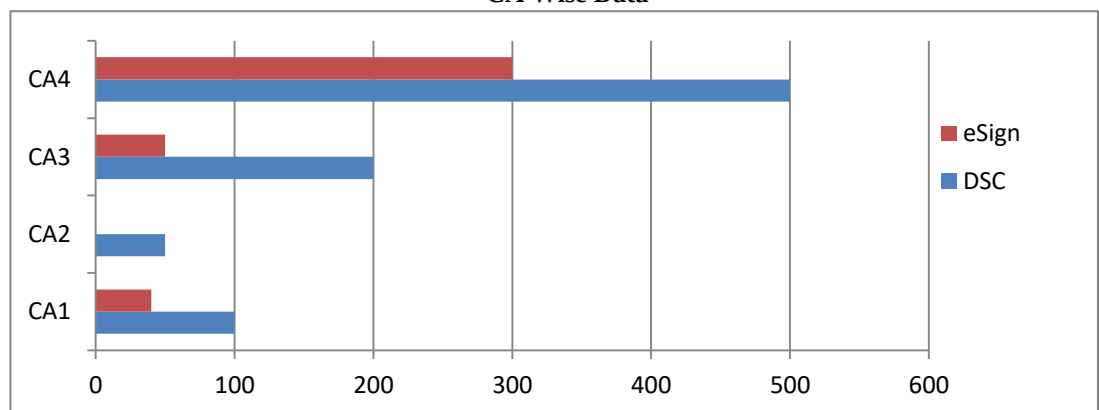
Month Wise Data



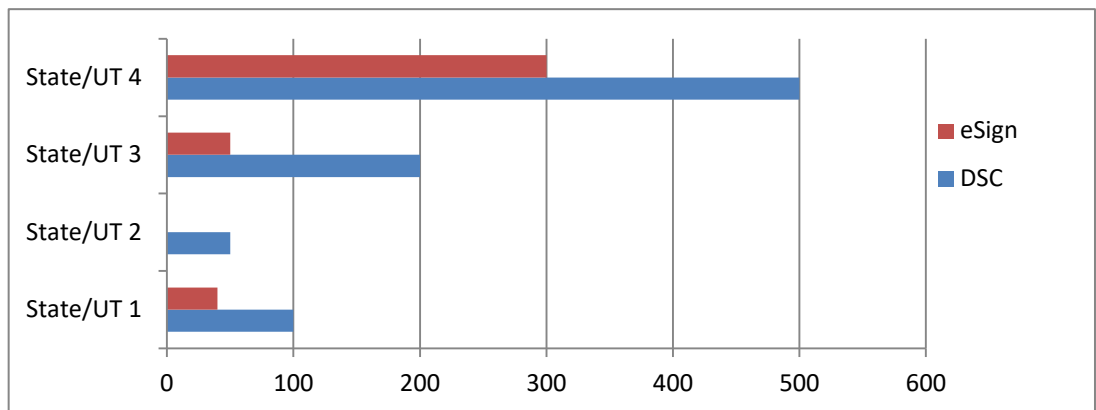
Year Wise Data



CA Wise Data



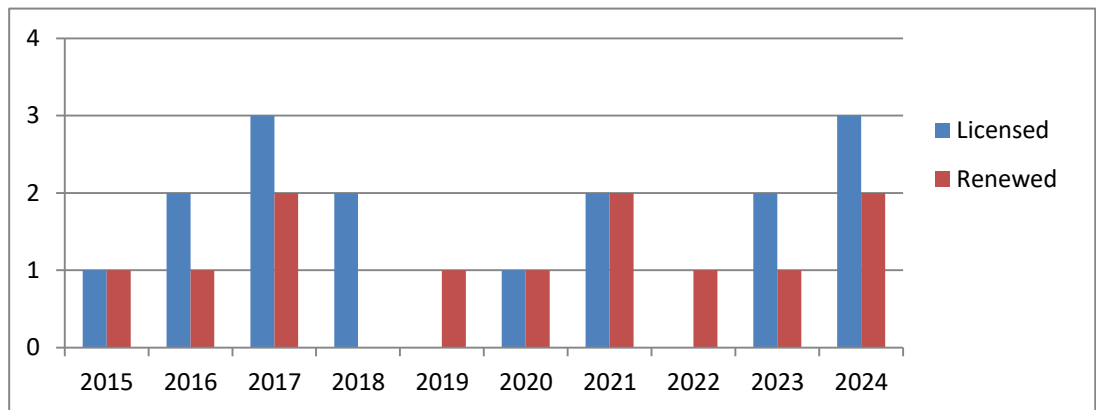
State/UT Wise Data



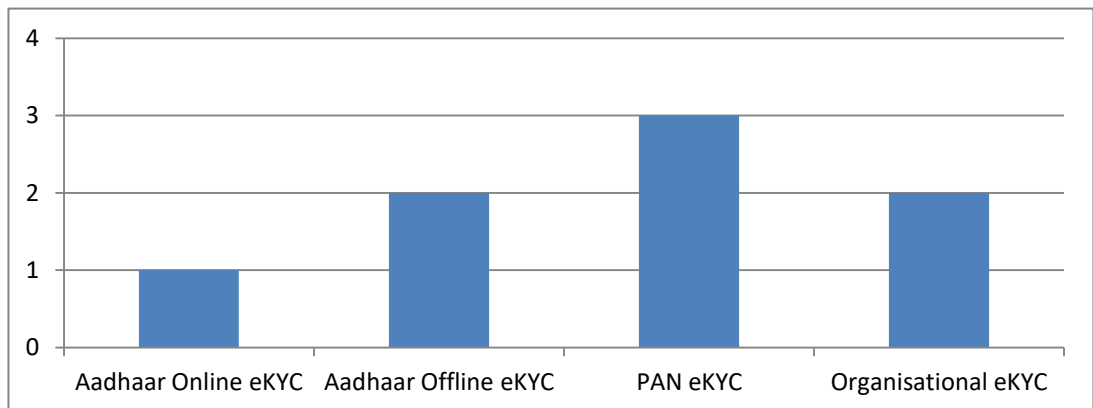
5. Authorized Officers of CCA can view the number of CAs with year of licensing for specific period/cumulative.

Filter Type:

a. Year: <Particular / Multiple/All>



6. Authorized Officers of CCA can view the number of ESPs and ESP-wise approval of eKYC modes



ii. Access by Licensees:

1. Licensees can view the number of eSign and DSCs issued by them only, on a monthly, yearly, and cumulative basis.

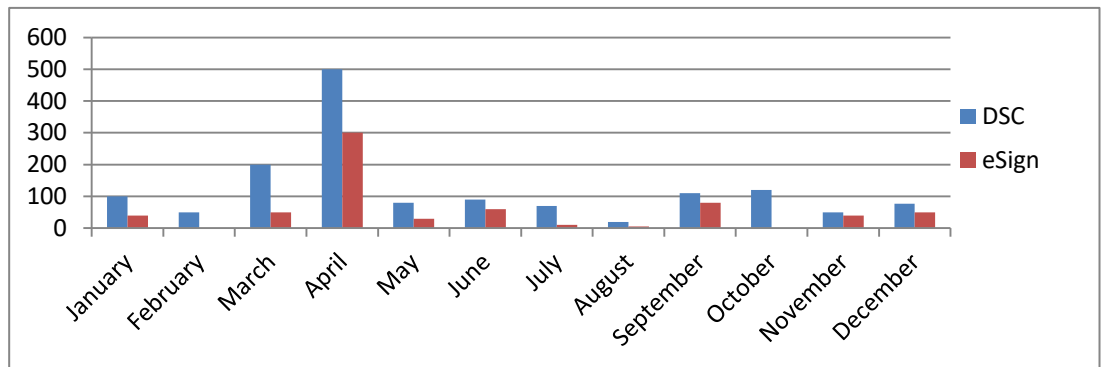
Monthly/Yearly data of eSign/DSC Issued by Individual CAs

Filter Type:

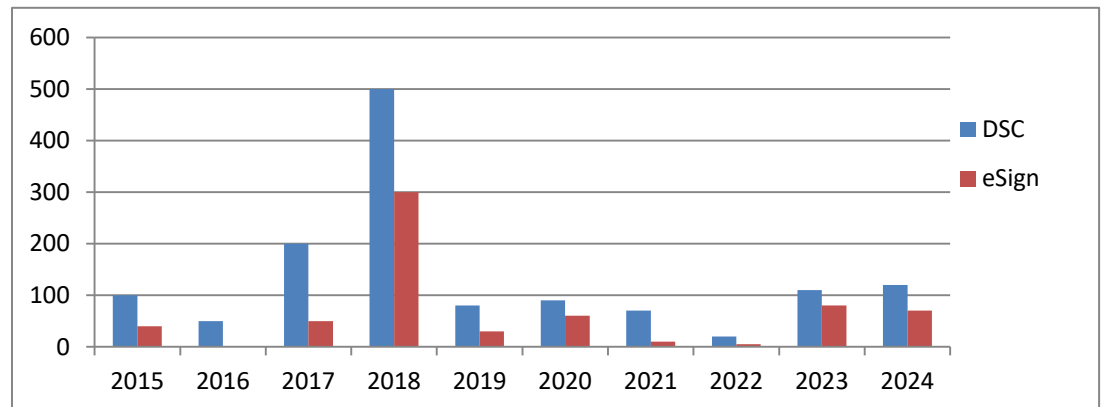
a. Year: <Particular / Multiple/All>

b. Month: <Particular /Multiple/All>

Month Wise Data



Year Wise Data



c. Reports

i. Generation of Reports by Authorized Officers of CCA:

1. Monthly Reports:

- Authorized Officers of CCA can generate Monthly report of eSign certificates issued by each CA.
- Authorized Officers of CCA can generate Monthly report of DSCs issued by each CA.

Controller of Certifying Authorities (CCA)			
<u>Monthly Report of eSign & DSC</u>			
As on dd.mm.yyyy			
Monthly report of eSign and Digital Signature Certificate(s) (DSCs) issued by each CA/ESP under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA/ESP Name	Month, Year (eSign)	Month, Year (DSC)

2. Yearly Reports:

- Authorized Officers of CCA can generate Yearly report of eSign certificates issued by each CA.
- Authorized Officers of CCA can generate Yearly report of Digital Signature Certificates issued by each CA.

Controller of Certifying Authorities (CCA)			
<u>Yearly Report of eSign & DSC</u>			
As on dd.mm.yyyy			
Yearly report of eSign and Digital Signature Certificate(s) (DSCs) issued by each CA/ESP under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA/ESP Name	Year (eSign)	Year (DSC)

3. Customized Period Reports:

- a. Authorized Officers of CCA can generate Customized report for a specific period detailing eSign certificates issued by each CA.
- b. Authorized Officers of CCA can generate Customized report for a specific period detailing Digital Signature Certificates issued by each CA.

Controller of Certifying Authorities (CCA) <u>Customized Report of eSign & DSC</u> As on dd.mm.yyyy			
Report of eSign and Digital Signature Certificate(s) (DSCs) from <dd/mm/yyyy> to <dd/mm/yyyy> issued by each CA/ESP under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA/ESP Name	Total eSign Issued	Total DSC Issued

4. Cumulative Reports:

- a. Authorized Officers of CCA can generate Aggregate report on the total eSign certificates issued by all CAs over time.
- b. Authorized Officers of CCA can generate Aggregate report on the total Digital Signature Certificates issued by all CAs over time.

Controller of Certifying Authorities (CCA) <u>Cumulative Report of eSign & DSC</u> As on dd.mm.yyyy			
Report of eSign and Digital Signature Certificate(s) (DSCs) from <start date> to <current date> issued by each CA/ESP under the Controller of Certifying Authorities (CCA)			
Sl. No.	State/Union Territory	Total eSign Issued	Total DSC Issued

5. Customized Cumulative Reports:

- a. Authorized Officers of CCA can generate Customized cumulative report for a specified period detailing total eSign certificates issued by CAs.
- b. Authorized Officers of CCA can generate Customized cumulative report for a specified period detailing total DSCs issued by CAs.

Controller of Certifying Authorities (CCA) <u>Customized Cumulative Report of eSign & DSC</u> As on dd.mm.yyyy			
Report of eSign and Digital Signature Certificate(s) (DSCs) from dd/mm/yyyy to dd/mm/yyyy issued by each CA/ESP under the Controller of Certifying Authorities (CCA)			
Sl. No.	State/Union Territory	Total eSign Issued	Total DSC Issued

6. Licensing Reports:

- a. Authorized Officers of CCA can generate Report on CAs and ESPs who obtained licenses during a specified period or since the beginning.

Controller of Certifying Authorities (CCA) <u>Licensing Report of CAs and ESPs from beginning</u> As on dd.mm.yyyy					
Report on CAs and ESPs that obtained licenses from the beginning under the Controller of Certifying Authorities (CCA).					
Sl. No.	CA/ESP Name	Type (CA/ESP)	License Issue Date	License Valid Up To	ESP Empanelment Date

Controller of Certifying Authorities (CCA) <u>Customized Licensing Report of CAs and ESPs</u> As on dd.mm.yyyy						
Report on CAs and ESPs that obtained licenses from <dd/mm/yyyy> to <dd/mm/yyyy> under the Controller of Certifying Authorities (CCA).						
Sl. No.	CA/ESP Name	Type (CA/ESP)	License Type (New/Renewed)	License Issued / Renewed Date	License Valid Up To	ESP Empanelment Date

7. Audit Timeliness Reports:

- a. Authorized Officers of CCA can generate Report on audits initiated within the scheduled timeframe.

Controller of Certifying Authorities (CCA) <u>Report on Annual Audits Initiated on Time</u> As on dd.mm.yyyy			
Report on Annual Audits initiated on time for each CA under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA Name	Scheduled Audit Date	Audit Start Date

- b. Authorized Officers of CCA can generate Report on audits that were delayed.

Controller of Certifying Authorities (CCA) <u>Report on Annual Audits Delayed</u> As on dd.mm.yyyy				
Report on Annual Audits delayed for each CA under the Controller of Certifying Authorities (CCA)				
Sl. No.	CA Name	Scheduled Audit Date	Audit Started (Yes/No)	Audit Start Date

- c. Authorized Officers of CCA can generate Report on audit reports submitted on time.

Controller of Certifying Authorities (CCA) <u>Report on Annual Audits Submitted on Time</u> As on dd.mm.yyyy			
Report on Annual Audits submitted on time for each CA under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA Name	Scheduled Audit Submission Date	Date of Submission

Controller of Certifying Authorities (CCA) <u>Report on Annual Audits Submitted Late</u> As on dd.mm.yyyy			
Report on Annual Audits submitted Late for each CA under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA	Scheduled Audit Submission Date	Date of Submission

8. Audit Closure Reports:

- a. Authorized Officers of CCA can generate closure audit report for each CA.

Controller of Certifying Authorities (CCA) <u>Annual Audits Closure Report</u> As on dd.mm.yyyy						
Closure Report on Annual Audits for each CA under the Controller of Certifying Authorities (CCA)						
Sl. No.	CA	Scheduled Audit Date	Audit Start Date	Scheduled Audit Submission Date	Audit Submission Date	Audit Completion Date

9. Auditor-Specific Reports:

- a. Authorized Officers of CCA can generate Report detailing audits conducted by a specific auditor within a given period.

Controller of Certifying Authorities (CCA) <u>Report on Audits Conducted By <Name of Agency></u> As on dd.mm.yyyy					
Report on Audits conducted from <dd/mm/yyyy> to <dd/mm/yyyy> by <Name of Agency> empanelled under the Controller of Certifying Authorities (CCA)					
Sl. No.	Auditors	Audit Conducted For	Audit Type (Annual Audit/ New Application Audit/ Renewal Application Audit)	Audit Initiated On	Audit Completed On

- a. Authorized Officers of CCA can generate Report on CAs whose licenses are due for renewal.

Controller of Certifying Authorities (CCA) <u>Report on CAs licenses due for renewal</u> As on dd.mm.yyyy		
Report on Certifying Authorities (CAs) Licenses Pending For Renewal under the Controller of Certifying Authorities (CCA)		
Sl. No.	CA Name	License Expiry Date

11. Contact Information Reports:

- a. Authorized Officers of CCA can generate Report on list of telephone numbers and email addresses of all licensed CAs and auditors.

Controller of Certifying Authorities (CCA) <u>Report on CA Details</u> As on dd.mm.yyyy			
Report on CA details under the Controller of Certifying Authorities (CCA)			
Sl. No.	CA Name	Telephone/Mobile Number(s)	Email Id (s)

Controller of Certifying Authorities (CCA) <u>Report on Audit Agency Details</u> As on dd.mm.yyyy			
Report on Empanelled Audit Agency details under the Controller of Certifying Authorities (CCA)			
Sl. No.	Audit Agency Name	Telephone/Mobile Number(s)	Email Id (s)

12. Physical Location Reports:

- a. Authorized Officers of CCA can generate Report on the physical locations of CA sites, including primary and disaster recovery (DR) sites.

Controller of Certifying Authorities (CCA) <u>Report on CA Site Location(s)</u> As on dd.mm.yyyy				
Report on Physical Locations of CA sites under the Controller of Certifying Authorities (CCA)				
Sl. No.	CA Name	Primary Site	DR Site	Other Site

2.3 User Classes and Characteristics

The above mentioned software will be used by following user classes with given characteristics:

1. O/o CCA

- Authorized Officer of CCA can generate a Unique Code for Intent Registration by the interested user (Individual/ Company/ Firm/ Government Agency) to become a licensed Certifying Authority (CA) upon assessment of ability & capacity of the user to become a licensed CA.
- Authorized Officer of CCA can register the Empaneled Audit Agency by submitting the Audit Agency Registration Form.
- Authorized Officer of CCA first verifies the receipt of payment and uploads the proof of payment received.
- Authorized Officer of CCA can scrutinize the application and supporting documents including CPS. Any discrepancies found are notified to the applicant.
- Authorized Officer of CCA can scrutinize and review the submitted eSign application, along with supporting documents and applicable fee, if any.
- Authorized Officer of CCA can prepare and publish a schedule for regular annual audits of the Licensees.
- Authorized Officer of CCA first verifies the receipt of bank guarantee and uploads the proof of confirmation received.
- Authorized Officers of CCA can view the number of eSign and DSCs issued by CAs on a monthly, yearly, and cumulative basis.
- Authorized Officers of CCA can view state-wise data of eSign and DSCs issued by CAs on a monthly, yearly, and cumulative basis.
- Authorized Officers of CCA can view individual CAs' data of issued eSign and DSCs on a state-wise, monthly, yearly, and cumulative basis.
- Authorized Officers of CCA can view the number of issued eSign and DSCs for a specific period.
- Authorized Officers of CCA can view the number of CAs and the years in which CAs have been licensed or renewed enabling identification of the duration each CA has been licensed, for a specific period or cumulatively.
- Authorized Officers of CCA can view the number of ESPs and ESP-wise approval of eKYC modes.
- Authorized Officers of CCA can generate Monthly report of eSign certificates issued by each CA.
- Authorized Officers of CCA can generate Monthly report of DSCs issued by each CA.
- Authorized Officers of CCA can generate Yearly report of eSign certificates issued by each CA.

- Authorized Officers of CCA can generate Yearly report of Digital Signature Certificates issued by each CA.
- Authorized Officers of CCA can generate Customized report for a specific period detailing eSign certificates issued by each CA.
- Authorized Officers of CCA can generate Customized report for a specific period detailing Digital Signature Certificates issued by each CA.
- Authorized Officers of CCA can generate Aggregate report on the total eSign certificates issued by all CAs over time.
- Authorized Officers of CCA can generate Aggregate report on the total Digital Signature Certificates issued by all CAs over time.
- Authorized Officers of CCA can generate Customized cumulative report for a specified period detailing total eSign certificates issued by CAs.
- Authorized Officers of CCA can generate Customized cumulative report for a specified period detailing total DSCs issued by CAs.
- Authorized Officers of CCA can generate Report on CAs and ESPs who obtained licenses during a specified period or since the beginning.
- Authorized Officers of CCA can generate Report on audits initiated within the scheduled timeframe.
- Authorized Officers of CCA can generate Report on audits that were delayed.
- Authorized Officers of CCA can generate Report on audit reports submitted on time.
- Authorized Officers of CCA can generate Report on audit reports that were submitted late.
- Authorized Officers of CCA can generate Report on the completion and closure of audits for each CA.
- Authorized Officers of CCA can generate Report detailing audits conducted by a specific auditor within a given period.
- Authorized Officers of CCA can generate Report on CAs whose licenses are due for renewal.
- Authorized Officers of CCA can generate Report on list of telephone numbers and email addresses of all licensed CAs and auditors.
- Authorized Officers of CCA can generate Report on the physical locations of CA sites, including primary and disaster recovery (DR) sites.

2. Applicant (Individual/ Company/ Firm/ Government Agency)

- Applicant can Submit/ Register her/ his Intent to become a CA using the Unique Code generated & shared by the O/o CCA.
- Logged-in applicant can submit the application based on the selected category of application and services to be offered by paying the applicable fee either online or offline, uploading payment proof and uploading required necessary documents, including CPS, to support the

application along with any third-party testing report.

- The applicant can rectify the discrepancies notified and attach fresh supporting documents, if required. This process will continue for maximum of 3 (three) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- The applicant can initiate the process of Audit at his/ her own end from one of the Empaneled Auditors, after due approval from the CCA.
- The applicant can appoint audit agency offline and inform the O/o CCA by entering the details of the appointed agency in the software for the said audit.
- The applicant can submit the action taken report along with the supporting documents against the issued NCs, if any.
- The applicant can fulfill the shortcomings and submit documents in support, if required. This correction process is limited to a maximum of 2 (two) attempts. Failure to address shortcomings within these attempts may lead to rejection of application or more attempts may be given to the applicant depending upon the Approval of CCA.
- The Applicant can submit the required Bank Guarantee, CSR, Agreement etc.
- The applicant can endorse/ digitally sign the application, documents, reports, etc.
- The applicant can track the submitted application.
- The applicant will be notified about application rejection.

3. Application Review Committee

- Application Review Committee can review the application & supporting documents including CPS, notify the applicant about discrepancies in application, if any.
- Application Review Committee can recommend the CCA for initiation of Audit upon successful review of the application.

4. Audit Agency/ Auditor

- Audit Agency can nominate the names of minimum 2 (two) Auditors for the conduction of said audit out of which 1 (one) must be having a valid CISA certification while the other should be at least a valid ISO 27001 certified professional. The Agency must upload documents in this regard.
- Audit Agency can submit undertakings, define the scope of audit and provide a schedule for the audit.
- On receipt of approval from the CCA regarding appointment of the Audit Agency and Audit Team, the Audit Team will commence the audit process.
- The Auditor can view the application & supporting documents, including CPS, for the audit.
- The Auditor can view previous audit reports, if any.

- On completion of audit, audit agency can submit the audit report along with list of NCs, if any.
- The auditor can verify action taken report and supporting documents submitted by the applicant against the issued NCs.
- Audit Agency can endorse/ digitally sign the submitted reports.

5. Report Review Committee

- Report Review Committee can review the uploaded audit report, NCs and action taken report, notifying the applicant of any identified shortcomings or pitfalls in the report.
- Report Review Committee can recommend for the issuance/ renewal of license to the CCA upon successful review of the Audit Report.

6. CCA

- The CCA can issue “in-principle approval” for grant of license to the applicant upon the conformance of the audit report by the Committee.
- Upon verification of bank guarantee, the CCA can issue CA License to the applicant on receipt of required CSR, Agreement, etc.
- The CCA can issue approval for providing eSign services to the Licensee on receipt of necessary approvals.
- The CCA can issue notice for suspension of license to the licensee upon non-acceptance of reply submitted by the Licensee.
- The CCA can issue an “in-principle approval” for the grant of license renewal to the licensee upon conformance of the Audit report by the Committee.
- Upon verification of bank guarantee, the CCA can issue Renewed License to the Licensee on receipt of required Agreement, and CSR etc.
- The CCA can issue approval for renewal of eSign services being provided by the Licensee on receipt of necessary approvals.
- The CCA can have a detailed graphical representation of all DSC and eSign certificate related data submitted by the licensees.

7. Licensee

- After becoming a CA, Licensee is required to submit a disclosure record.
- Licensee can submit application for becoming eSign Service Provider by selecting their eKYC method(s) and uploading the necessary approvals for these methods, downloading a system-generated cover letter, digitally signing it and uploading the same. Additionally, the Licensee must indicate the API versions they intend to use and accordingly upload the corresponding audit report & documents, including CPS.

- Upon successful scrutiny and review, the Licensee can submit required e-KYC approval documents obtained from concerned authorities, whichever be applicable.
- The Licensee can get performed the half yearly internal audits either by its own team or through an external agency and submit the reports.
- The Licensee can initiate the annual audit process as per the schedule shared by O/o CCA.
- The Licensee can pay the applicable fee and submit the application for renewal of the CA License by uploading the necessary required documents, including CPS & any third-party testing report, to support the application in between 45 to 90 days before the validity expiry of the existing license.
- The Licensee can rectify the discrepancies notified and attach fresh supporting documents, if required. This process will continue for maximum of 2 (two) attempts. Failure to address discrepancies within these attempts may lead to rejection of application or more attempts may be given to the licensee depending upon the Approval of CCA.
- The Licensee can appoint audit agency offline by entering the details of the appointed agency in the software for the said audit with the condition that same audit agency can't be engaged in consecutive years.
- The Licensee can submit the action taken report along with the supporting documents against the issued NCs, if any.
- The Licensee can fulfill the shortcomings and submit supporting documents, if required. This corrective process is limited to a maximum of 2 (two) attempts. Failure to address shortcomings within these attempts may lead to any stringent action like show-cause notice and/ or suspension of license from O/o CCA or more attempts may be given depending upon the Approval of CCA.
- The Licensee can submit reply to the show-cause notice issued by the O/o CCA.
- The Licensee can submit the required Bank Guarantee, Agreement, CSR etc., as applicable.
- After the renewal of CA License, Licensee can submit application for renewal of eSign Service choosing the appropriate API version and accordingly uploading the corresponding audit report.
- The Licensee can submit the application for cessation of license, issued by CCA, by uploading the necessary required documents (as per Section 21 of the IT Act), to support the application in between 60 to 90 days before the validity expiry of the license.
- The Licensees can submit their monthly data regarding Digital Signature Certificates (DSCs) & eSign certificates issued and Application Service Provider (ASP) on-boarded by them.
- The Licensee can endorse/ digitally sign the application, documents, reports, etc.
- The Licensee can track the submitted application.
- The Licensee will be notified about application rejection.
- The licensee can also request for additional services, such as relocating disaster recovery

(DR) sites, by submitting an ad hoc form.

- After becoming an eSign Service Provider (ESP), ESP is required to submit the disclosure record.
- Licensees can view the number of eSign and DSCs issued by them only, on a monthly, yearly, and cumulative basis.

2.4 Use Case Scenario

1. Intent for License

ID	UC01
Name	Unique Intent Code generation by Authorized Officer of CCA
Actor	Authorized Officer of CCA
Description	This use case allows the Authorized Officer of CCA to generate a unique code for intent registration by the interested user.
Business Rule	All the details submitted must be accurate
Special Requirement	The software must validate the completeness of the Intent Code generation form.
Assumption	The authorized officer fills the details correctly
Pre-Conditions	• The authorized officer successfully logged in to the software.
Dependencies	User database for login & Intent database for storage.
Post-Conditions	The intent code is generated & saved in the database.
Alternate Flow	NA

ID	UC02
Name	Intent Registration by the Applicant
Actor	Applicant/ Interested User
Description	This use case allows the Applicant to submit the intent registration form.
Business Rule	All the details submitted must be accurate
Special Requirement	The software must validate the completeness of the Intent registration form.
Assumption	The applicant fills the details correctly
Pre-Conditions	• The applicant has received the Unique Intent Code from the O/o CCA.
Dependencies	Intent database for storage.
Post-Conditions	The intent is successfully submitted and saved in the database.
Alternate Flow	NA

2. Verification & Login

ID	UC03
Name	Generate Login Credentials
Actor	System
Description	Login credentials will be generated for the interested user and shared over the registered email address/mobile.
Business Rule	• Credentials must be securely generated and stored. • Notification must be sent to the user's registered email address/mobile.
Special Requirement	• The software must have a secure method of generating and storing credentials (Username and Password). • The software must have the ability to send notifications.
Assumption	The user's verification has been completed successfully.
Pre-Conditions	• The authorized officer successfully logged in to the software. • The user has been verified by the Authorized Officer
Dependencies	• User database for storing credentials.

Software Requirements Specification (SRS) for
Licensing of Certifying Authorities and Dashboard for DSCs

	Email/SMS integration for sending notifications.
Post-Conditions	- Login credentials are generated and stored. - The user receives their credentials via email/SMS.
Alternate Flow	NA

ID	UC04
Name	Empanelled Audit Agency Registration
Actor	Authorized Officer of CCA
Description	The Authorized Officer of CCA can register the Empanelled Audit Agency by submitting the Audit Agency Registration Form.
Business Rule	The registration form must be complete and accurate.
Special Requirement	The software must validate the completeness of the registration form.
Assumption	The Authorized Officer has already received details of the empanelled Audit Agency.
Pre-Conditions	- The authorized officer successfully logged in to the software. - The Audit Agency Registration form is completed.
Dependencies	Audit Agency database for storing registration details.
Post-Conditions	The Audit Agency registration details have been successfully stored in the database.
Alternate Flow	NA

ID	UC05
Name	Generate Login Credentials
Actor	System
Description	Login credentials will be generated and shared over Audit Agencies' registered email address/mobile.
Business Rule	- Credentials must be securely generated and stored. - Notification must be sent to the Audit Agency's registered email address/mobile.
Special Requirement	The software must support secure credential generation processes.
Assumption	The Audit Agency details have already been submitted.
Pre-Conditions	- The authorized officer successfully logged in to the software. - The Audit Agency Registration form has been submitted.
Dependencies	- Audit Agency database for storing registration details. - Email/SMS integration for sending notifications.
Post-Conditions	- Login credentials are generated and stored. - The Audit Agency receives their credentials via email/SMS.
Alternate Flow	NA

ID	UC06
Name	User Login and Access via 2-Factor Authentication
Actor	Any User
Description	Only authorized users can login and access the software via a 2-factor authentication process and mechanism.
Business Rule	- Users must provide valid credentials. 2-factor authentication must be completed for access.
Special Requirement	The software must generate OTP and send it to the users via email/SMS.
Assumption	The user's account has already been created.
Pre-Conditions	Users have valid login credentials.
Dependencies	- User database for credentials. - Email/SMS integration for sending notifications.
Post-Conditions	The user is authenticated and granted access to the software based on their role.
Alternate Flow	NA

3. License Application

ID	UC07
Name	Application Submission and Payment Verification
Actor	Logged-in Applicant, Authorized Officer of CCA
Description	The logged-in applicant submits the application based on the selected category and services, pays the applicable fee either online or offline, uploads payment proof and required necessary documents, including CPS, to support the application, along with any third-party testing report. The O/o CCA is notified via email/SMS. The Authorized Officer of CCA verifies the receipt of payment and uploads the proof of payment received.
Business Rule	- Payment must be verified before further processing. - The applicant must submit all necessary documents.
Special Requirement	The system must support online payment and document uploads.
Assumption	- The applicant is logged into the system. - The applicant has selected the appropriate category. - The applicant has the necessary documents ready for upload.
Pre-Conditions	The applicant is logged into the system.
Dependencies	- Payment integration. - Email/SMS integration for sending notification. - Application database to store applicant application details.
Post-Conditions	- Application and payment details are stored in the system. - The O/o CCA is notified of the application submission.
Alternate Flow	NA

ID	UC08
Name	Application Scrutiny and Discrepancy Notification
Actor	Authorized Officer of CCA
Description	Upon verification of payment, the Authorized Officer of CCA scrutinizes the application and supporting documents, including CPS. Any discrepancies found are notified to the applicant via email/SMS. If the application is rejected, the applicant will be notified about the application rejection via email/SMS.
Business Rule	- Discrepancies must be notified to the applicant. - The applicant has a maximum of three attempts to rectify discrepancies.
Special Requirement	The system must track the number of attempts to rectify discrepancies. Notifications must be sent to the applicant regarding application status.
Assumption	The application and supporting documents are available for scrutiny.
Pre-Conditions	- Payment verification is completed. - Application and supporting documents are available for scrutiny.
Dependencies	Email/SMS integration for sending notification.
Post-Conditions	An application number will be generated and acknowledgement will be sent to applicant upon completion of scrutiny.
Alternate Flow	NA

ID	UC09
Name	Application Review and Discrepancy Notification
Actor	Application Review Committee
Description	Upon successful scrutiny of the application by the Authorized Officer of CCA, the Application Review Committee reviews the application and supporting documents, including CPS. Any discrepancies found are notified to the applicant via email/SMS. If the application is rejected, the applicant will be notified about the application rejection via email/SMS.
Business Rule	- Discrepancies must be notified to the applicant. - The applicant has a maximum of three attempts to rectify discrepancies.

Special Requirement	- The system must track the number of attempts to rectify discrepancies. - Notifications must be sent to the applicant regarding application status.
Assumption	The application has been scrutinized and discrepancies resolved.
Pre-Conditions	Application and supporting documents are available for review.
Dependencies	Email/SMS integration for sending notification.
Post-Conditions	Applicant is notified to initiate the audit process.
Alternate Flow	NA

4. Applicant Audit

ID	UC10
Name	Appointment of Audit Agency and Audit Team
Actor	Applicant, Audit Agency
Description	The Applicant appoints an audit agency offline and enters the details of the appointed agency in the software. The Audit Agency nominates a minimum of 2 auditors, uploads the required documents, and submits undertakings, defines the audit scope, and provides an audit schedule. Notifications are sent to relevant parties via email/SMS.
Business Rule	- One nominated auditor must have a valid CISA certification. - Another auditor must be a valid ISO 27001 certified professional.
Special Requirement	The system must allow entry and notification of audit agency and auditor details.
Assumption	The Applicant selects a qualified audit agency and the Audit Agency has qualified auditors.
Pre-Conditions	- Applicant is logged into the system. - Empaneled Audit Agency details are available.
Dependencies	Integration of email/SMS. Database to store appointed audit agency and auditor details.
Post-Conditions	Audit Agency and auditor details are recorded in the system.
Alternate Flow	NA

ID	UC11
Name	Audit Process and Report Submission
Actor	Audit Agency, Auditor, Applicant
Description	Upon receiving approval from the CCA, the Audit Team commences the audit process. The Auditor can view the application and supporting documents. After completing the audit, the Audit Agency submits the audit report along with any NCs. The Applicant submits an action taken report with supporting documents against the NCs, and the Auditor verifies these submissions. Notifications are sent via email/SMS.
Business Rule	Action taken reports must address all issued NCs.
Special Requirement	- The system must support document viewing and submission. - The system must track the audit process and NC resolution.
Assumption	- The Audit Team conducts the audit thoroughly. - The Applicant responds to NCs promptly.
Pre-Conditions	Approval from CCA for Audit Agency.
Dependencies	Integration of email/SMS
Post-Conditions	Audit report and action taken reports are submitted and verified.
Alternate Flow	NA

ID	UC12
Name	Report Review and Application Recommendation
Actor	Report Review Committee, Applicant
Description	The Report Review Committee reviews the uploaded audit report, NCs, and action taken reports. They notify the applicant of any shortcomings. The applicant addresses these shortcomings and resubmits supporting documents. If all shortcomings are resolved, the application is recommended for Issuance of License. Notifications are sent via email/SMS.
Business Rule	Shortcomings must be addressed within a maximum of two attempts. Further attempts may be granted with CCA approval.
Special Requirement	Digital signature functionality is required.
Assumption	- The Report Review Committee conducts a thorough review. - The Applicant responds to shortcomings promptly.
Pre-Conditions	- Audit report and action taken reports are submitted. - Shortcomings are identified.
Dependencies	- Integration of email/SMS. - Digital signature system.
Post-Conditions	Application is recommended for Issuance of License if all shortcomings are resolved.
Alternate Flow	NA

5. License Issuance

ID	UC13
Name	Issuance of in-principle approval
Actor	CCA, Applicant
Description	The CCA issues “in-principle approval” for the grant of a license to the applicant upon the conformance of the audit report by the Committee. The Applicant is notified via email/SMS.
Business Rule	Approval is granted only after audit report conformance by the Committee.
Special Requirement	The system must support the issuance and notification process.
Assumption	The audit report has been passed by the Committee.
Pre-Conditions	Audit report conformance by the Committee.
Dependencies	Notification system for email/SMS.
Post-Conditions	In-principle approval is issued to the applicant and applicant is notified.
Alternate Flow	NA

ID	UC14
Name	Submission and Verification of Required Documents
Actor	Applicant, Authorized Officer of CCA
Description	The Applicant submits the required Bank Guarantee, CSR, and Agreement, as applicable. The Authorized Officer of CCA verifies the receipt of the bank guarantee and uploads the proof of confirmation received. The O/o CCA is notified via email/SMS.
Business Rule	- The submitted documents must meet the required standards. - Verification of bank guarantee must be completed before proceeding.
Special Requirement	The system must support document submission and verification.
Assumption	The Applicant submits all required documents correctly.
Pre-Conditions	In-principle approval is granted.
Dependencies	Integration of email/SMS.
Post-Conditions	- Required documents are submitted and verified. - Proof of bank guarantee confirmation is uploaded.
Alternate Flow	NA

6. eSign Application

ID	UC16
Name	Application for becoming eSign Service Provider
Actor	Licensee
Description	After becoming a CA, the Licensee submits an application for becoming an eSign Service Provider by selecting their eKYC method(s), uploading necessary approvals, downloading a system-generated cover letter, digitally signing it, and uploading it. The Licensee must indicate the API versions they intend to use and upload the corresponding audit report & documents, including CPS. The O/o CCA is notified via email/SMS.
Business Rule	- The Licensee must submit all required documents and approvals. - Digital signing of the cover letter is mandatory.
Special Requirement	The system must support document upload, cover letter generation, and digital signing.
Assumption	The Licensee has all necessary approvals and documents ready for submission.
Pre-Conditions	The Licensee has obtained a CA license.
Dependencies	- Database to store eSign application details. - Integration of email/SMS.
Post-Conditions	The application for becoming an eSign Service Provider is submitted and the O/o CCA is notified.
Alternate Flow	NA

ID	UC17
Name	Scrutiny and Review of eSign Service Provider Application
Actor	Authorized Officer of CCA, Licensee
Description	The Authorized Officer of CCA scrutinizes and reviews the submitted application, along with supporting documents, including CPS, and applicable fee, if any. Any discrepancies found are notified to the licensee via email/SMS.
Business Rule	All submitted documents and fees must be reviewed for accuracy and completeness and discrepancies must be notified promptly.
Special Requirement	Notifications must be sent to the licensee.
Assumption	The Licensee has submitted all required documents and fees.
Pre-Conditions	The application for becoming an eSign Service Provider has been submitted.
Dependencies	Integration of email/SMS.
Post-Conditions	Discrepancies are recorded and notified to the licensee.
Alternate Flow	NA

ID	UC18
Name	Discrepancy Rectification and Application Approval
Actor	Licensee, Authorized Officer of CCA
Description	The Licensee rectifies notified discrepancies and attaches supporting documents, if any. The O/o CCA is notified via email/SMS. This process continues for a maximum of 2 attempts. Failure to address discrepancies within these attempts may lead to application rejection or more attempts may be given based on CCA approval. Upon successful rectification, the CCA issues approval for providing eSign services to the Licensee.
Business Rule	The Licensee has a maximum of 2 attempts to rectify discrepancies. Further attempts may be granted with CCA approval.
Special Requirement	The system must track the number of rectification attempts.
Assumption	The Licensee promptly addresses and rectifies discrepancies.
Pre-Conditions	Discrepancies have been notified to the Licensee.
Dependencies	Integration of email/SMS
Post-Conditions	Discrepancies are rectified, or the application is rejected.

	The Licensee is notified of the final decision.
Alternate Flow	NA

7. Internal & Annual Audit

ID	UC19
Name	Internal & Annual Audit
Actor	Authorized Officer of CCA, Licensee, Audit Agency, Auditor, Report Review Committee, CCA
Description	The Authorized Officer of CCA prepares and publishes a schedule for annual audits. The Licensee conducts half-yearly internal audits and submits reports. The Licensee initiates the annual audit, appoints an audit agency, and the Audit Agency nominates auditors, defines scope and schedule, and commences the audit. Audit reports are submitted, and discrepancies are addressed. The CCA reviews and marks the audit process complete or issues a notice if shortcomings are not resolved. Notifications are sent via email/SMS.
Business Rule	- Internal audits must be performed every six months. - For annual audit same audit agency or auditors cannot be engaged in consecutive years. - Shortcomings must be addressed within two attempts.
Special Requirement	- The system must support schedule preparation, report submission, document upload, and notification processes. - Digital signature functionality is required.
Assumption	The Authorized Officer prepares schedule for annual audit.
Pre-Conditions	- A list of Licensees is available. - The audit schedule has been published.
Dependencies	- Integration of email/SMS. - Digital Signature integration.
Post-Conditions	- The audit schedule is published and Licensees are notified. - Internal audit reports are submitted and the O/o CCA is notified. - The annual audit process is initiated and conducted as per schedule. - The audit report and action taken report are submitted, reviewed, and verified. - The audit process is marked complete or the license is suspended.
Alternate Flow	NA

8. Renewal Application

ID	UC20
Name	Application Submission and Payment Verification
Actor	Licensee, Authorized Officer of CCA
Description	The Licensee submits the application based on the selected category and services, pays the applicable fee either online or offline, uploads payment proof and required necessary documents, including CPS, to support the application, along with any third-party testing report. The O/o CCA is notified via email/SMS. The Authorized Officer of CCA verifies the receipt of payment and uploads the proof of payment received.
Business Rule	- Payment must be verified before further processing. - The Licensee must submit all necessary documents.
Special Requirement	The system must support online payment and document uploads.
Assumption	- The Licensee is logged into the system. - The Licensee has selected the appropriate category. - The Licensee has the necessary documents ready for upload.
Pre-Conditions	The Licensee is logged into the system.
Dependencies	Payment integration.

	- Email/SMS integration for sending notification. - Application database to store Licensee application details.
Post-Conditions	- Application and payment details are stored in the system. - The O/o CCA is notified of the application submission.
Alternate Flow	NA

ID	UC21
Name	Application Scrutiny and Discrepancy Notification
Actor	Authorized Officer of CCA
Description	Upon verification of payment, the Authorized Officer of CCA scrutinizes the application and supporting documents, including CPS. Any discrepancies found are notified to the applicant via email/SMS. If the application is rejected, the applicant will be notified about the application rejection via email/SMS.
Business Rule	- Discrepancies must be notified to the Licensee. - The applicant has a maximum of three attempts to rectify discrepancies.
Special Requirement	The system must track the number of attempts to rectify discrepancies. Notifications must be sent to the Licensee regarding application status.
Assumption	The application and supporting documents are available for scrutiny.
Pre-Conditions	- Payment verification is completed. - Application and supporting documents are available for scrutiny.
Dependencies	Email/SMS integration for sending notification.
Post-Conditions	An application number will be generated and acknowledgement will be sent to licensee upon completion of scrutiny.
Alternate Flow	NA

ID	UC22
Name	Application Review and Discrepancy Notification
Actor	Application Review Committee
Description	Upon successful scrutiny of the application by the Authorized Officer of CCA, the Application Review Committee reviews the application and supporting documents, including CPS. Any discrepancies found are notified to the applicant via email/SMS. If the application is rejected, the applicant will be notified about the application rejection via email/SMS.
Business Rule	- Discrepancies must be notified to the licensee. - The licensee has a maximum of three attempts to rectify discrepancies.
Special Requirement	- The system must track the number of attempts to rectify discrepancies. - Notifications must be sent to the licensee regarding application status.
Assumption	The application has been scrutinized and discrepancies resolved.
Pre-Conditions	Application and supporting documents are available for review.
Dependencies	Email/SMS integration for sending notification.
Post-Conditions	Licensee is notified to initiate the audit process.
Alternate Flow	NA

9. Renewal Audit

ID	UC23
Name	Appointment of Audit Agency and Audit Team
Actor	Licensee, Audit Agency
Description	The Licensee appoints an audit agency offline and enters the details of the appointed agency in the software. The Audit Agency nominates a minimum of 2 auditors, uploads the required documents, and submits undertakings, defines the audit scope, and provides an audit schedule. Notifications are sent to relevant parties via email/SMS.
Business Rule	- One nominated auditor must have a valid CISA certification. - Another auditor must be a valid ISO 27001 certified professional.

Special Requirement	The system must allow entry and notification of audit agency and auditor details.
Assumption	The Licensee selects a qualified audit agency and the Audit Agency has qualified auditors.
Pre-Conditions	• Licensee is logged into the system. • Empaneled Audit Agency details are available.
Dependencies	Integration of email/SMS. Database to store appointed audit agency and auditor details.
Post-Conditions	Audit Agency and auditor details are recorded in the system.
Alternate Flow	NA

ID	UC24
Name	Audit Process and Report Submission
Actor	Audit Agency, Auditor, Licensee
Description	Upon receiving approval from the CCA, the Audit Team commences the audit process. The Auditor can view the application and supporting documents. After completing the audit, the Audit Agency submits the audit report along with any NCs. The Licensee submits an action taken report with supporting documents against the NCs, and the Auditor verifies these submissions. Notifications are sent via email/SMS.
Business Rule	Action taken reports must address all issued NCs.
Special Requirement	• The system must support document viewing and submission. • The system must track the audit process and NC resolution.
Assumption	• The Audit Team conducts the audit thoroughly. • The Licensee responds to NCs promptly.
Pre-Conditions	Approval from CCA for Audit Agency.
Dependencies	Integration of email/SMS
Post-Conditions	Audit report and action taken reports are submitted and verified.
Alternate Flow	NA

ID	UC25
Name	Report Review and Application Recommendation
Actor	Report Review Committee, Licensee
Description	The Report Review Committee reviews the uploaded audit report, NCs, and action taken reports. They notify the Licensee of any shortcomings. The applicant addresses these shortcomings and resubmits supporting documents. If all shortcomings are resolved, the application is recommended for Issuance of License. Notifications are sent via email/SMS.
Business Rule	Shortcomings must be addressed within a maximum of two attempts. Further attempts may be granted with CCA approval.
Special Requirement	Digital signature functionality is required.
Assumption	• The Report Review Committee conducts a thorough review. • The Licensee responds to shortcomings promptly.
Pre-Conditions	• Audit report and action taken reports are submitted. • Shortcomings are identified.
Dependencies	• Integration of email/SMS. • Digital signature system.
Post-Conditions	Application is recommended for Issuance of License if all shortcomings are resolved.
Alternate Flow	NA

10. License Renewal

ID	UC26
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Name	Issuance of in-principle approval
Actor	CCA, Licensee
Description	The CCA issues “in-principle approval” for the renewal of a license to the Licensee upon the conformance of the audit report by the Committee. The Licensee is notified via email/SMS.
Business Rule	Approval is granted only after audit report conformance by the Committee.
Special Requirement	The system must support the issuance and notification process.
Assumption	The audit report has been passed by the Committee.
Pre-Conditions	Audit report conformance by the Committee.
Dependencies	Notification system for email/SMS.
Post-Conditions	In-principle approval is issued to the licensee and the licensee is notified.
Alternate Flow	NA

ID	UC27
Name	Submission and Verification of Required Documents
Actor	Licensee, Authorized Officer of CCA
Description	The Licensee submits the required Bank Guarantee and Agreement, as applicable. The Authorized Officer of CCA verifies the receipt of the bank guarantee and uploads the proof of confirmation received. The O/o CCA is notified via email/SMS.
Business Rule	• The submitted documents must meet the required standards. • Verification of bank guarantee must be completed before proceeding.
Special Requirement	The system must support document submission and verification.
Assumption	The Licensee submits all required documents correctly.
Pre-Conditions	In-principle approval is granted.
Dependencies	Integration of email/SMS.
Post-Conditions	• Required documents are submitted and verified. • Proof of bank guarantee confirmation is uploaded.
Alternate Flow	NA

ID	UC28
Name	Renewal of CA License
Actor	CCA, Licensee
Description	Upon verification of the bank guarantee, the CCA renews the CA License to the Licensee on receipt of the required agreement. The Licensee is notified via email/SMS.
Business Rule	License is issued only after receipt and verification of all required documents.
Special Requirement	The system must support the issuance and notification process.
Assumption	All required documents are verified and received.
Pre-Conditions	Required documents are submitted and verified.
Dependencies	• Database to store license details. • Integration of email/SMS
Post-Conditions	CA License renewed and the Licensee is notified.
Alternate Flow	NA

11. Renewal of eSign Service

ID	UC29
Name	Application to renew eSign Service
Actor	Licensee
Description	Upon renewal of the License, the Licensee submits an application for renewal of eSign Service by selecting their eKYC method(s), uploading necessary approvals, downloading a system-generated cover letter, digitally signing it, and uploading it.

	The Licensee must indicate the API versions they intend to use and upload the corresponding audit report & documents, including CPS. The O/o CCA is notified via email/SMS.
Business Rule	- The Licensee must submit all required documents and approvals. - Digital signing of the cover letter is mandatory.
Special Requirement	The system must support document upload, cover letter generation, and digital signing.
Assumption	The Licensee has all necessary approvals and documents ready for submission.
Pre-Conditions	The Licensee has renewed.
Dependencies	- Database to store renewal of eSign application details. - Integration of email/SMS.
Post-Conditions	The application for renewal of eSign Service is submitted and the O/o CCA is notified.
Alternate Flow	NA

ID	UC30
Name	Scrutiny and Review of eSign Service Provider Renewal Application
Actor	Authorized Officer of CCA, Licensee
Description	The Authorized Officer of CCA scrutinizes and reviews the submitted application, along with supporting documents, including CPS, and applicable fee, if any. Any discrepancies found are notified to the licensee via email/SMS.
Business Rule	All submitted documents and fees must be reviewed for accuracy and completeness and discrepancies must be notified promptly.
Special Requirement	Notifications must be sent to the licensee.
Assumption	The Licensee has submitted all required documents and fees.
Pre-Conditions	The application for renewal of eSign Service Provider has been submitted.
Dependencies	Integration of email/SMS.
Post-Conditions	Discrepancies are recorded and notified to the licensee.
Alternate Flow	NA

ID	UC31
Name	Discrepancy Rectification and Application Approval
Actor	Licensee, Authorized Officer of CCA
Description	The Licensee rectifies notified discrepancies and attaches supporting documents, if any. The O/o CCA is notified via email/SMS. This process continues for a maximum of 2 attempts. Failure to address discrepancies within these attempts may lead to application rejection or more attempts may be given based on CCA approval. Upon successful rectification, the CCA issues approval for providing eSign services to the Licensee.
Business Rule	The Licensee has a maximum of 2 attempts to rectify discrepancies. Further attempts may be granted with CCA approval.
Special Requirement	The system must track the number of rectification attempts.
Assumption	The Licensee promptly addresses and rectifies discrepancies.
Pre-Conditions	Discrepancies have been notified to the Licensee.
Dependencies	Integration of email/SMS
Post-Conditions	- Discrepancies are rectified, or the application is rejected. - The Licensee is notified of the final decision.
Alternate Flow	NA

12. Cessation of License

ID	UC32
Name	Application for Cessation of License
Actor	Licensee, Authorized Officer of CCA

Description	The Licensee submits an application for cessation of license by uploading necessary documents (as per Section 21 of the IT Act) between 60 to 90 days before license expiry. The Authorized Officer scrutinizes the application and documents, notifying the Licensee of any discrepancies. The Licensee rectifies discrepancies and resubmits documents. Upon successful review, an application number is generated, and the license is ceased upon expiry after CCA approval. Notifications are sent via email/SMS.
Business Rule	- Application must be submitted between 60 to 90 days before license expiry. - All discrepancies must be addressed within 2 attempts or as approved by CCA.
Special Requirement	- Application must be submitted between 60 to 90 days before license expiry. - All discrepancies must be addressed within 2 attempts or as approved by CCA.
Assumption	The Licensee submits all required documents accurately and within the specified timeframe.
Pre-Conditions	The Licensee has a valid CA license nearing expiry.
Dependencies	- Database to store application details. - Integration of email/SMS for providing notification.
Post-Conditions	An application number is generated and the license is ceased upon expiry after CCA approval.
Alternate Flow	NA

13. Dashboard & Reports

ID	UC33
Name	Dashboard & Reports
Actor	Licensee, Authorized Officer(O/o CCA, CCA)
Description	Licensees can submit their monthly data regarding DSCs and eSign certificates issued, and can view this data graphically using charts and graphs. The Authorized Officer can view graphical representations of data regarding DSC and eSign certificates issued, as well as generate different types of reports.
Business Rule	- Monthly data must be submitted accurately and on time. - The system should provide accurate and up-to-date graphical representations.
Special Requirement	Monthly data must be submitted accurately and on time.
Assumption	The Licensee submits accurate monthly data.
Pre-Conditions	- The Licensee submits accurate monthly data. - The Authorized Officer has the necessary access to view and generate reports.
Dependencies	Database to get all the data's regarding DSCs & eSign.
Post-Conditions	- Monthly data is submitted by the Licensee. - Graphical representations of the data are available. - Reports are generated by the Authorized Officer as required.
Alternate Flow	NA

2.5 Operating Environment

Hardware platform:

Processor – Octa core processor with min. 2GHz base frequency & 12MB cache memory

RAM – min. 16 GB

Hard Disk – Free disk space of min. 1 TB

Software platform:

Operating System – Red Hat 8 or advanced

Web Server – Tomcat 9 or advanced

Database Server – PostgreSQL 14 or advanced

Communication platform:

24x7 Network Connection with High bandwidth

2.6 Design and Implementation Constraints

Design and Implementation constraints are put in place to ensure that the project will be delivered within the agreed timeline, budget and the final software meets the specified requirements without scope creep or feature bloat.

Technological Constraint:

The software must be compatible with the designated technology stack, including programming languages, frameworks, libraries, and development tools specified in the project requirements. Compatibility issues with third-party integrations will be addressed to ensure seamless interoperability. Integration with external systems, services, or APIs may pose technical challenges, such as data format mismatches, authentication requirements or communication protocols. Compatibility testing and validation may be conducted to ensure smooth integration and interoperability with third-party components.

Resource Constraint:

The scope of work will be strictly as per the approved Software Requirements Specification (SRS) document. The SRS document serves as a blueprint for the development team, detailing the functional and non-functional requirements, user interfaces and other essential aspects of the software to be developed. Adhering to the approved SRS ensures that the final product meets the expectations of CCA and fulfils the intended purpose of the software. Any deviation from the SRS will require formal approval through the change management process.

The design, development and implementation phase of the project is to be completed within one year duration from the date of award of project. The duration of one year provides a timeline for completing the initial development efforts and delivering a functional software product ready for deployment. During this period, the project team will work on conceptualizing, designing, implementing, and testing the software solution as per the requirements specified in the SRS.

During O&M phase, the project team will ensure the continuous operation of the software, address any issues that arise, and implement necessary updates or enhancements to maintain its functionality and performance.

External Constraint:

The software must deal with various types of data generated or provided by users during its operation which may include personal information, financial records, transaction details, preferences and any other data relevant to the software's functionality.

The software must implement robust data handling practices including encryption, access controls, authentication mechanisms, audit trails and data retention policies, to ensure the confidentiality, integrity, and availability of user data throughout its lifecycle.

Privacy policies and terms of service must be clearly communicated to users, outlining the purposes for which their data will be collected, processed, and shared, as well as their rights regarding data access, correction, and deletion.

Operational Constraint:

The software should be designed to accommodate various user roles, each with distinct permissions and access privileges based on their functional responsibilities and organizational hierarchy. User interfaces should be designed to reflect different access levels and functionalities available to each user role.

The software must implement robust access control mechanism to enforce security policies and regulate user interactions based on their assigned roles and permissions. The software must incorporate authentication mechanisms to verify the identity of users accessing the software. Authentication methods may include username/ password authentication and multi-factor authentication (MFA). Password policies, such as minimum length, complexity requirements, and expiration periods, should be enforced to ensure the strength and integrity of user credentials.

Once users are authenticated, the software must enforce authorization rules to determine the actions they are permitted to perform based on their roles and privileges. Authorization mechanisms validate whether a user has the necessary permissions to access specific resources or perform certain operations within the software.

Performance Constraint:

The software architecture must be designed to scale horizontally and vertically to accommodate increasing loads and user traffic. Load balancing techniques may be implemented to distribute incoming

requests evenly across multiple servers or instances, ensuring optimal resource utilization and preventing overloading of any single component.

The software must be capable of handling multiple concurrent requests from users or external systems without experiencing performance degradation or resource contention. Caching mechanism may be employed to store frequently accessed data or pre-computed results in memory or disk cache, reducing the need for repeated computation and database queries.

The software must implement fault-tolerant mechanism to ensure software reliability and availability in the event of failures or errors. The software must incorporate robust error handling strategies to gracefully manage exceptions and prevent software crashes that could impact performance.

Implementation Constraint:

The software must be implemented using specific programming languages, frameworks, libraries, and development tools as specified in the project requirements. The software development process must follow predefined coding standards and best practices established by the organization or industry guidelines.

Robust quality assurance (QA) processes, including code reviews, automated testing, manual testing, and user acceptance testing (UAT), must be implemented to identify and address defects, bugs, and usability issues throughout the development lifecycle.

2.7 User Documentation

User documentation will be available in the form of a User Manual and will be provided upon completion of the development and implementation works.

2.8 Assumptions and Dependencies

Assumptions and dependencies play a crucial role in shaping the design, functionality, and usability of a software. These assumptions provide a foundation for the software's requirements and help in setting expectations regarding its usage and performance.

Technical Assumption:

The software assumes compatibility with the latest version of popular operating systems such as Windows, macOS, and Linux distributions. It also assumes compatibility with modern web browsers including but not limited to Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, ensuring a consistent user experience across different platforms and devices.

The software assumes a minimum hardware configuration of 16GB RAM, 1 TB of hard disk space and an octa-core processor to ensure adequate performance and responsiveness under typical usage scenarios. While the software may run on lower-spec hardware configurations, optimal performance and user experience are expected on systems meeting or exceeding the specified hardware requirements.

The software assumes reliable internet connectivity to access remote resources, cloud services, or external APIs as required by its functionality. It also assumes support for standard network protocols such as HTTP/ HTTPS for web communication, ensuring seamless integration with other web-based services and platforms.

The software assumes support for a minimum screen resolution of 1024x768 pixels to accommodate its user interface (UI) components and content layout. It also assumes compatibility with various display sizes and aspect ratios, including desktop monitors, laptops, tablets, and mobile devices, through responsive design or adaptive layout techniques.

Operational Assumption:

The software assumes that users possess basic computer literacy skills, including familiarity with common input devices such as keyboard and mouse, as well as basic knowledge of operating system navigation and file management. Users are expected to understand fundamental concepts such as clicking, dragging, typing, and navigating through graphical user interfaces (GUIs) to interact with the software effectively.

The software assumes that users have a basic understanding of the English language, as the user interface, error messages, and documentation are predominantly presented in English. Users are assumed to have prior experience or familiarity with similar digital tool that share common features, functionalities, or workflows. This familiarity enables users to leverage their existing knowledge and skills to adapt quickly to the software's interface, functionality, and usage patterns, minimizing the learning curve and increasing user productivity.

Business Assumption:

Users are expected to have access to reliable internet connectivity, either through wired or wireless connections, to ensure uninterrupted access to the software's features and functionalities. Inadequate or unstable internet connectivity may impede the user experience, result in data transmission errors, and disrupt critical business operations reliant on the software. The availability of a stable internet connection is crucial for accessing online services, data exchange, and transaction processing within the software.

The assumption of stable internet connectivity enables users to access the software from various locations, devices, and network environments, enhancing flexibility and convenience in usage.

The software assumes that users will rely on online services and transactions as part of their routine activities or business operations. This includes activities such as accessing digital content, submitting forms, making payments, communicating with other users or stakeholders, and retrieving real-time information from external sources. The assumption of online service availability and transactional capability aligns with modern business practices and consumer behaviour, facilitating digital interactions and transactions in an increasingly connected world.

Environmental Assumption:

The software assumes that users, particularly those involved in licensing and audit-related works in connection to the Controller of Certifying Authorities (CCA), possess familiarity with relevant acts, policies, procedures, regulations, and guidelines governing digital certification and authentication. This includes but is not limited to acts such as the Information Technology (IT) Act, 2000, policies and procedures established by the CCA, regulations governing digital signatures and certificates, and guidelines issued by regulatory authorities or industry bodies. Users are expected to have a basic understanding of the legal and regulatory framework surrounding digital certification and authentication, including compliance requirements, obligations, and best practices for ensuring the security and integrity of digital transactions.

The software assumes that users responsible for licensing and audit-related activities will adhere to relevant licensing requirements and undergo periodic audits to ensure compliance with regulatory standards and industry guidelines. Users are expected to follow established procedures for obtaining licenses, certifications, or authorizations from regulatory authorities, as well as maintaining accurate records, conducting internal audits, and cooperating with external auditors as necessary.

3. External Interface Requirements

3.1 User Interfaces

Below are few basic details regarding user interfaces of the Portal:-

1. Sign In Form:

- The sign-in form facilitates the sign-in process for users with existing accounts within the software. Users can access their account by entering their username, password, and completing the captcha verification.

Sr. No.	Column Name	Data Type	Validation
1	Username	VARCHAR(20)	Required
2	Password	VARCHAR(75)	Required, Length(8-75), Complexity(at least one uppercase letter, one lowercase letter, one number, and one special character)
3	Captcha	VARCHAR(6)	Required, Must match with the generated captcha

2. OTP Form:

- The OTP form ensures additional security by dispatching a one-time password (OTP) to the user's registered email address and mobile number. Users are required to enter the OTP to proceed with the sign-in process.

Sr. No.	Column Name	Data Type	Validation
1	OTP	VARCHAR(6)	Required, Numeric, Fixed Length (6 digits)
2	Captcha	VARCHAR(6)	Required, Must match with the generated captcha

3. Intent Submission

- The intent form captures the basic details of users (Individual/ Company/ Firm/ Government Agency) interested in becoming a licensed Certifying Authority (CA).

i) Personal Details

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR(6)	Required
2	First Name	VARCHAR(30)	Required, Alphabets
3	Middle Name	VARCHAR(30)	Alphabets
4	Last Name/ Surname	VARCHAR(45)	Alphabets
5	Mobile Number	VARCHAR(10)	Required, Number
6	Email Id	VARCHAR(50)	Required, Alphanumeric

ii) Address

Sr. No.	Column Name	Data Type	Validation
1	Flat/ Door/ Block No	VARCHAR(15)	Required, Alphanumeric
2	Name of Premises/ Building/ Village	VARCHAR(30)	Required, Alphabets
3	Road/ Street/ Lane/ Post Office	VARCHAR(30)	Required, Alphanumeric
4	Area/ Locality/ Taluka/ Sub-Division	VARCHAR(30)	Required, Alphabets
5	Town/ City/ District	VARCHAR(30)	Required, Alphabets
6	State/ Union Territory	VARCHAR(40)	Required, Alphabets
7	Pincode	VARCHAR(6)	Required, Numeric
8	Mobile No.	VARCHAR (10)	Required, Numeric

4. New License Application Form:

- The Application Form streamlines the submission process for new license applications requests. It features meticulously structured form fields to capture all pertinent information accurately.
- Users have the capability to seamlessly upload supporting documents, enhancing the completeness of the application process.

Sr. No.	Column Name	Data Type	Validation
1	Application Category	VARCHAR(17)	Required

a. Form for Individual

i) Personal Details

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR (5)	Required
2	First Name	VARCHAR(30)	Required, Only Alphabets
3	Middle Name	VARCHAR(30)	Only Alphabets
4	Last Name/ Surname	VARCHAR(45)	Only Alphabets
5	Salutation (Other)	VARCHAR (5)	Required
6	First Name (Other)	VARCHAR(30)	Only Alphabets
7	Middle Name (Other)	VARCHAR(30)	Only Alphabets
8	Last Name/ Surname (Other)	VARCHAR(45)	Only Alphabets

ii) Residence Address

Sr. No.	Column Name	Data Type	Validation
1	Flat/ Door/ Block No	VARCHAR(15)	Required, Alphanumeric
2	Name of Premises/ Building/ Village	VARCHAR(30)	Required, Alphabets
3	Road/ Street/ Lane/ Post Office	VARCHAR(30)	Required, Alphanumeric
4	Area/ Locality/ Taluka/ Sub-Division	VARCHAR(30)	Required, Alphabets
5	Town/ City/ District	VARCHAR(30)	Required, Alphabets
6	State/ Union Territory	VARCHAR(40)	Required, Alphabets
7	Pincode	VARCHAR(6)	Required, Number
8	Telephone No.	VARCHAR(10)	Number
9	Fax	VARCHAR(12)	Number
10	Mobile No.	VARCHAR (10)	Required, Number

iii) Office Address

Sr. No.	Column Name	Data Type	Validation
1	Name of Office	VARCHAR (75)	Required, Alphanumeric
2	Flat/ Door/ Block No	VARCHAR(15)	Required, Alphanumeric
3	Name of Premises/ Building/ Village	VARCHAR(30)	Required, Alphabets
4	Road/ Street/ Lane/ Post Office	VARCHAR(30)	Required, Alphanumeric
5	Area/ Locality/ Taluka/ Sub-Division	VARCHAR(30)	Required, Alphabets
6	Town/ City/ District	VARCHAR(30)	Required, Alphabets
7	State/ Union Territory	VARCHAR(40)	Required, Alphabets
8	Pincode	VARCHAR(6)	Required, Numeric
9	Telephone No.	VARCHAR(10)	Required, Numeric
10	Fax	VARCHAR(12)	Numeric

iv) Communication Address

Sr. No.	Column Name	Data Type	Validation
1	Address For Communication (Residence/ Office Address)	LONG(9)	Required

v) Additional Details

Sr. No.	Column Name	Data Type	Validation
1	Father's Salutation	VARCHAR (5)	Required
2	Father's First Name	VARCHAR(30)	Required, Alphabets
3	Father's Middle Name	VARCHAR(30)	Alphabets
4	Father's Last Name/ Surname	VARCHAR(45)	Alphabets
5	Sex (Male/ Female)	VARCHAR(6)	Required, Alphabets
6	Date of Birth	DATE	Required, DATE
7	Nationality	VARCHAR(56)	Required, Alphabets
8	Email Id	VARCHAR(50)	Alphanumeric

vi) Financial and additional Details

Sr. No.	Column Name	Data Type	Validation
1	Credit Card Type	VARCHAR(30)	Alphabets
2	Credit Card No.	VARCHAR(20)	Number
3	Issued By	VARCHAR(75)	Alphabets
4	Web URL address	VARCHAR(50)	Alphanumeric
5	Passport No.	VARCHAR(10)	Alphanumeric, any one
6	Passport Issuing Authority	VARCHAR(75)	Alphabets
7	Passport Expiry Date	DATE	DATE
8	Voter's Identity Card No.	VARCHAR(20)	Alphanumeric, any one
9	Income Tax PAN No.	VARCHAR(10)	Alphanumeric, any one
10	ISP Name	VARCHAR(35)	Required, Alphabet
11	ISP's Website Address	VARCHAR(50)	Alphanumeric
12	ISP Username	VARCHAR(50)	Alphanumeric
13	Personal Web Page URL	VARCHAR(50)	Alphanumeric
14	Capital in the business or profession	VARCHAR(10)	Required, Numeric

vii) Form for Bank Details & Location of facility in India

Sr. No.	Column Name	Data Type	Validation
1	Bank Name	VARCHAR(75)	Required, Alphabets
2	Branch	VARCHAR(50)	Required, Alphanumeric
3	Bank Account No.	VARCHAR(18)	Required, Alphanumeric
4	Type of Bank Account	VARCHAR(50)	Required, Alphabets

viii) Location of facility in India for generation of DSC

Sr. No.	Column Name	Data Type	Validation
1	Location 1 (Primary)	VARCHAR(50)	Required, Alphanumeric
2	Location 2 (DR Site)	VARCHAR(50)	Alphanumeric
3	Location 3 (Other)	VARCHAR(50)	Alphanumeric

ix) CPS Document Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	CPS Document	VARCHAR(20)	Doc File, Required

x) Necessary Documents Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	Other Documents (numbers)	VARCHAR(20)	Doc/ PDF file

xi) Bank Draft/ Pay Order Form

Sr. No.	Column Name	Data Type	Validation
1	Name of Bank	VARCHAR(75)	Required, Alphabets
2	Draft/ Pay order No.	VARCHAR(10)	Required, Alphanumeric
3	Date of Issue	DATE	Required
4	Amount	VARCHAR(6)	Required, Number

b. Form For Company/ Firm/ Body of Individuals/ Association of Persons/ Local Authority

i) Business Details Form

Business Details

Sr. No.	Column Name	Data Type	Validation
1	Registration Number	VARCHAR(21)	Required, Alphanumeric
2	Date of Incorporation/ Agreement/ Partnership	DATE	Required

Particulars of Business

Sr. No.	Column Name	Data Type	Validation
1	Head Office	VARCHAR(75)	Required, Alphabets
2	Name of Office	VARCHAR (75)	Required, Alphabets
3	Flat/ Door/ Block No	VARCHAR(15)	Required, Alphanumeric
4	Name of Premises/ Building/ Village	VARCHAR(30)	Required, Alphabets
5	Road/ Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
6	Area/ Locality/ Taluka/ Sub-Division	VARCHAR(30)	Required, Alphabets
7	Town/ City/ District	VARCHAR(30)	Required, Alphabets
8	State/ Union Territory	VARCHAR(40)	Required, Alphabets
9	Pincode	VARCHAR(6)	Required, Numeric
10	Telephone No.	VARCHAR(10)	Required, Numeric
11	Fax	VARCHAR(12)	Numeric
12	Web Page URL address	VARCHAR(50)	Alphanumeric
13	No. of Branches	VARCHAR(6)	Required, Numeric
14	Nature of Business	VARCHAR(35)	Required, Alphabets

Additional Details

Sr. No.	Column Name	Data Type	Validation
1	PAN No.	VARCHAR(10)	Required, Alphanumeric
2	Turnover (last financial year)	VARCHAR(16)	Required, Numeric
3	Net Worth	VARCHAR(16)	Required, Numeric
4	Paid up Capital	VARCHAR(16)	Required, Numeric
5	Insurance Policy No.	VARCHAR(16)	Required, Alphanumeric
6	Insurance Company	VARCHAR(75)	Required, Alphabet

ii) Details of Partners/ Members/ Directors

Personal Details

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR (5)	Required
2	First Name	VARCHAR(30)	Required, Alphabets
3	Middle Name	VARCHAR(30)	Alphabets
4	Last Name/ Surname	VARCHAR(45)	Alphabets

Address

Sr. No.	Column Name	Data Type	Validation
1	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric

2	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
3	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
4	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
5	Town/City/District	VARCHAR(30)	Required, Alphabets
6	State/Union Territory	VARCHAR(40)	Required, Alphabets
7	Pincode	VARCHAR(6)	Required, Numeric
8	Telephone No.	VARCHAR(10)	Numeric
9	Fax	VARCHAR(12)	Numeric
10	Mobile No.	VARCHAR (10)	Required, Numeric

Other Details

Sr. No.	Column Name	Data Type	Validation
1	Nationality	VARCHAR(56)	Required, Alphabet
2	Visa Type (For Foreigner)	VARCHAR(15)	Alphabets
3	Visa Number (For Foreigner)	VARCHAR(8)	Number
4	Visa Issue Date (For Foreigner)	DATE	DATE
5	Visa Expiry Date (For Foreigner)	DATE	DATE
6	Passport No.	VARCHAR(8)	Alphanumeric
7	Passport Issuing Authority	VARCHAR(75)	Alphabet
8	Passport Expiry Date	DATE	DATE
9	Voter's Identity Card No.	VARCHAR(10)	Alphanumeric
10	PAN No.	VARCHAR(10)	Alphanumeric
11	Email Id	VARCHAR(50)	Required, Alphanumeric
12	Personal Web Page URL	VARCHAR(50)	Alphanumeric

iii) Authorized Representative Form

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR (5)	Required
2	First Name	VARCHAR(30)	Required, Alphabets
3	Middle Name	VARCHAR(30)	Alphabets
4	Last Name/ Surname	VARCHAR(45)	Alphabets
5	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
6	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabet
7	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphabet
8	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabet
9	Town/City/District	VARCHAR(30)	Required, Alphabet
10	State/Union Territory	VARCHAR(40)	Required, Alphabet
11	Pincode	VARCHAR(6)	Required, Numeric
12	Telephone No.	VARCHAR(10)	Numeric
13	Fax	VARCHAR(12)	Numeric
14	Nature of Business	VARCHAR (35)	Required, Alphabet

iv) Form for Bank Details & Location of facility in India

Sr. No.	Column Name	Data Type	Validation
1	Bank Name	VARCHAR(75)	Required, Alphabets
2	Branch	VARCHAR(50)	Required, Alphanumeric
3	Bank Account No.	VARCHAR(18)	Required, Alphanumeric
4	Type of Bank Account	VARCHAR(50)	Required, Alphabets

v) Location of facility in India for generation of DSC

Sr. No.	Column Name	Data Type	Validation
1	Location 1 (Primary)	VARCHAR(50)	Required, Alphanumeric
2	Location 2 (DR Site)	VARCHAR(50)	Alphanumeric
3	Location 3 (Other)	VARCHAR(50)	Alphanumeric

vi) CPS Document Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	CPS Document	VARCHAR(20)	Required, Doc File

vii) Necessary Documents Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	Other Documents (numbers)	VARCHAR(20)	Doc/ PDF file

viii) Bank Draft/ Pay Order Form

Sr. No.	Column Name	Data Type	Validation
1	Name of Bank	VARCHAR(75)	Required, Alphabets
2	Draft/ Pay order No.	VARCHAR(10)	Required, Alphanumeric
3	Date of Issue	DATE	Required
4	Amount	VARCHAR(6)	Required, Number

c. Form For Government Ministry/ Department/ Agency/ Authority

i) Organization Details Form

Particulars of Organization

Sr. No.	Column Name	Data Type	Validation
1	Organization Name	VARCHAR(75)	Required, Alphabets
2	Administrative Ministry/ Department	VARCHAR(75)	Required, Alphabets
3	Under State/ Central Government	VARCHAR(20)	Required, Alphabets
4	Flat/Door/ Block No	VARCHAR(15)	Required, Alphanumeric
5	Name of Premises/ Building/ Village	VARCHAR(30)	Required, Alphabets
6	Road/ Street/ Lane/ Post Office	VARCHAR(30)	Required, Alphanumeric
7	Area/ Locality/ Taluka/ Sub-Division	VARCHAR(30)	Required, Alphabets
8	Town/ City/ District	VARCHAR(30)	Required, Alphabets
9	State/ Union Territory	VARCHAR(40)	Required, Alphabets
10	Pincode	VARCHAR(6)	Required, Numeric
11	Telephone No.	VARCHAR(10)	Required, Numeric
12	Fax	VARCHAR(12)	Numeric
13	Web Page URL address	VARCHAR(50)	Alphanumeric
14	Name of Head of Organization	VARCHAR(75)	Required, Alphabets
15	Designation	VARCHAR(35)	Required, Alphabets
16	Email Address	VARCHAR(50)	Required, Alphanumeric

ii) Other Details

Sr. No.	Column Name	Data Type	Validation
1	Public Key @	-	-
2	Other Information	TEXT	Alphanumeric

iii) Form for Bank Details & Location of facility in India

Sr. No.	Column Name	Data Type	Validation
1	Bank Name	VARCHAR(75)	Required, Alphabets
2	Branch	VARCHAR(50)	Required, Alphabets & Digits
3	Bank Account No.	VARCHAR(20)	Required, Alphabets & Digits
4	Type of Bank Account	VARCHAR(50)	Required, Alphabets

iv) Location of facility in India for generation of DSC

Sr. No.	Column Name	Data Type	Validation
1	Location 1 (Primary)	VARCHAR(50)	Required, Alphanumeric
2	Location 2 (DR Site)	VARCHAR(50)	Alphanumeric

3	Location 3 (Other)	VARCHAR(50)	Alphanumeric
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v) CPS Document Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	CPS Document	VARCHAR(20)	Required, Doc File

vi) Necessary Documents Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	Other Documents (numbers)	VARCHAR(20)	Doc/PDF file

vii) Bank Draft/Pay Order Form

Sr. No.	Column Name	Data Type	Validation
1	Name of Bank	VARCHAR(75)	Required, Alphabets
2	Draft/ Pay order No.	VARCHAR(10)	Required, Alphanumeric
3	Date of Issue	DATE	Required
4	Amount	VARCHAR(6)	Required, Number

5. Renewal License Application Form:

- The Renewal Application Form streamlines the submission process for renewal of license applications requests. It features meticulously structured form fields to capture all pertinent information accurately.
- Users have the capability to seamlessly upload supporting documents, enhancing the completeness of the application process.

Sr. No.	Column Name	Data Type	Validation
1	Application Category	VARCHAR(17)	Required

a. Form for Individual

i) Personal Details

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR (5)	Required
2	First Name	VARCHAR(30)	Required, Only Alphabets
3	Middle Name	VARCHAR(30)	Only Alphabets
4	Last Name/Surname	VARCHAR(45)	Only Alphabets
5	Salutation (Other)	VARCHAR (5)	Only Alphabets
6	First Name (Other)	VARCHAR(30)	Only Alphabets
7	Middle Name (Other)	VARCHAR(30)	Only Alphabets
8	Last Name / Surname (Other)	VARCHAR(45)	Only Alphabets

ii) Residence Address

Sr. No.	Column Name	Data Type	Validation
1	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
2	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
3	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
4	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
5	Town/City/District	VARCHAR(30)	Required, Alphabets
6	State/Union Territory	VARCHAR(40)	Required, Alphabets
7	Pincode	VARCHAR(6)	Required, Number
8	Telephone No.	VARCHAR(10)	Number
9	Fax	VARCHAR(12)	Number

10	Mobile No.	VARCHAR(10)	Required, Number
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iii) Office Address

Sr. No.	Column Name	Data Type	Validation
1	Name of Office	VARCHAR (75)	Required, Alphabets
2	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
3	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
4	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
5	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
6	Town/City/District	VARCHAR(30)	Required, Alphabets
7	State/Union Territory	VARCHAR(40)	Required, Alphabets
8	Pincode	VARCHAR(6)	Required, Numeric
9	Telephone No.	VARCHAR(10)	Required, Numeric
10	Fax	VARCHAR(12)	Numeric

iv) Communication Address

Sr. No.	Column Name	Data Type	Validation
1	Address For Communication (Residence/ Office Address)	LONG(10)	Required

v) Additional Details

Sr. No.	Column Name	Data Type	Validation
1	Father's Salutation	VARCHAR (5)	Required
2	Father's First Name	VARCHAR(30)	Required, Alphabets
3	Father's Middle Name	VARCHAR(30)	Alphabets
4	Father's Last Name/Surname	VARCHAR(45)	Alphabets
5	Sex (Male/Female)	VARCHAR(6)	Required, Alphabets
6	Date of Birth	DATE	Required
7	Nationality	VARCHAR(56)	Required, Alphabets
8	Email Id	VARCHAR(50)	Alphanumeric

vi) Financial and additional Details

Sr. No.	Column Name	Data Type	Validation
1	Credit Card Type	VARCHAR(30)	Alphabets
2	Credit Card No.	VARCHAR(20)	Number
3	Issued By	VARCHAR(75)	Alphabets
4	Web URL address	VARCHAR(50)	Alphanumeric
5	Passport No.	VARCHAR(10)	Alphanumeric
6	Passport Issuing Authority	VARCHAR(75)	Alphabets
7	Passport Expiry Date	DATE	DATE
8	Voter's Identity Card No.	VARCHAR(20)	Alphanumeric
9	Income Tax PAN No.	VARCHAR(10)	Alphanumeric
10	ISP Name	VARCHAR(35)	Required, Alphabets
11	ISP's Website Address	VARCHAR(50)	Alphanumeric
12	ISP Username	VARCHAR(50)	Alphanumeric
13	Personal Web Page URL	VARCHAR(50)	Alphanumeric
14	Capital in the business or profession	VARCHAR(16)	Required, Numeric

vii) Form for Bank Details & Location of facility in India

Sr. No.	Column Name	Data Type	Validation
1	Bank Name	VARCHAR(75)	Required, Alphabets
2	Branch	VARCHAR(50)	Required, Alphanumeric
3	Bank Account No.	VARCHAR(18)	Required, Alphanumeric
4	Type of Bank Account	VARCHAR(50)	Required, Alphabets

viii) Location of facility in India for generation of DSC

Sr. No.	Column Name	Data Type	Validation
1	Location 1 (Primary)	VARCHAR(50)	Required, Alphanumeric
2	Location 2 (DR Site)	VARCHAR(50)	Alphanumeric
3	Location 3 (Other)	VARCHAR(50)	Alphanumeric
4	Public Key @	-	-

ix) CPS Document Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	CPS Document	VARCHAR(20)	Doc File, Required

x) Necessary Documents Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	Other Documents (numbers)	VARCHAR(20)	Doc/PDF file

xi) Bank Draft/Pay Order Form

Sr. No.	Column Name	Data Type	Validation
1	Name of Bank	VARCHAR(75)	Required, Alphabets
2	Draft/ Pay order No.	VARCHAR(10)	Required, Alphanumeric
3	Date of Issue	DATE	Required
4	Amount	VARCHAR(6)	Required, Number

b. Form For Company/Firm/Body of Individuals/Association of Persons/Local Authority

i) Business Details Form

Business Details

Sr. No.	Column Name	Data Type	Validation
1	Registration Number	VARCHAR(21)	Required, Alphanumeric
2	Date of Incorporation/ Agreement/ Partnership	DATE	Required

Particulars of Business

Sr. No.	Column Name	Data Type	Validation
1	Head Office	VARCHAR(75)	Required, Alphabets
2	Name of Office	VARCHAR(75)	Required, Alphabets
3	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
4	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
5	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
6	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
7	Town/City/District	VARCHAR(30)	Required, Alphabets
8	State/Union Territory	VARCHAR(40)	Required, Alphabets
9	Pincode	VARCHAR(6)	Required, Numeric
10	Telephone No.	VARCHAR(10)	Required, Numeric
11	Fax	VARCHAR(12)	Numeric
12	Web Page URL address	VARCHAR(50)	Alphanumeric
13	No. of Branches	VARCHAR(6)	Required, Numeric
14	Nature of Business	VARCHAR(35)	Required, Alphabets

Additional Details

Sr. No.	Column Name	Data Type	Validation
1	PAN No.	VARCHAR(10)	Required, Alphanumeric
2	Turnover (last financial year)	VARCHAR(16)	Required, Numeric
3	Net Worth	VARCHAR(16)	Required, Numeric
4	Paid up Capital	VARCHAR(16)	Required, Numeric

5	Insurance Policy No.	VARCHAR(16)	Required, Alphanumeric
6	Insurance Company	VARCHAR(75)	Required, Alphabet

ii) Details of Partners/Members/Directors

Personal Details

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR (5)	Required
2	First Name	VARCHAR(30)	Required, Alphabets
3	Middle Name	VARCHAR(30)	Alphabets
4	Last Name/ Surname	VARCHAR(45)	Alphabets

Address

Sr. No.	Column Name	Data Type	Validation
1	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
2	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
3	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
4	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
5	Town/City/District	VARCHAR(30)	Required, Alphabets
6	State/Union Territory	VARCHAR(40)	Required, Alphabets
7	Pincode	VARCHAR(6)	Required, Numeric
8	Telephone No.	VARCHAR(10)	Numeric
9	Fax	VARCHAR(12)	Numeric
10	Mobile No.	VARCHAR (10)	Required, Numeric

Other Details

Sr. No.	Column Name	Data Type	Validation
1	Nationality	VARCHAR(56)	Required, Alphabet
2	Visa Type (For Foreigner)	VARCHAR(15)	Alphabet
3	Visa Number (For Foreigner)	VARCHAR(8)	Number
4	Visa Issue Date (For Foreigner)	DATE	DATE
5	Visa Expiry Date (For Foreigner)	DATE	DATE
6	Passport No.	VARCHAR(8)	Alphanumeric
7	Passport Issuing Authority	VARCHAR(75)	Alphabet
8	Passport Expiry Date	DATE	DATE
9	Voter's Identity Card No.	VARCHAR(10)	Alphanumeric
10	PAN No.	VARCHAR(10)	Alphanumeric
11	Email Id	VARCHAR(254)	Required, Alphanumeric
12	Personal Web Page URL	VARCHAR(253)	Alphanumeric

iii) Authorized Representative Form

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR (5)	Required
2	First Name	VARCHAR(30)	Required, Alphabets
3	Middle Name	VARCHAR(30)	Alphabets
4	Last Name/ Surname	VARCHAR(45)	Alphabets
5	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
6	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabet
7	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
8	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabet
9	Town/City/District	VARCHAR(30)	Required, Alphabet
10	State/Union Territory	VARCHAR(40)	Required, Alphabet
11	Pincode	VARCHAR(6)	Required, Numeric
12	Telephone No.	VARCHAR(10)	Numeric

13	Fax	VARCHAR(12)	Numeric
14	Nature of Business	VARCHAR (35)	Required, Alphabet

iv) Form for Bank Details

Sr. No.	Column Name	Data Type	Validation
1	Bank Name	VARCHAR(75)	Required, Alphabets
2	Branch	VARCHAR(50)	Required, Alphanumeric
3	Bank Account No.	VARCHAR(20)	Required, Alphanumeric
4	Type of Bank Account	VARCHAR(50)	Required, Alphabets

v) Location of facility in India for generation of DSC & Public Key

Sr. No.	Column Name	Data Type	Validation
1	Location 1 (Primary)	VARCHAR(50)	Required, Alphanumeric
2	Location 2 (DR Site)	VARCHAR(50)	Alphanumeric
3	Location 3 (Other)	VARCHAR(50)	Alphanumeric
4	Public Key @	-	-

vi) CPS Document Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	CPS Document	VARCHAR(20)	Required, Doc File

vii) Necessary Documents Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	Other Documents (numbers)	VARCHAR(20)	Doc/PDF file

viii) Bank Draft/Pay Order Form

Sr. No.	Column Name	Data Type	Validation
1	Name of Bank	VARCHAR(75)	Required, Alphabets
2	Draft/ Pay order No.	VARCHAR(10)	Required, Alphanumeric
3	Date of Issue	DATE	Required
4	Amount	VARCHAR(6)	Required, Number

c. Form For Government Ministry/ Department/ Agency/ Authority

i) Organization Details Form

Particulars of Organization

Sr. No.	Column Name	Data Type	Validation
1	Organization Name	VARCHAR(75)	Required, Alphabets
2	Administrative Ministry/Department	VARCHAR(75)	Required, Alphabets
3	Under State/ Central Government	VARCHAR(20)	Required, Alphabets
4	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
5	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
6	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric
7	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
8	Town/City/District	VARCHAR(30)	Required, Alphabets
9	State/Union Territory	VARCHAR(40)	Required, Alphabets
10	Pincode	VARCHAR(6)	Required, Numeric
11	Telephone No.	VARCHAR(10)	Required, Numeric
12	Fax	VARCHAR(12)	Numeric
13	Web Page URL address	VARCHAR(50)	Alphanumeric
14	Name of Head of Organization	VARCHAR(75)	Required, Alphabets
15	Designation	VARCHAR(35)	Required, Alphabets
16	Email Address	VARCHAR(50)	Required, Alphanumeric

ii) Other Details

Sr. No.	Column Name	Data Type	Validation
1	Public Key @	-	-
2	Other Information	TEXT	Alphanumeric

iii) Form for Bank Details

Sr. No.	Column Name	Data Type	Validation
1	Bank Name	VARCHAR(75)	Required, Alphabets
2	Branch	VARCHAR(50)	Required, Alphanumeric
3	Bank Account No.	VARCHAR(20)	Required, Alphanumeric
4	Type of Bank Account	VARCHAR(50)	Required, Alphabets

iv) Location of facility in India for generation of DSC

Sr. No.	Column Name	Data Type	Validation
1	Location 1 (Primary)	VARCHAR(50)	Required, Alphanumeric
2	Location 2 (DR Site)	VARCHAR(50)	Alphanumeric
3	Location 3 (Other)	VARCHAR(50)	Alphanumeric

v) CPS Document Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	CPS Document	VARCHAR(20)	Required, Doc File

vi) Necessary Documents Uploading Form

Sr. No.	Column Name	Data Type	Validation
1	Other Documents (numbers)	VARCHAR(20)	Doc/PDF file

vii) Bank Draft/Pay Order Form

Sr. No.	Column Name	Data Type	Validation
1	Name of Bank	VARCHAR(75)	Required, Alphabets
2	Draft/ Pay order No.	VARCHAR(10)	Required, Alphanumeric
3	Date of Issue	DATE	Required
4	Amount	VARCHAR(6)	Required, Number

6. Empanelled Audit Agency Registration Form:

- The Empanelled Audit Agency Form collects and checks important details of audit agencies that want to be listed for auditing Certifying Authorities (CAs). This form ensures that only qualified and competent audit agencies are authorized to perform these critical audits.

i) Basic Details

Sr. No.	Column Name	Data Type	Validation
1	Name	VARCHAR(100)	Required
2	Mobile Number(s)	VARCHAR(10)	Required, Number
3	Email Id(s)	VARCHAR(50)	Required, Alphanumeric
4	Telephone Number (s)	VARCHAR(10)	Number
5	Effective From	DATE	Required
6	Effective To	DATE	Required

ii) Address

Sr. No.	Column Name	Data Type	Validation
1	Flat/Door/Block No	VARCHAR(15)	Required, Alphanumeric
2	Name of Premises/Building/Village	VARCHAR(30)	Required, Alphabets
3	Road/Street/Lane/Post Office	VARCHAR(30)	Required, Alphanumeric

4	Area/Locality/Taluka/Sub-Division	VARCHAR(30)	Required, Alphabets
5	Town/City/District	VARCHAR(30)	Required, Alphabets
6	State/Union Territory	VARCHAR(40)	Required, Alphabets
7	Pincode	VARCHAR(6)	Required, Numeric

iii) Empanelled Auditors

Sr. No.	Column Name	Data Type	Validation
1	Salutation	VARCHAR(6)	Required
2	First Name	VARCHAR(30)	Required, Alphabets
3	Middle Name	VARCHAR(30)	Required, Alphabets
4	Last Name/ Surname	VARCHAR(30)	Required, Alphabets

7. License Issuance Form:

- The License Issuance Form captures the basic details of the issued license.

Sr. No.	Column Name	Data Type	Validation
1	License Serial Number	VARCHAR(25)	Required, Numeric
2	Date of Issuance	DATE	Required
3	Date of Expiry	DATE	Required
4	Certificate	VARCHAR(30)	Required, PDF/Image

8. eSign Application Form (New/ Renewal):

- The Application Form streamlines the submission process for new/renewal eSign applications requests. It features meticulously structured form fields to capture all pertinent information accurately.
- Users have the capability to seamlessly upload supporting documents, enhancing the completeness of the application process.

Sr. No.	Column Name	Data Type	Validation
1	eKYC Mode	VARCHAR(20)	Required, Alphabets
2	Cover Letter	VARCHAR(30)	Required, PDF
3	Approval(s)	VARCHAR(30)	Required, PDF
4	API Version(s)	VARCHAR(20)	Required
5	Audit Report(s)	VARCHAR(30)	Required, PDF
6	CPS Document	VARCHAR(30)	Required, Doc

3.2 Hardware Interfaces

Users should not need any special kind of external interfaces required to access the software. Users should only require a personal computer like hardware device with an Operating System supporting Graphical User Interfaces.

3.3 Software Interfaces

1. Web Pages:

- Developed primarily using React JS, CSS, JavaScript and HTML.

2. Accessibility:

- Accessible through web browsers on various Operating Systems, including Unix, Linux, Mac, and Windows.
- Provides flexibility and convenience for users across different platforms and browsers.

3. Micro-service Architecture:

- Implemented using Spring Boot framework.
- Enables the software to be modularized into smaller, independently deployable services.
- Enhances scalability, maintainability and flexibility.

4. Database:

- Utilizes PostgreSQL for storing user data, software data, etc.
- Offers robust data storage capabilities with support for relational database management.
- Ensures data integrity and reliability for the software.

5. Tools / IDEs:

- Spring tool suit
- Eclipse IDE

6. Third-Party Integrations

- Git SVN
- Jenkins
- SonarQube
- Kafka

3.4 Communications Interfaces

1. Internet Access:

- Users can access the software via internet.
- Provides universal accessibility for users from anywhere with internet connectivity.

2. HTTP / HTTPS Protocol Utilization:

- The software shall use the HTTP/ HTTPS protocol for internet communication.
- Ensures reliable and standardized data transmission between client and server.

3. Notification and Alert Delivery:

- Notifications and alerts are distributed to users via email and/or SMS.
- Keep users informed about important updates and events within the software.
- Enhances communication and ensures timely dissemination of critical information.

4. System Features

4.1 Responsive Design

The software UI should be interactive and adaptable to different screen sizes and devices. Menus & options should be intuitive & easily navigable.

4.2 2-Factor Authentication

The software should deploy 2-Factor Authentication mechanism for user login. Users have to first authenticate themselves with correct username & password, upon which an OTP should be generated & shared over their registered email address and/or mobile number, which they have to enter in the next step in due time before expiration of the OTP only then they should be able to login and get access to the software.

4.3 Role-based Authorization

The software should use Role-based Authorization to gain access to any resource in the software. Users should be able to access those resources only for which they have been given the access authorization based upon their user role and status.

4.4 Data Management

The software should implement proper data validation techniques ensuring accuracy and integrity of data entered into the system. The software should also implement regular backups and disaster recovery options.

4.5 Data Security & Privacy

Sensitive user data & information should be stored in a secure manner using proper Encryption mechanism.

User's personal data should not be displayed openly and should not be accessed by other than Authorized Officers of CCA.

4.6 System Integrity

Users should not be able to access those resources for which they have not been given permission.

Users should not be able to get access to other users' data & information.

Software should not display data & info of one user into the login of other user.

4.7 Bot Prevention

The software should implement Captcha to prevent bots from getting access to it.

4.8 Integration

The software should allow integration with other software applications through APIs. The software should seamlessly connect with popular third-party services.

4.9 Online Payment

The software should integrate & implement online payment gateway (Bharat Kosh) to enable users to pay the applicable license fee online with ease from their own location.

4.10 Instant Notification

The software should integrate & implement mechanism to send instant alerts & notification (Email & SMS) to users on happening/ processing of an event related to their activity on the software.

4.11 Document Authentication

The software should integrate & implement ways to enable users to be able to endorse/ authenticate the documents. Users should be able to digitally sign the documents using DSC or e-Sign service.

4.12 Scalability

The software should be scalable to handle increased loads by adding resources. The software should implement caching to improve performance by temporarily storing frequently accessed data.

4.13 Log Management

The software should implement log management & audit trail functionality to document evidence of certain events, procedures or operations, to reduce fraudulent activities, errors, and unauthorized use.

Logs should be gathered from various sources such as servers, applications & databases and logged in the database. Log data should be parsed and converted into a structured format to make it easier to search, filter, and analyze. Lifecycle of logs should be maintained regularly by archiving old logs and deleting unnecessary ones to save storage space and maintain system performance.

4.14 Visualization & Reporting

CCA and other Authorized Officers of CCA should be able to generate various types of charts, graphs & reports from the software with respect to the DSC and e-Sign related data entered by the Licensee from time to time.

5. Other Non-Functional Requirements

5.1 Performance Requirements

The software should be user friendly with an interactive user interface making it easy for the user to understand and navigate, also avoiding delays, if any. The software should load and update the interface on interaction quickly.

The software should implement mechanism & strategies for handling (detecting, managing & resolving) unexpected events (errors & exceptions) gracefully (without crashing or producing incorrect results) to ensure data integrity & reliability. Event occurrence should be logged for troubleshooting, relevant stakeholders should be notified for resolution and affected resources should be validated to ensure completeness & accuracy maintaining a high level of user trust & satisfaction.

5.2 Safety Requirements

The software should deploy proper input validation strategies so that unauthorized malicious input is detected & prevented well before processing. The software should check that data is both syntactically & semantically valid before using it in any way. Whitelisting and Blacklisting techniques may be used to perform input syntax validation. Moreover, validation techniques should be deployed both at the Client-side and Server-side as well.

Databases should use relational model & normalization to avoid redundancy. Database must be backed-up regularly to prevent loss of data. Database as well as the software should be behind a firewall.

5.3 Security Requirements

Data must be securely transmitted to server without any changes in information. All login credentials of the users must be protected for privacy using proper constraints as required in the software. There should implement such a login mechanism to avoid/ prevent chances of hacking along with proper Bot-prevention mechanism like Captcha at every possible form/ data submission step.

The software should implement data retention policy as per the organization's policy that clarifies what data should be stored or archived, where that should happen and for how long. Once a data set completes its retention period, it can be deleted or moved as historical data to secondary or tertiary storage, depending on business requirements.

5.4 Special Requirements

- a. Authorized Officer of CCA can only submit the intent received from interested users to become a licensed Certifying Authority (CA).
- b. Authorized Officer of CCA can only register the Empaneled Audit Agency.
- c. The applicant should be able to upload CPS document in both doc as well as pdf format.
- d. The applicant should get at least 3 attempts to rectify the discrepancies notified during

the new license application scrutiny phase.

- e. The applicant should get at least 3 attempts to rectify the discrepancies notified during the new license application review phase.
- f. The applicant should be able to upload the action taken report against the issued NCs.
- g. The applicant should get at least 2 attempts to rectify the discrepancies notified during the applicant audit report review phase and the annual audit report review case also.
- h. The licensee should get at least 2 attempts to rectify the discrepancies notified during the eSign application scrutiny cum review phase.
- i. The licensee should get at least 2 attempts to rectify the discrepancies notified during the license renewal application scrutiny phase.
- j. The licensee should get at least 2 attempts to rectify the discrepancies notified during the license renewal application review phase.
- k. The licensee should get at least 2 attempts to rectify the discrepancies notified during the license cessation application scrutiny cum review phase.
- l. The users should be able to digitally sign the uploaded documents.
- m. Concerned users should receive alerts & notifications regarding their application and/ or license.

5.5 Software Quality Attributes

5.5.1 Availability

The software must implement fault-tolerant mechanism to ensure software reliability and availability in the event of failures or errors. The software must incorporate robust error handling strategies to gracefully manage exceptions and prevent software crashes that could impact performance.

5.5.2 Compatibility

The software assumes compatibility with the latest version of popular operating systems such as Windows, macOS, and Linux distributions. It also assumes compatibility with modern web browsers including but not limited to Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari, ensuring a consistent user experience across different platforms and devices.

5.5.3 Durability

The software should work properly with many simultaneous users at the same time. The software must be capable of handling multiple concurrent requests from users or external systems without experiencing performance degradation or resource contention.

5.5.4 Accessibility

The software should be accessible on every device supporting GUI with latest web browser having internet connectivity.

5.5.5 Efficiency

The structure of the codebase should be logical & clear making it easy to understand, maintain, and extend. The codebase should be modular, broken down into smaller, reusable modules that encapsulate specific functionalities. Naming Conventions should be meaningful and descriptive to enhance readability. Comments and remarks should be used to explain the purpose, logic, and usage of modules, functions, and complex algorithms.

Thread pools or executors should be used/ employed to manage and reuse threads efficiently. Asynchronous programming techniques should be utilized to handle concurrent tasks without blocking. Synchronization mechanisms like locks, semaphores, and atomic variables should be used to control access to shared resources. Message-passing approaches should be preferred for communication between concurrent components.

5.5.6 Effectiveness

The software should effectively handle multiple operations concurrently without affecting/ disturbing another operation or user. The software should support a high throughput rate to process a large number of requests efficiently within a given timeframe.

5.5.7 Scalability

The software, designed using micro-services architecture, prioritizes scalability to effectively manage fluctuating workload demands. Employing both horizontal and vertical scaling strategies ensures optimal performance and responsiveness.

Micro-services offer many benefits over traditional monolithic software, including:

- **Scalability:** Micro-services can be scaled independently, making it easier to scale up or down as needed.
- **Resilience:** If one micro-service fails, the other micro-services can continue to operate, making micro-services based software more resilient to failure.
- **Agility:** Micro-services can be developed & deployed independently, making it easier to change the software without impacting the entire system.

There are several reasons required to scale micro-services. These include:

- **Handling increased traffic:** As the software becomes popular, need arises to scale micro-services to handle more traffic.
- **Improving performance:** Micro-services can be scaled independently, so specific micro-services that are becoming bottlenecks can only be scaled up.
- **Increasing reliability:** If one micro-service fails, the other micro-services can continue to operate, making the software more reliable.
- **Reducing costs:** By scaling micro-services efficiently, resources can be optimized to reduce infrastructure costs.

There are two main ways to scale micro-services:

- **Horizontal scaling:** This involves adding more micro-service instances to handle the increased load. It's the most common type of scaling micro-services. New containers or virtual machines can be added to distribute the load across multiple instances.
- **Vertical scaling:** This involves increasing the resources (CPU, memory, etc.) of an existing micro-service instance to handle additional load. While it can provide a short-term performance boost, it has limits. In the long term, it's more cost-effective to use horizontal scaling.

5.5.8 Portability

The software should be easily ported or migrated to other environments containing different hardware or operating system specifications. The portability problem is majorly faced in mobile native applications. As far as this software is concerned, the software can be very easily ported or migrated till the hardware, software and communication platform conditions specified under section 2.4 Operating Environment is successfully met.

5.5.9 Correctness

The software should be able to perform tasks correctly as specified by the predefined requirements when used under specified conditions. Regular Code review and Testing should be a part of the development. Unit testing will be done by the developer itself.

Integration testing will be done either by the developer or tester for ensuring the tested units work together as expected. System testing will be done to test the entire system together as a whole to ensure the system performs as expected.

5.5.10 Interoperability

The software should be compatible with other software/ applications. The software should provide APIs for integration. The software should adhere to industry standards like SOAP, XML, JSON, etc.

5.5.11 Compliance

The system should comply with relevant regulations. The system should provide audit logs for all critical actions and changes. The system should retain data as per data retention policy of the organization.

5.5.12 Maintainability

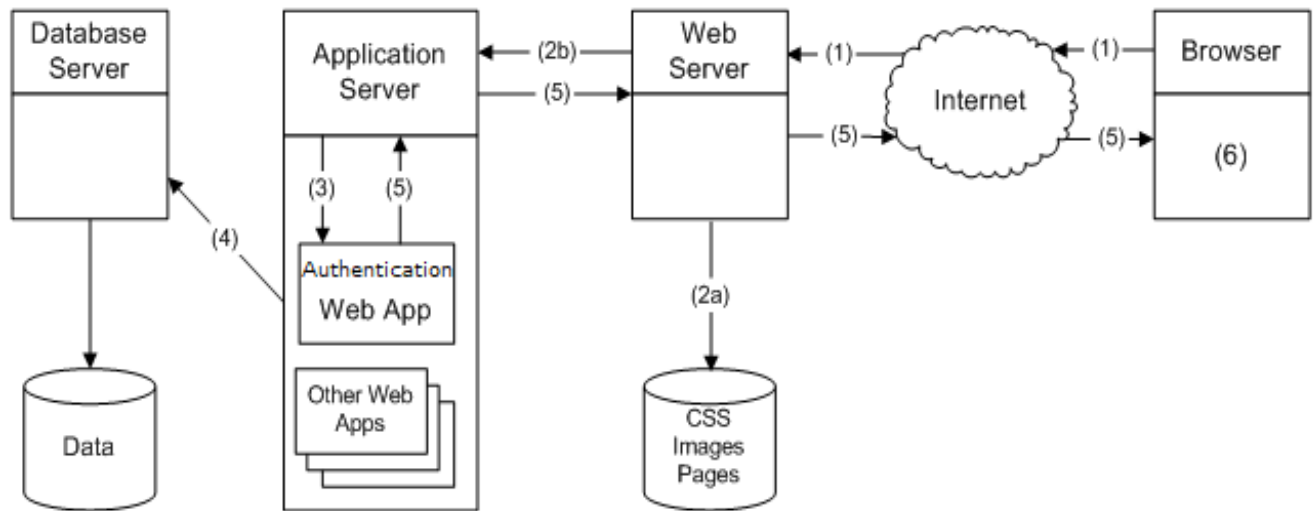
The software should be easy to maintain. There should be a clear separation between user interfaces & business logic and also between data access objects mapping the database & business logic.

The invocation of methods, functions & APIs should be in an organized manner to achieve a specific objective. These systematic calls should be well-documented and logically structured. Comments & remarks should be mentioned on each significant step to explain why it is necessary and what it accomplishes.

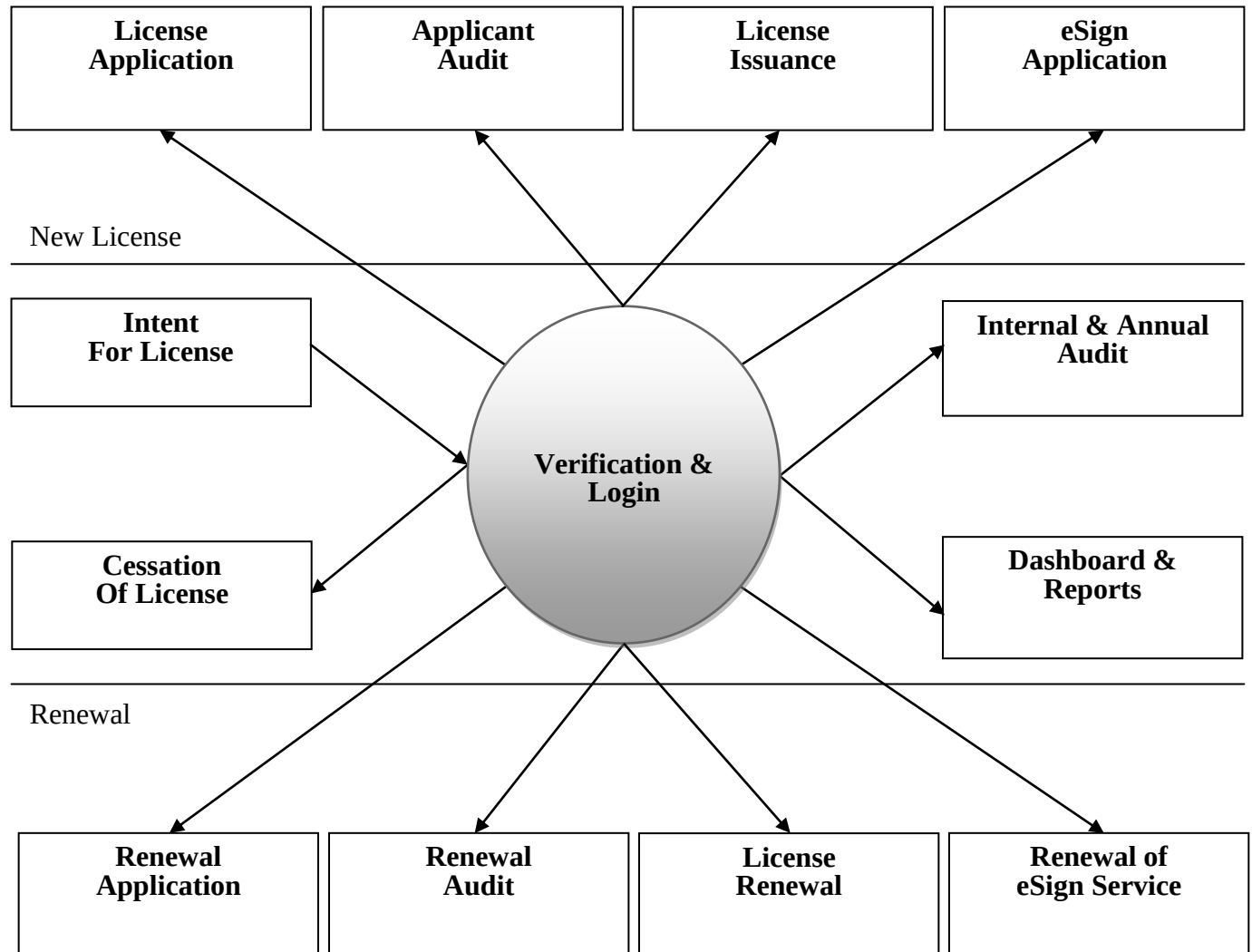
5.6 Business Rules

- a. Only registered & verified users will be allowed to login & access the software.
- b. Only authenticated users will be permitted to perform actions based on their roles, authorizations & privileges.

Appendix A: System Architecture

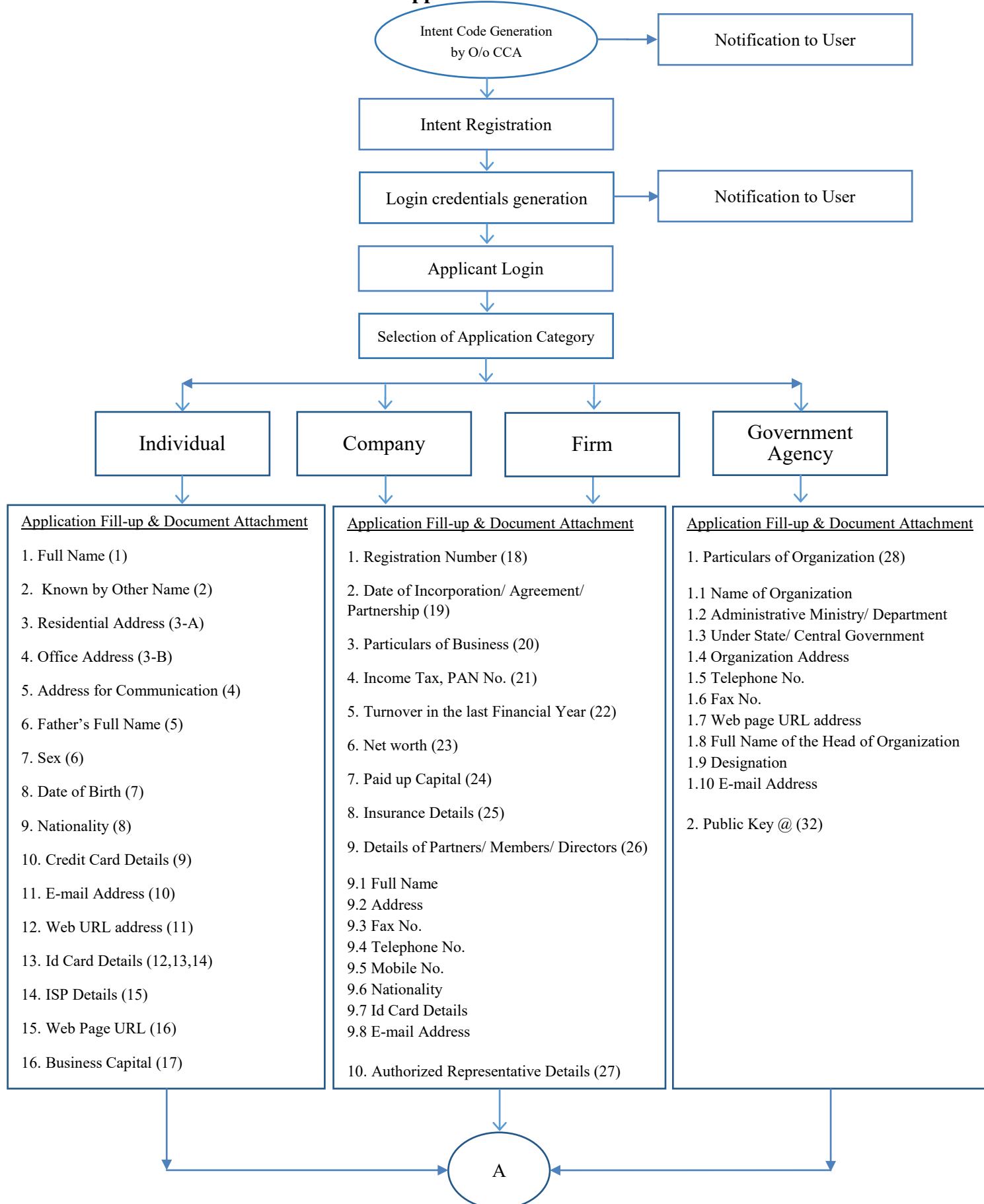


Appendix B: Application Architecture

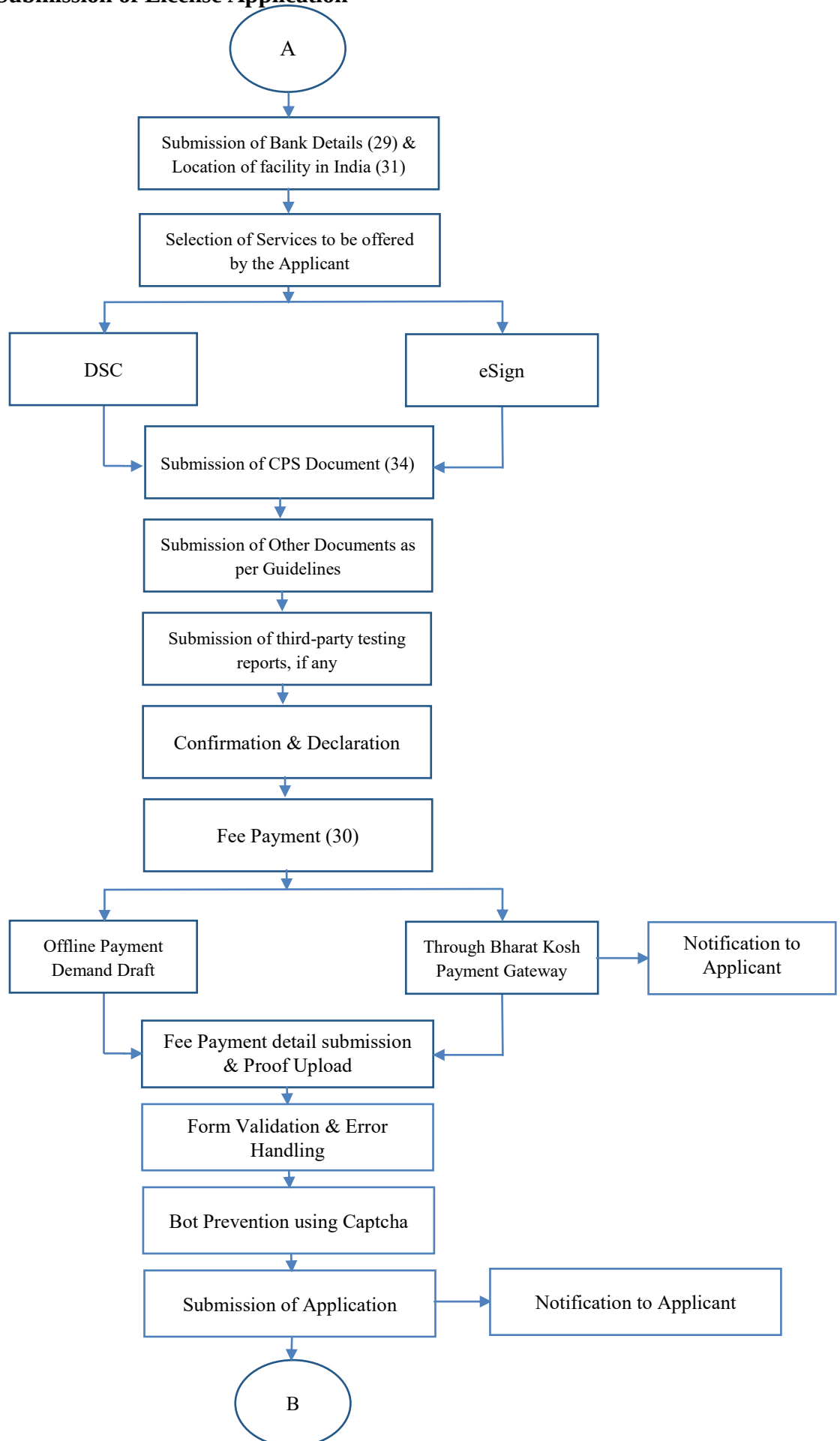


Appendix C: Process Flow Diagram

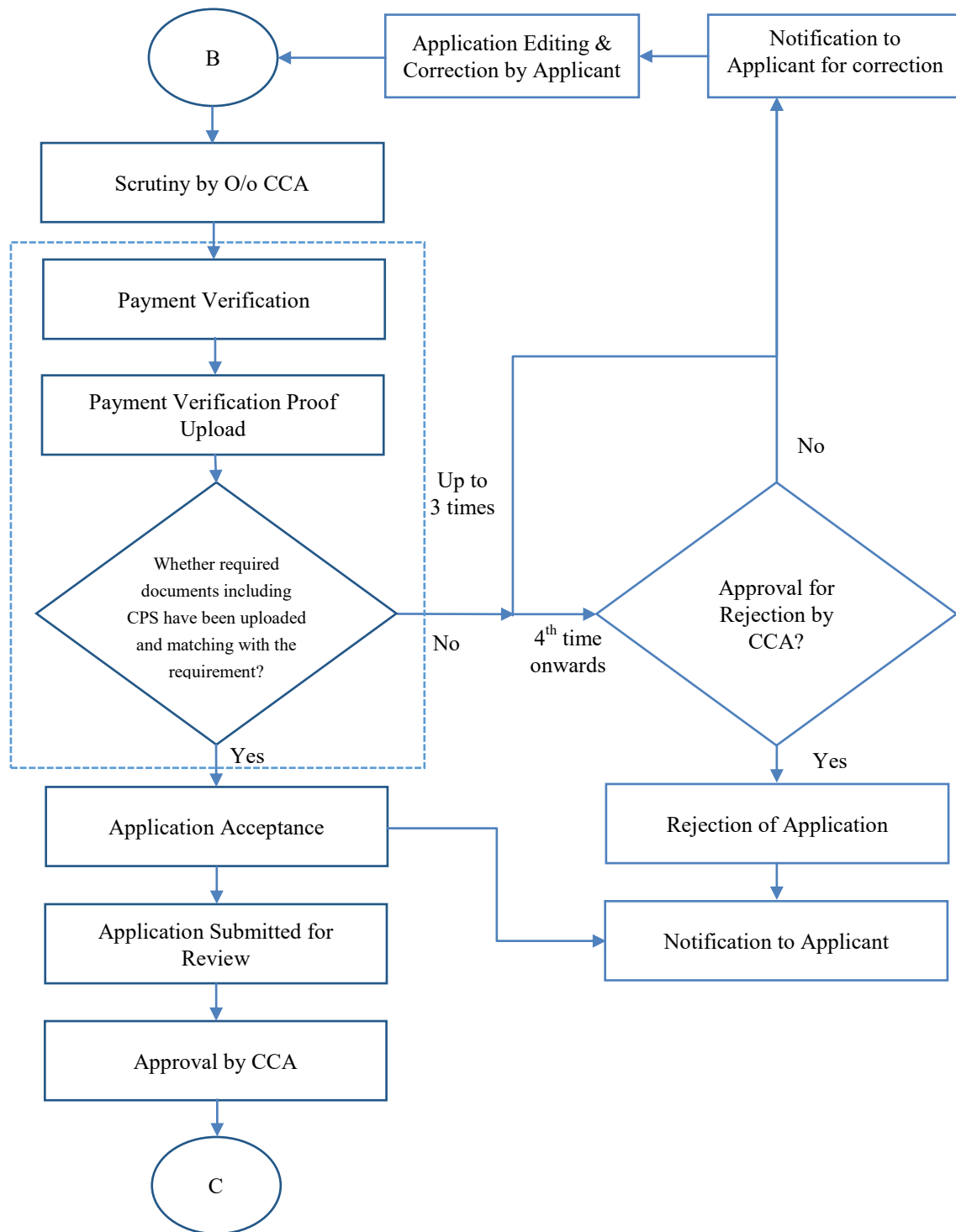
1. Process for Submission of License Application



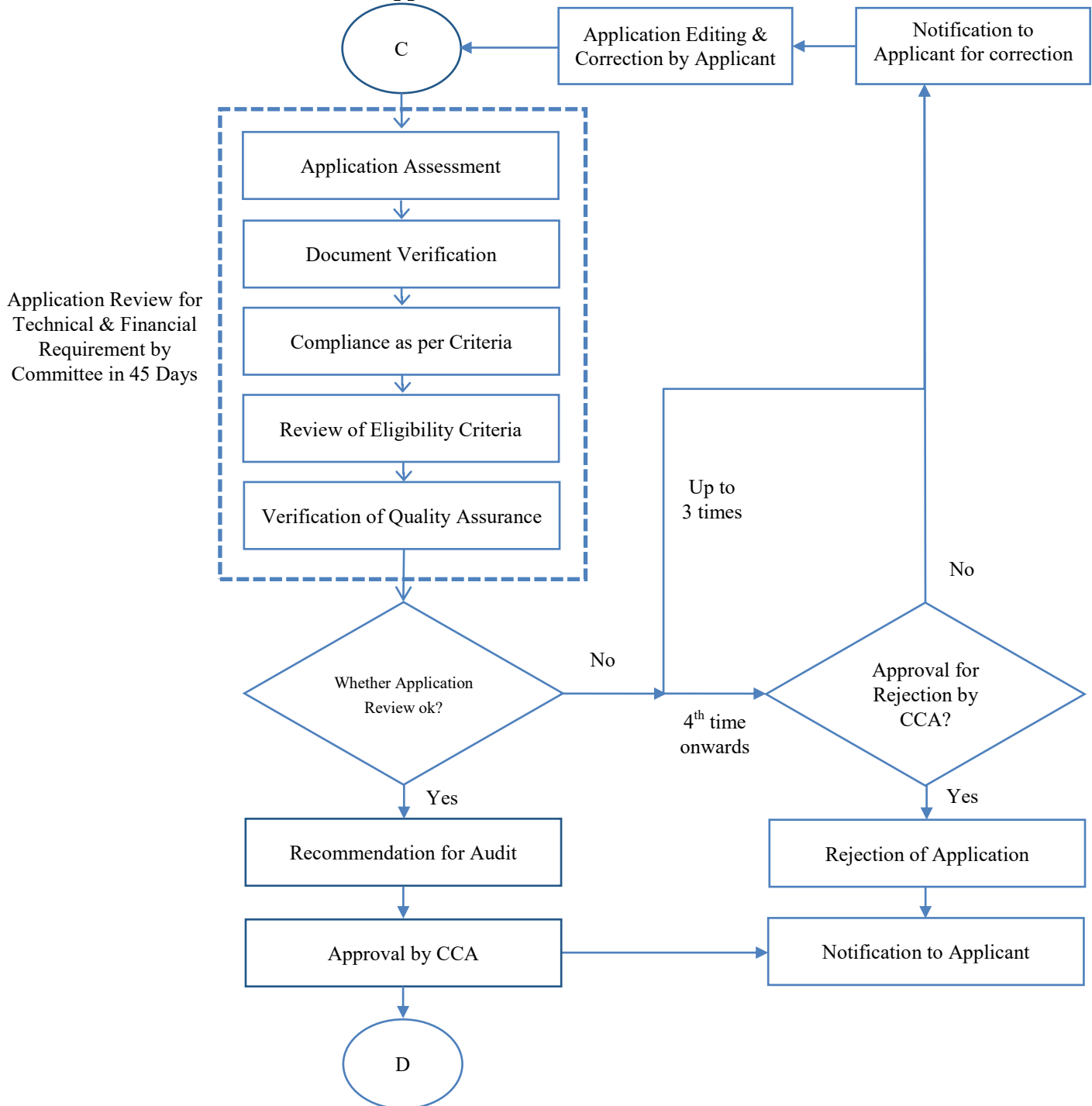
2. Process for Submission of License Application



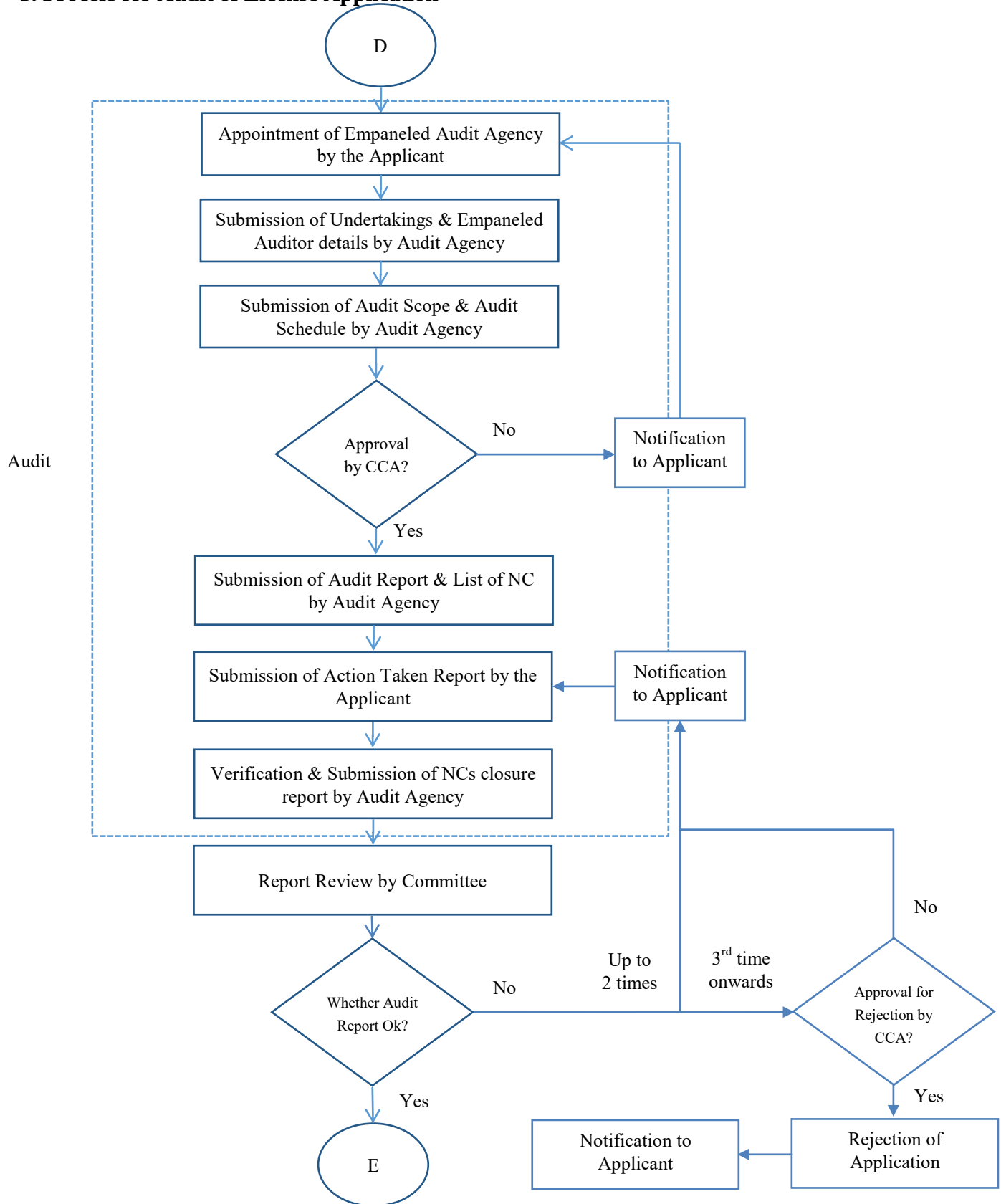
3. Process for Scrutiny of License Application



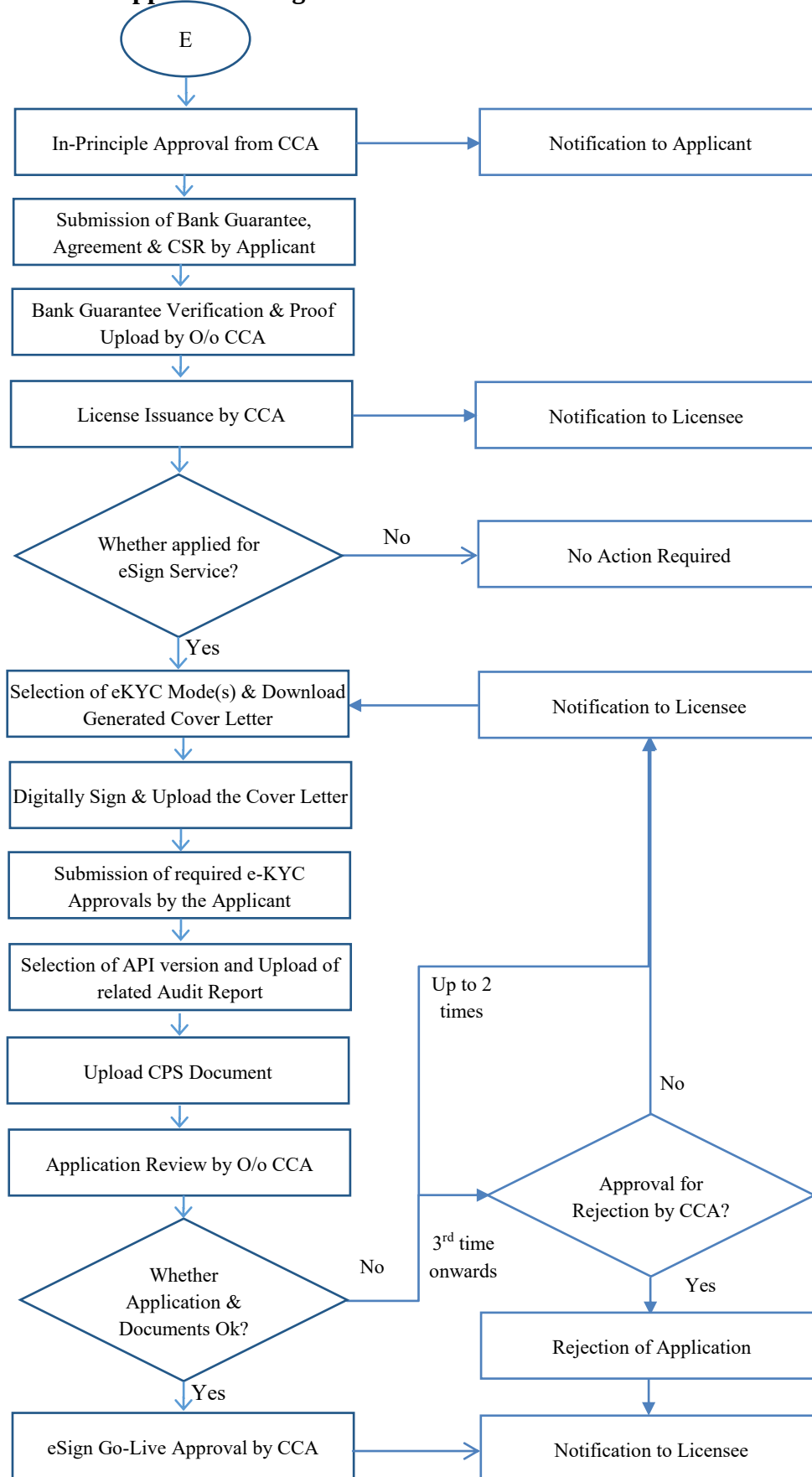
4. Process for Review of License Application



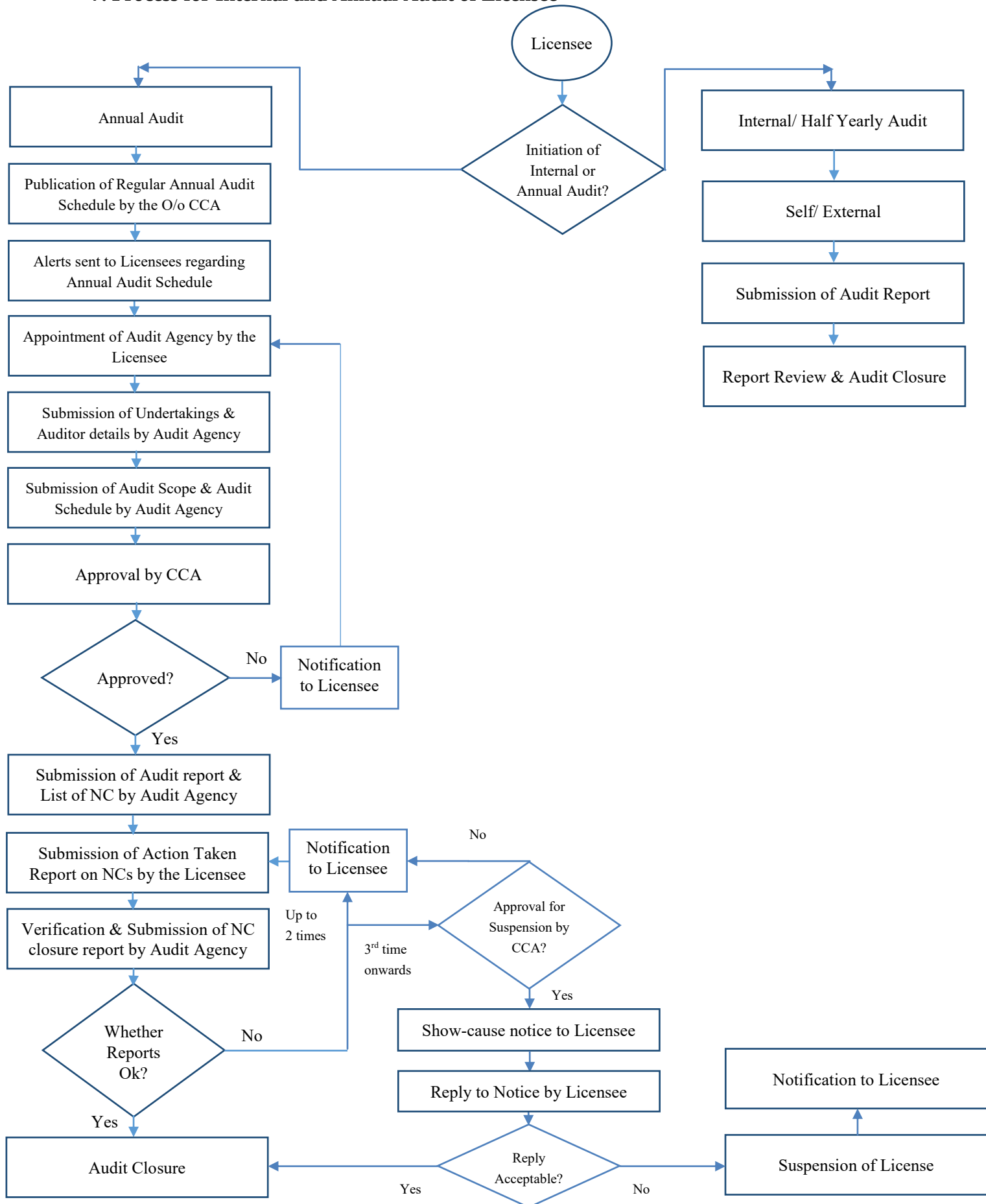
5. Process for Audit of License Application



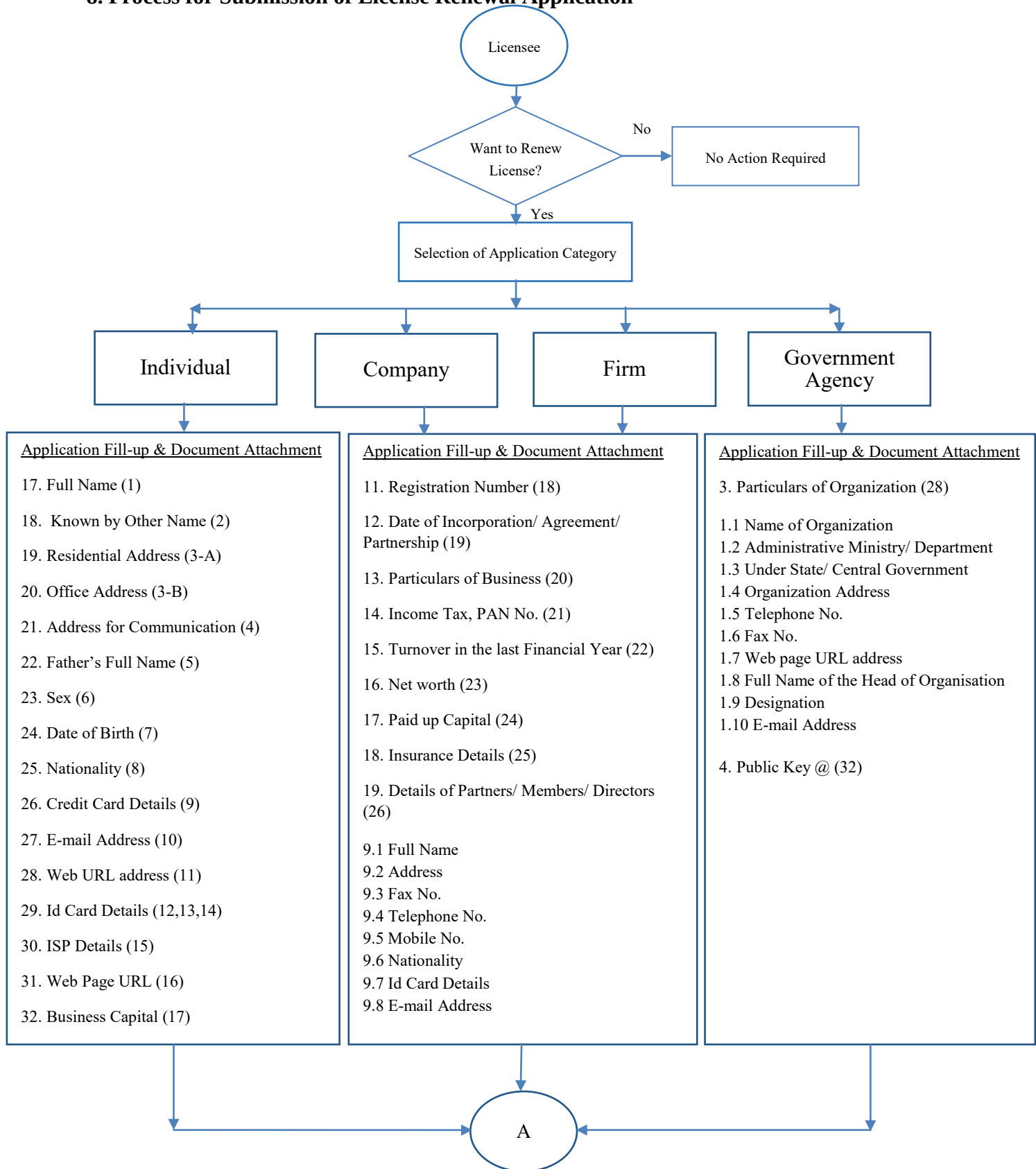
6. Process for License Approval & eSign Service



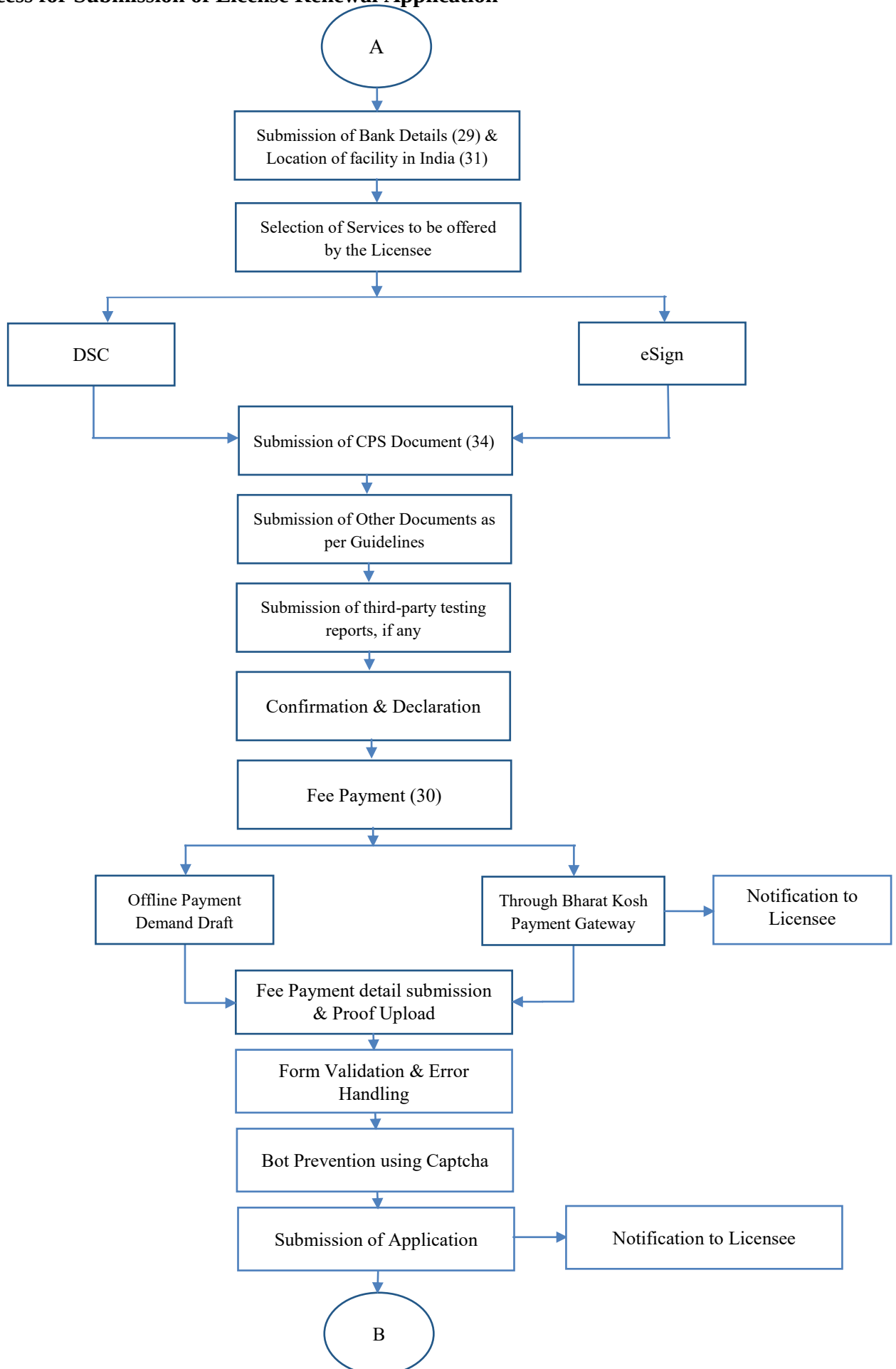
7. Process for Internal and Annual Audit of Licensee



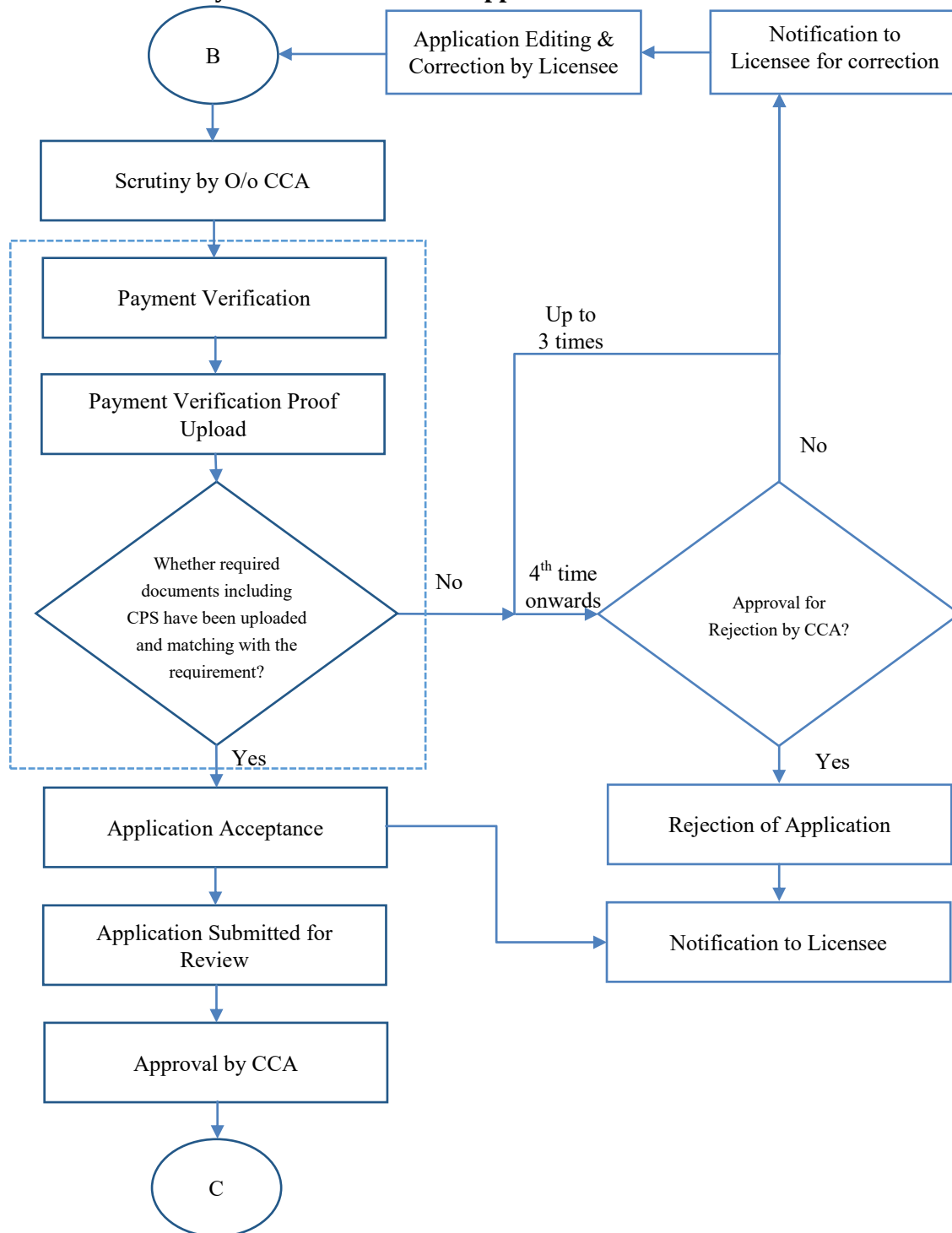
8. Process for Submission of License Renewal Application



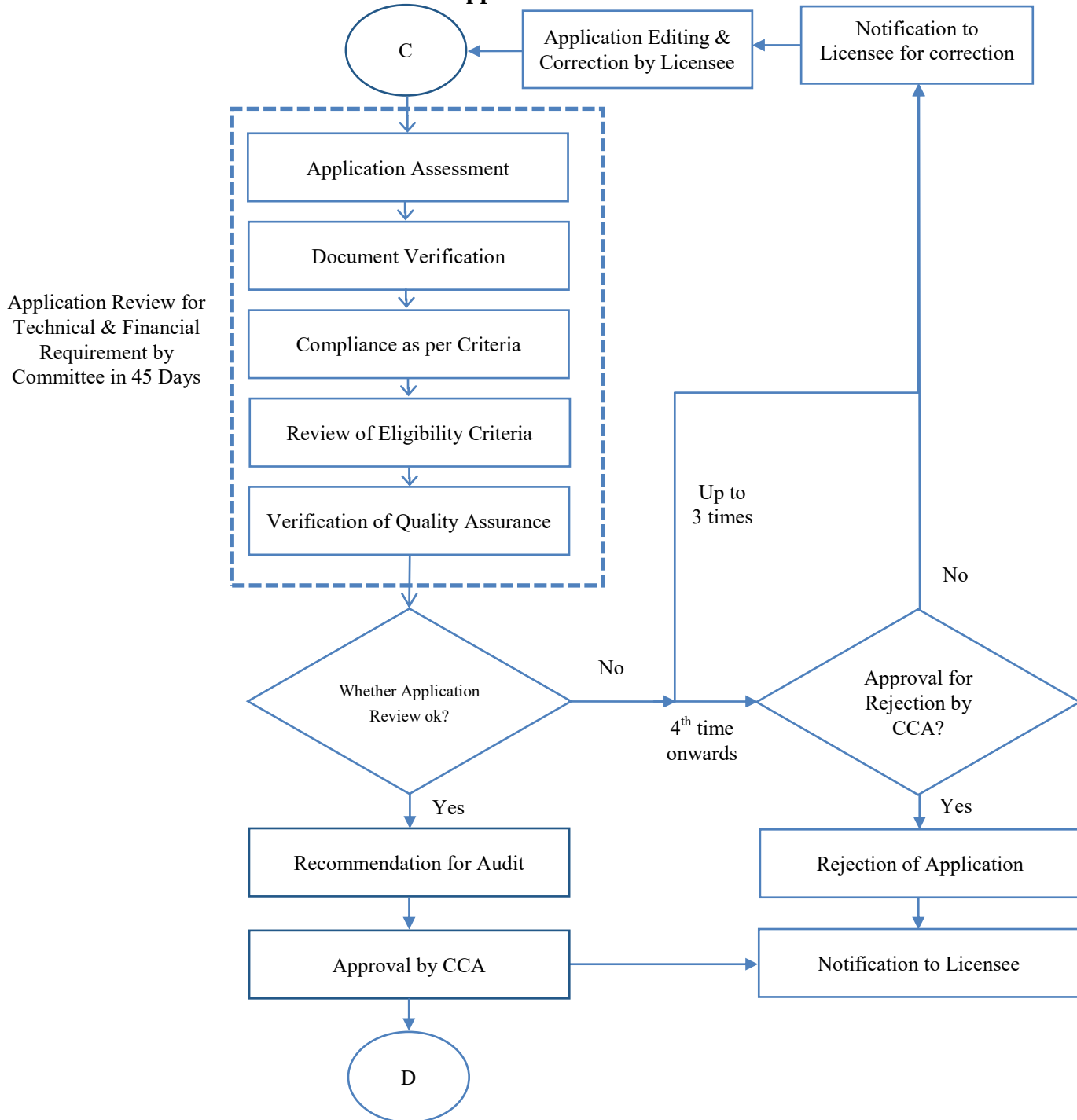
9. Process for Submission of License Renewal Application



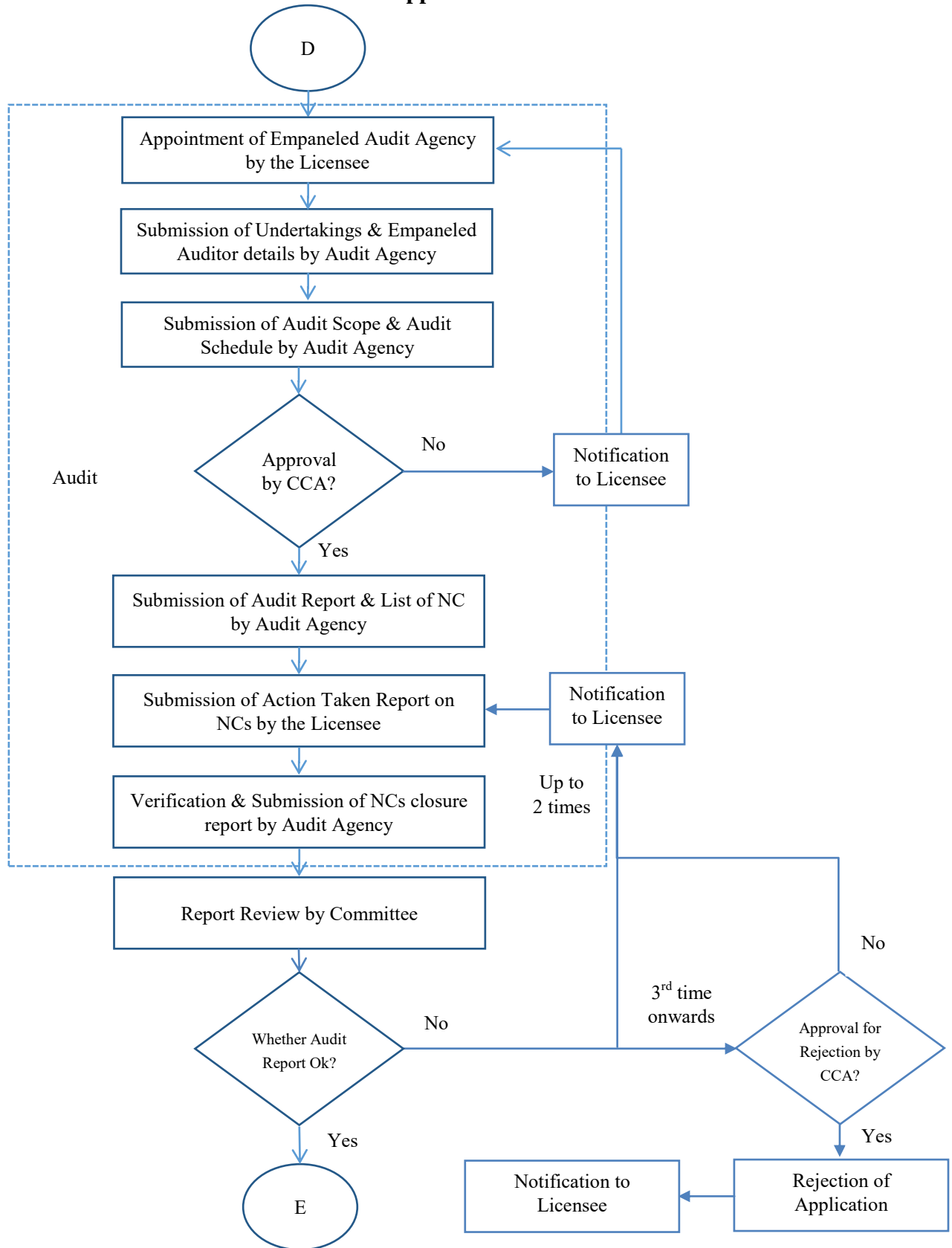
10. Process for Scrutiny of License Renewal Application



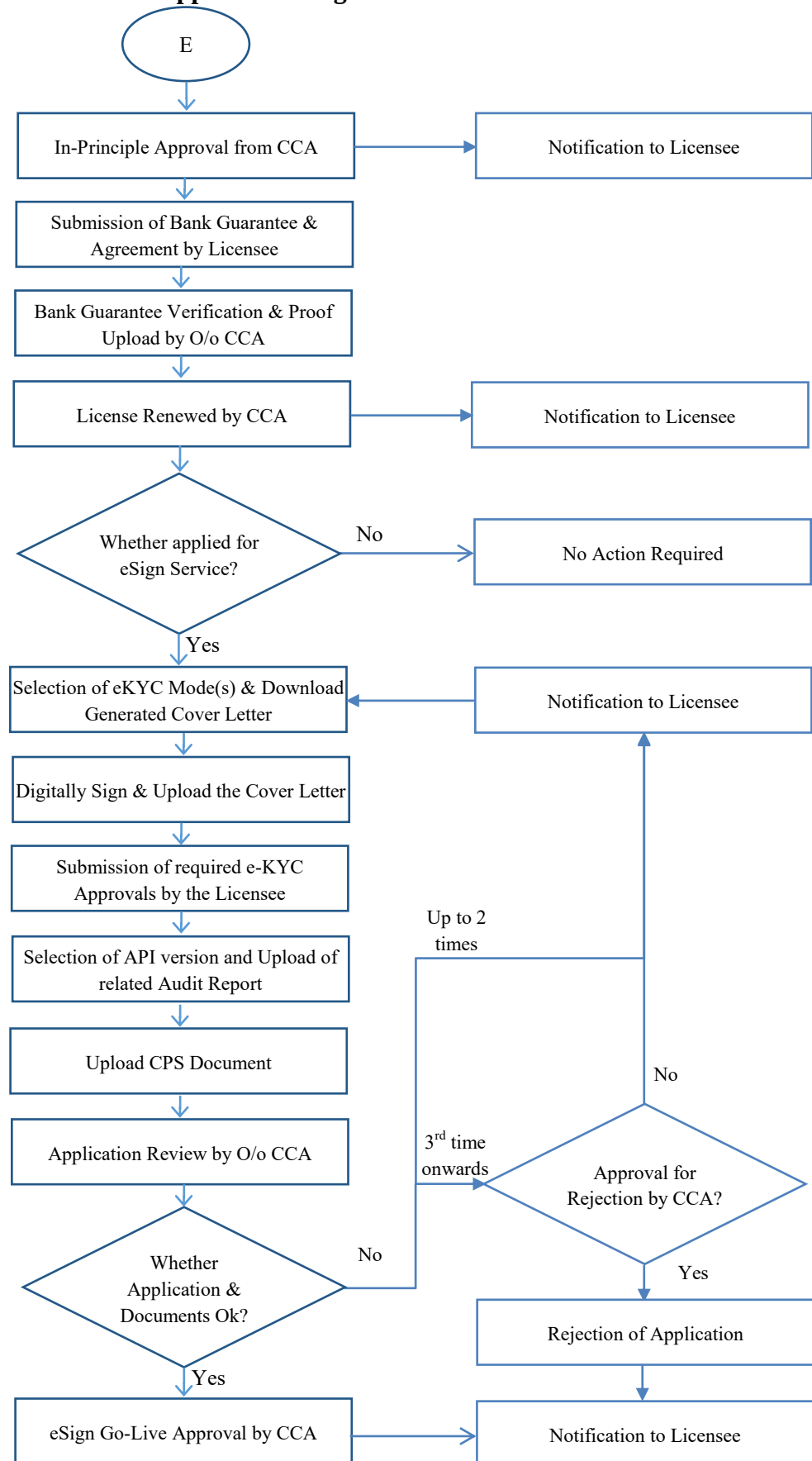
11. Process for Review of License Renewal Application



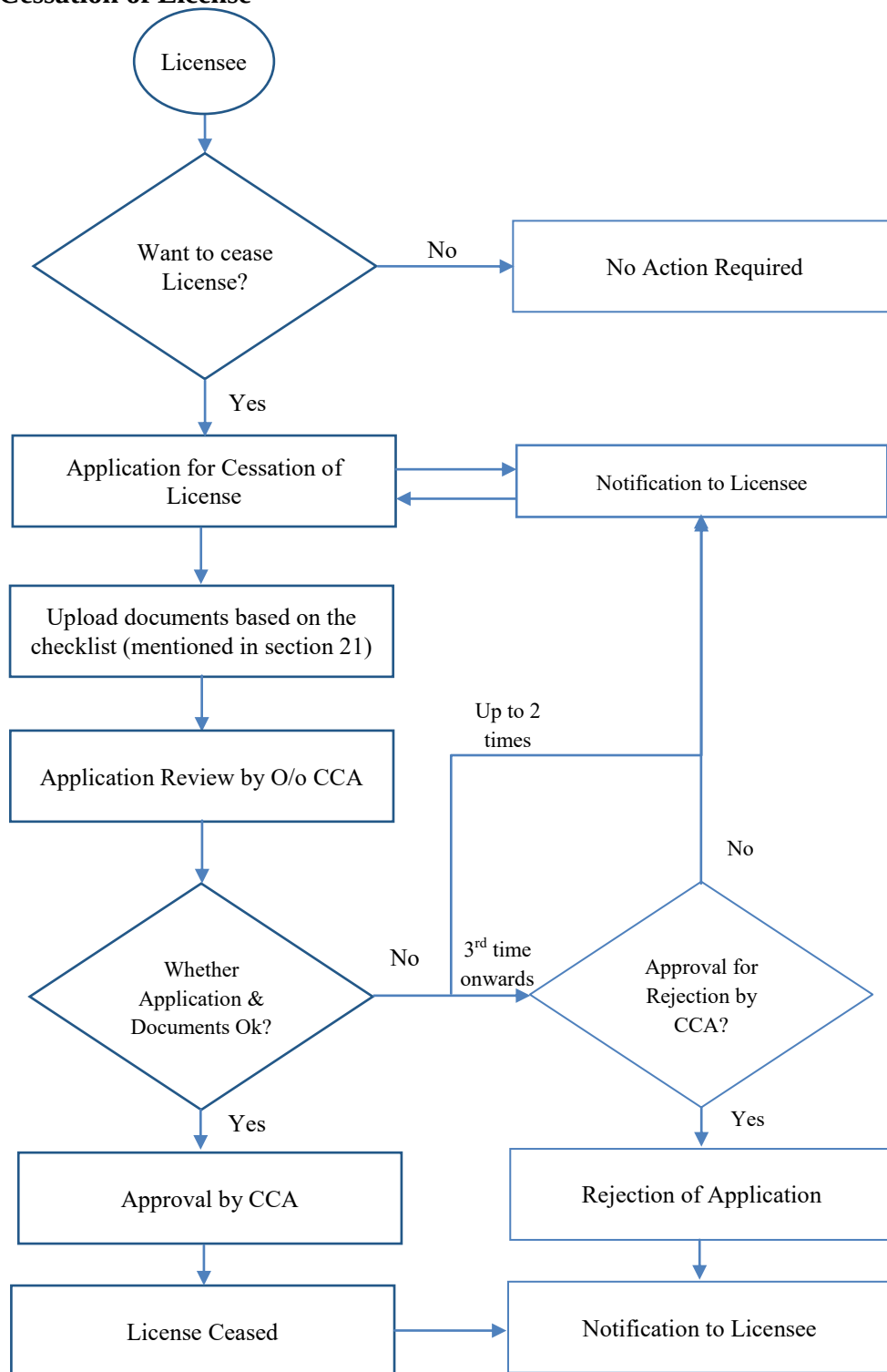
12. Process for Audit of License Renewal Application



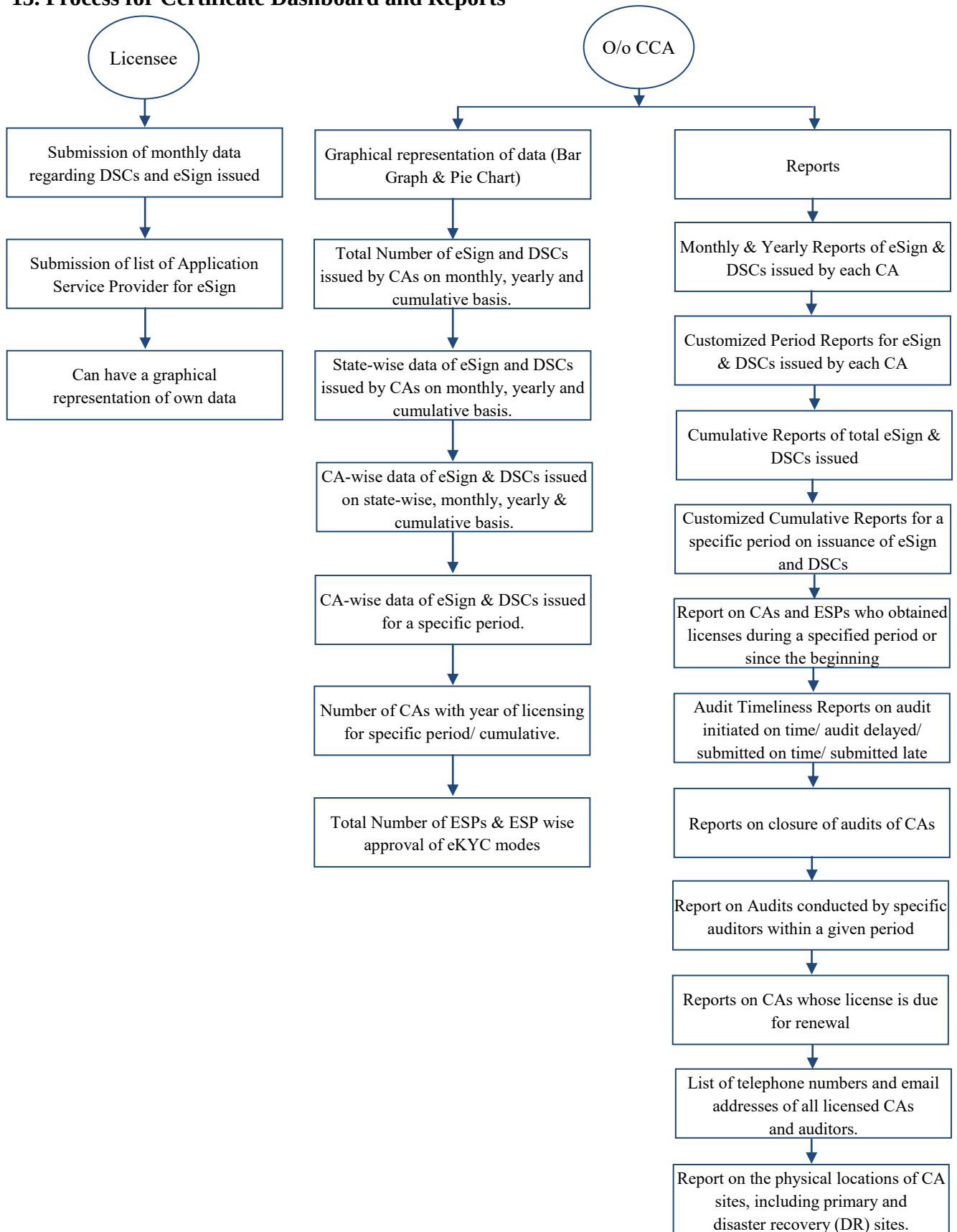
13. Process for License Renewal Approval & eSign Service



14. Process for Cessation of License

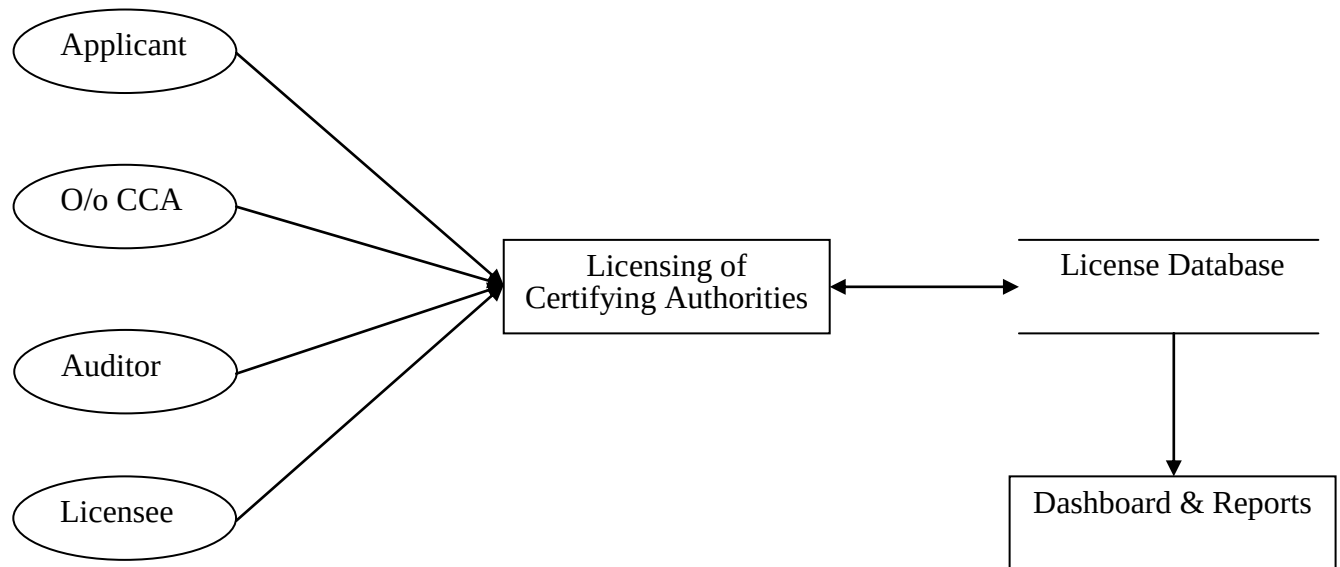


15. Process for Certificate Dashboard and Reports

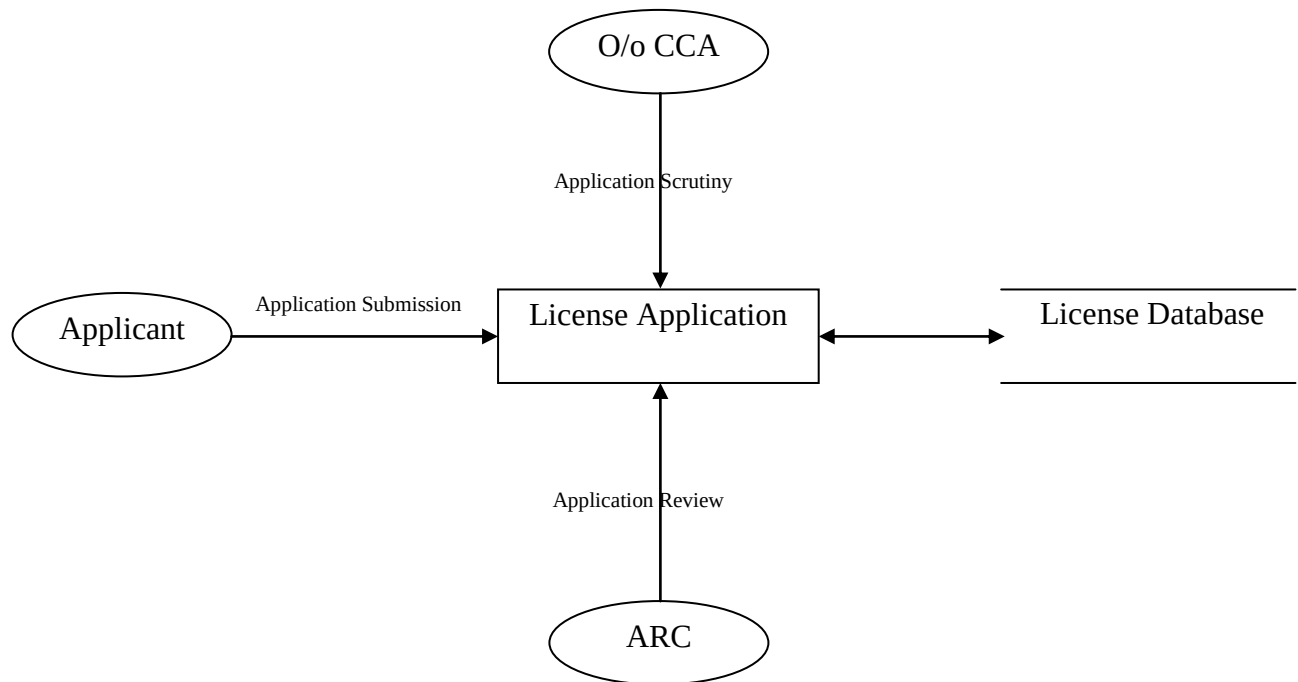


Appendix D: Data Flow Diagram

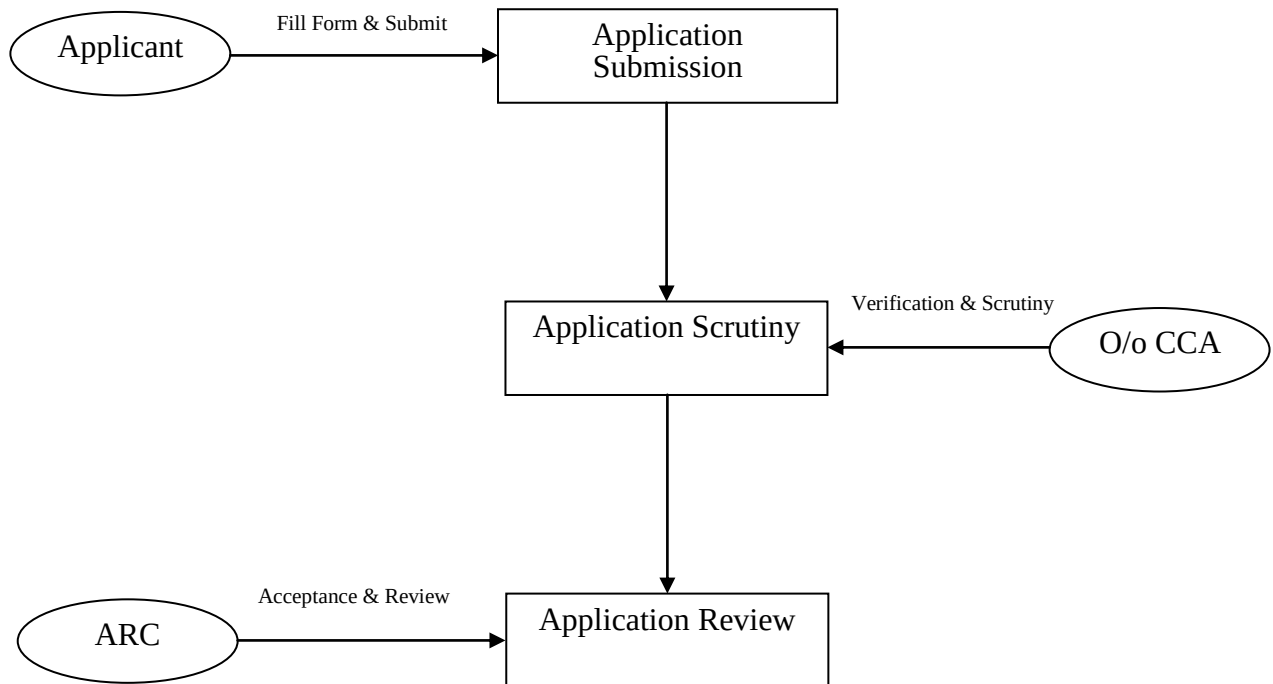
1. Zero Level DFD – Licensing of Certifying Authorities and Dashboard for DSCs



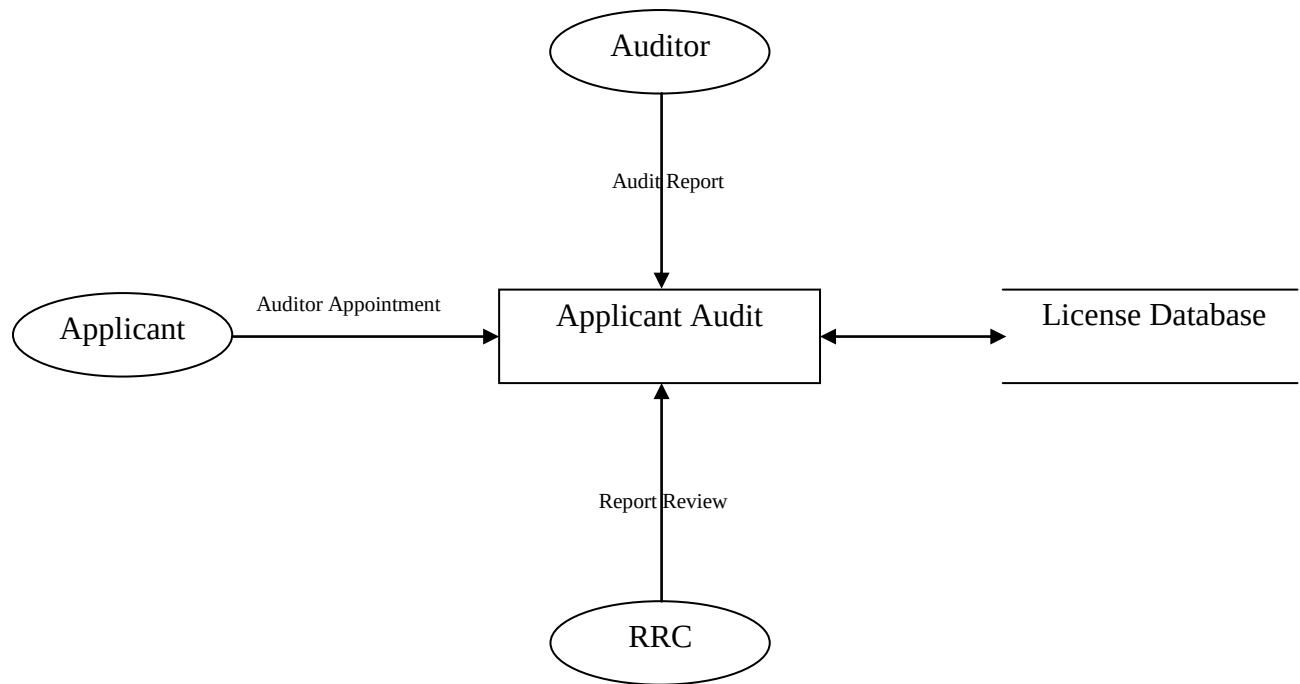
2. Level 1 DFD – License Application



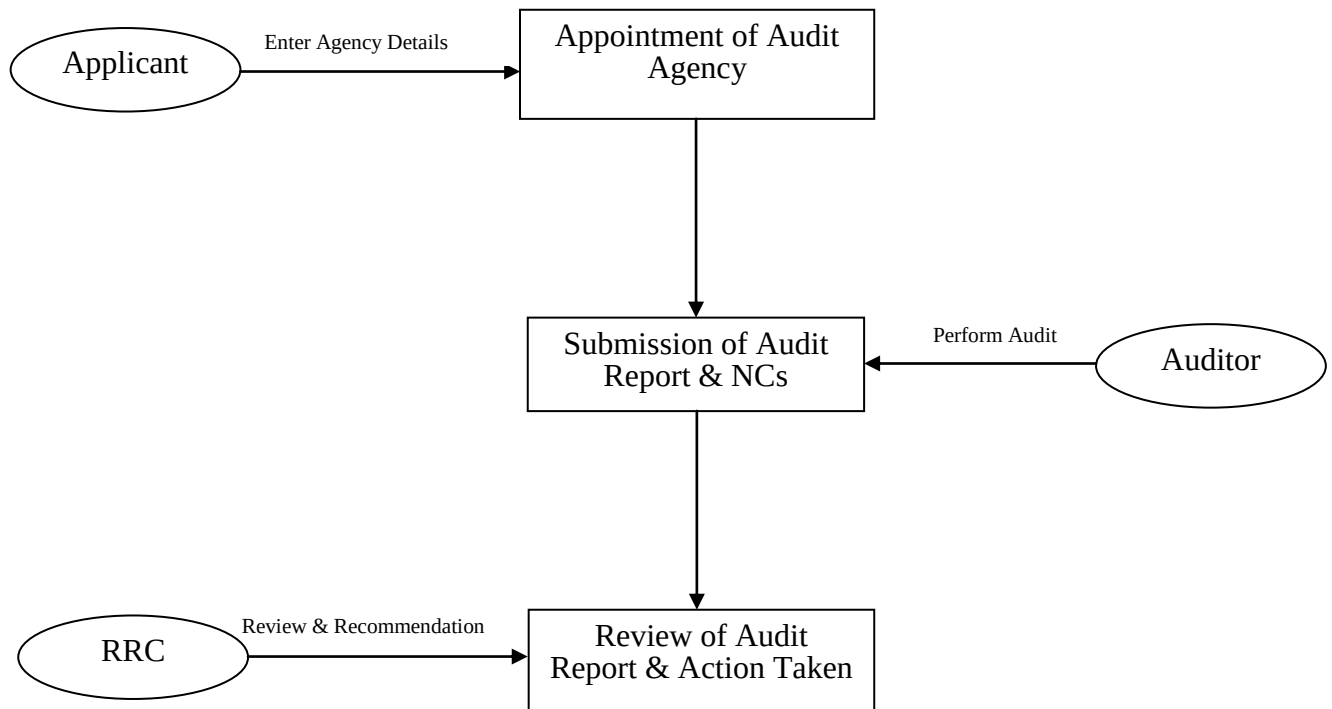
3. Level 2 DFD – License Application



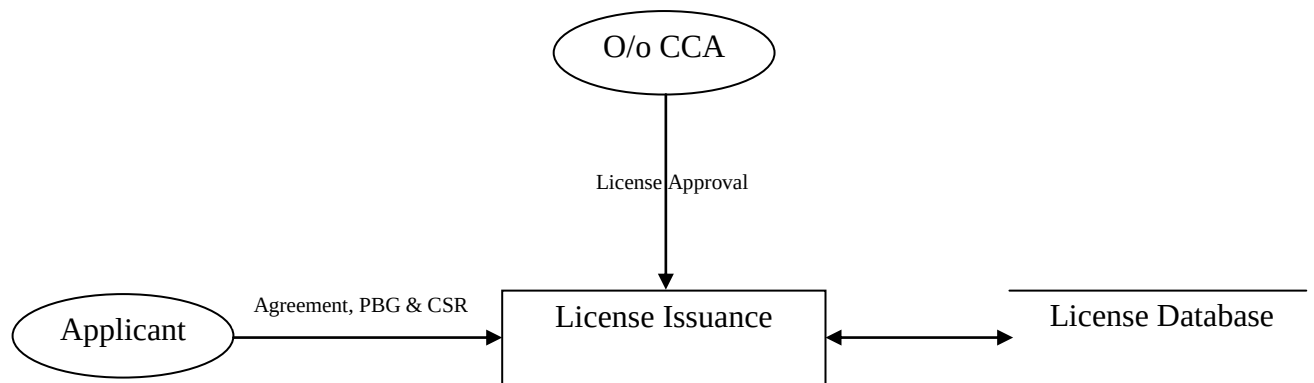
4. Level 1 DFD – Applicant Audit



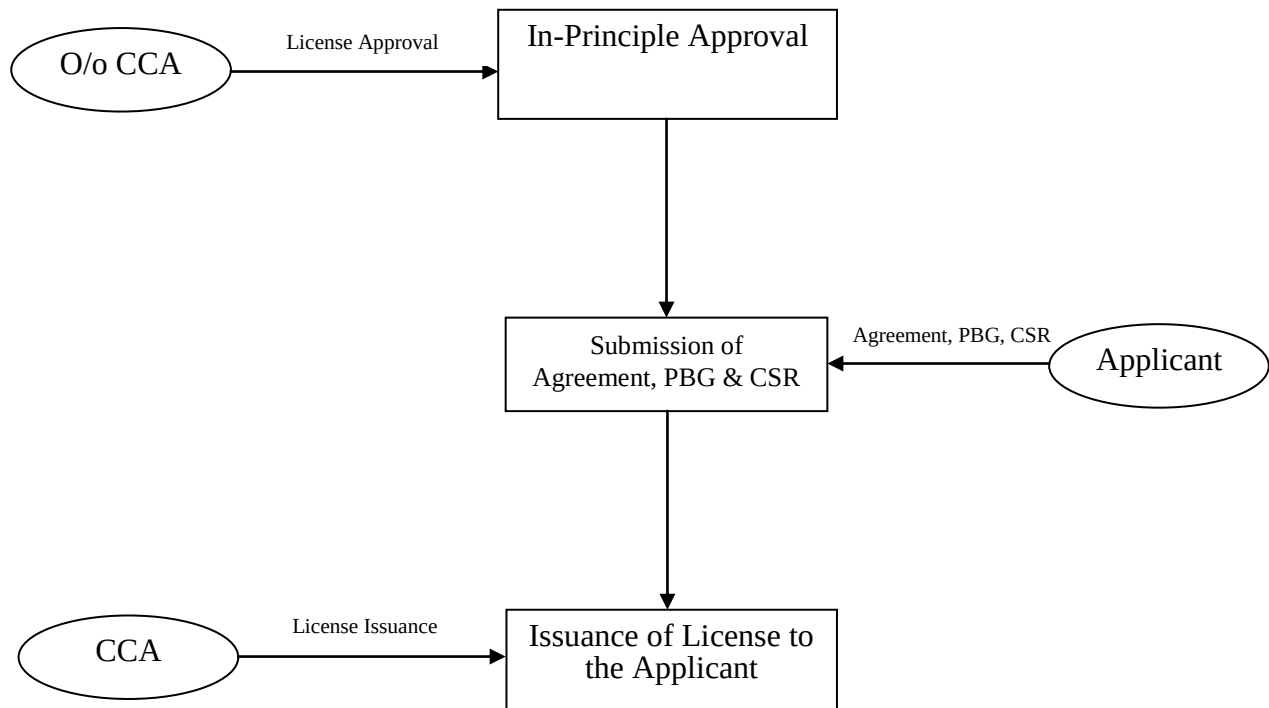
5. Level 2 DFD – Applicant Audit



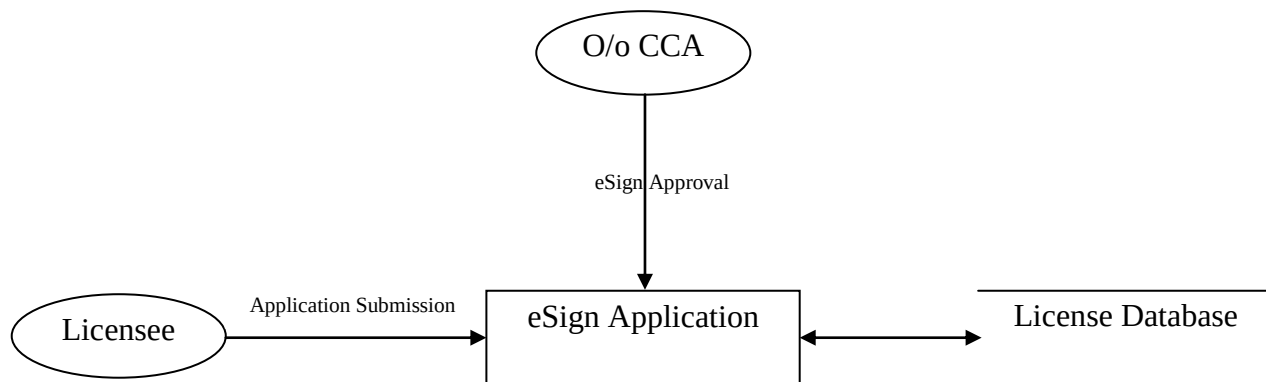
6. Level 1 DFD – License Issuance



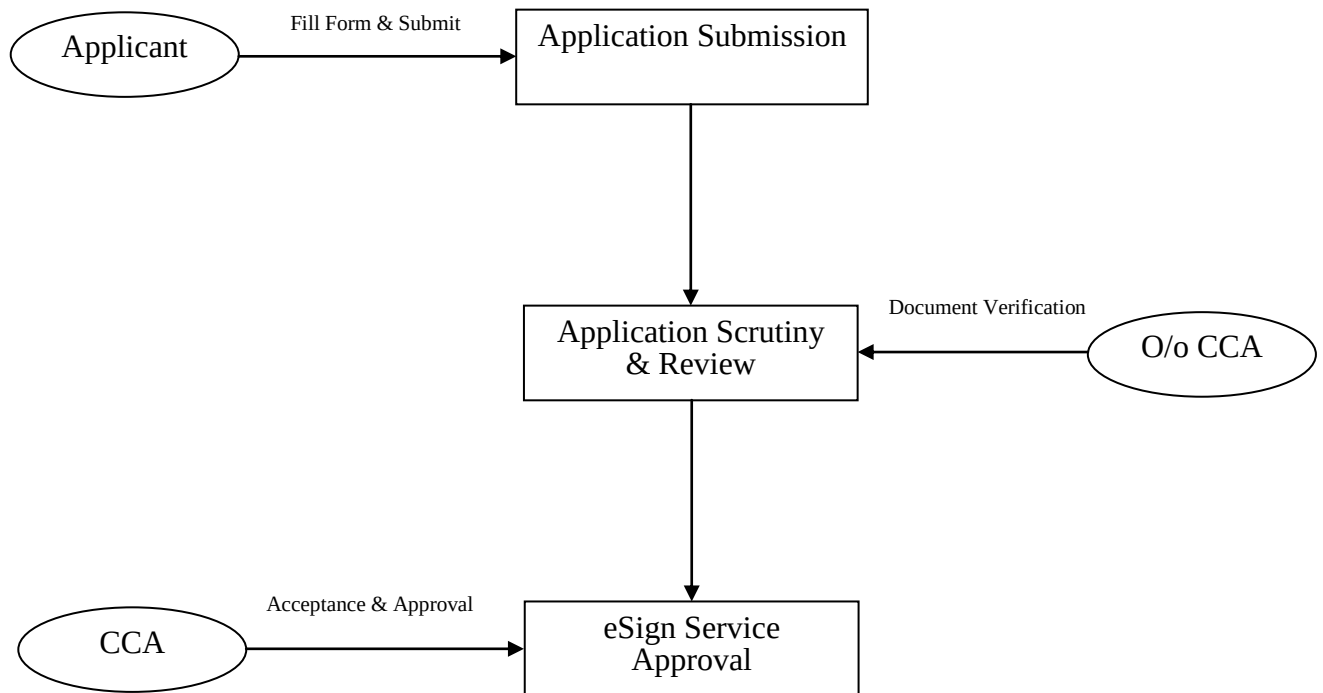
7. Level 2 DFD – License Issuance



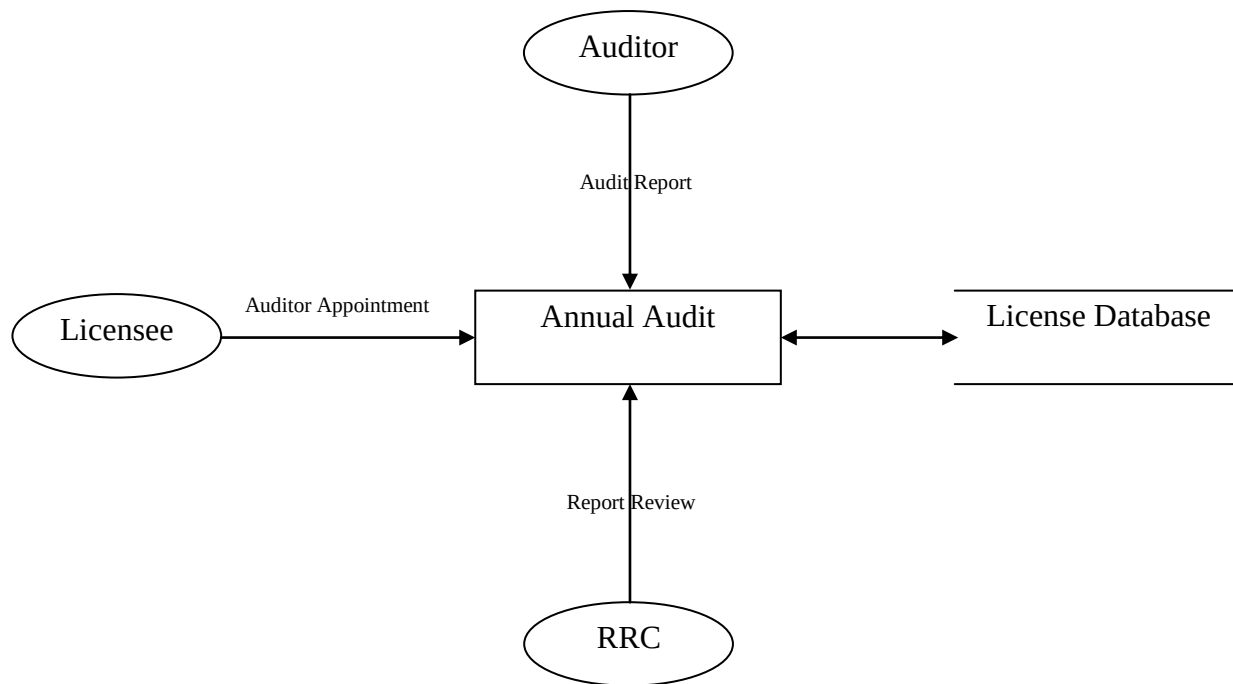
8. Level 1 DFD – eSign Application



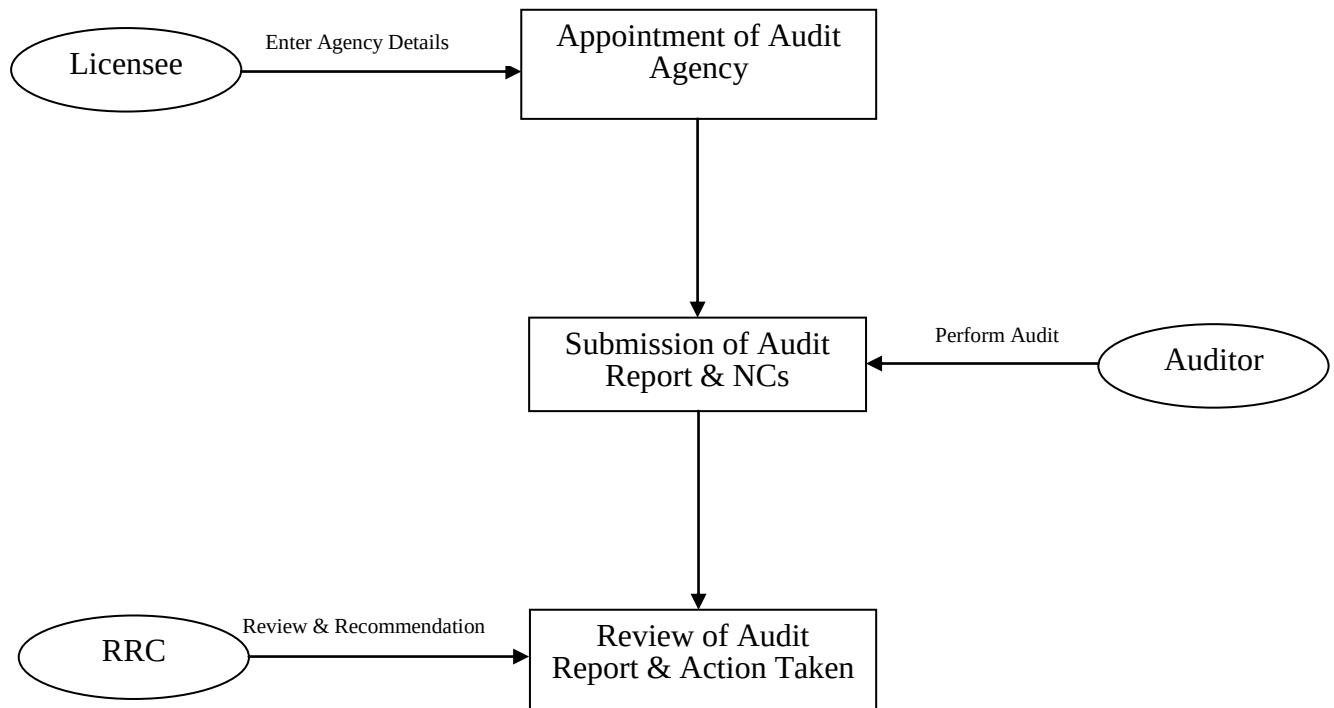
9. Level 2 DFD – eSign Application



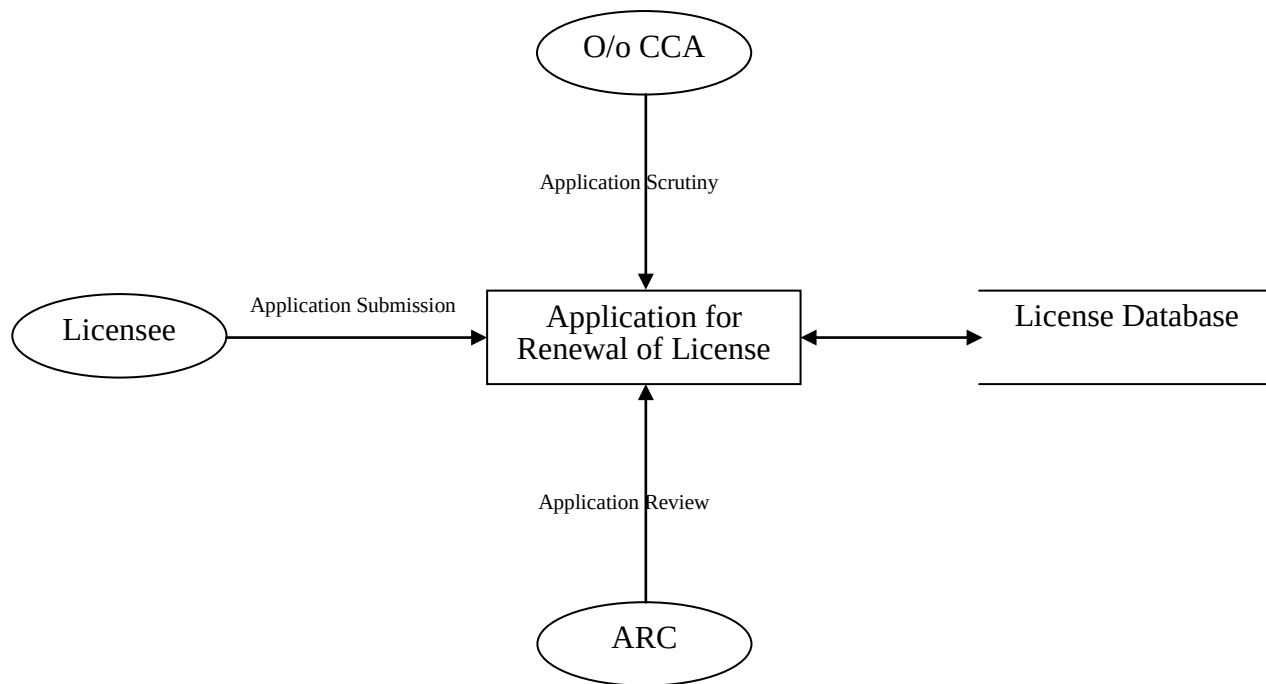
10. Level 1 DFD – Internal & Annual Audit



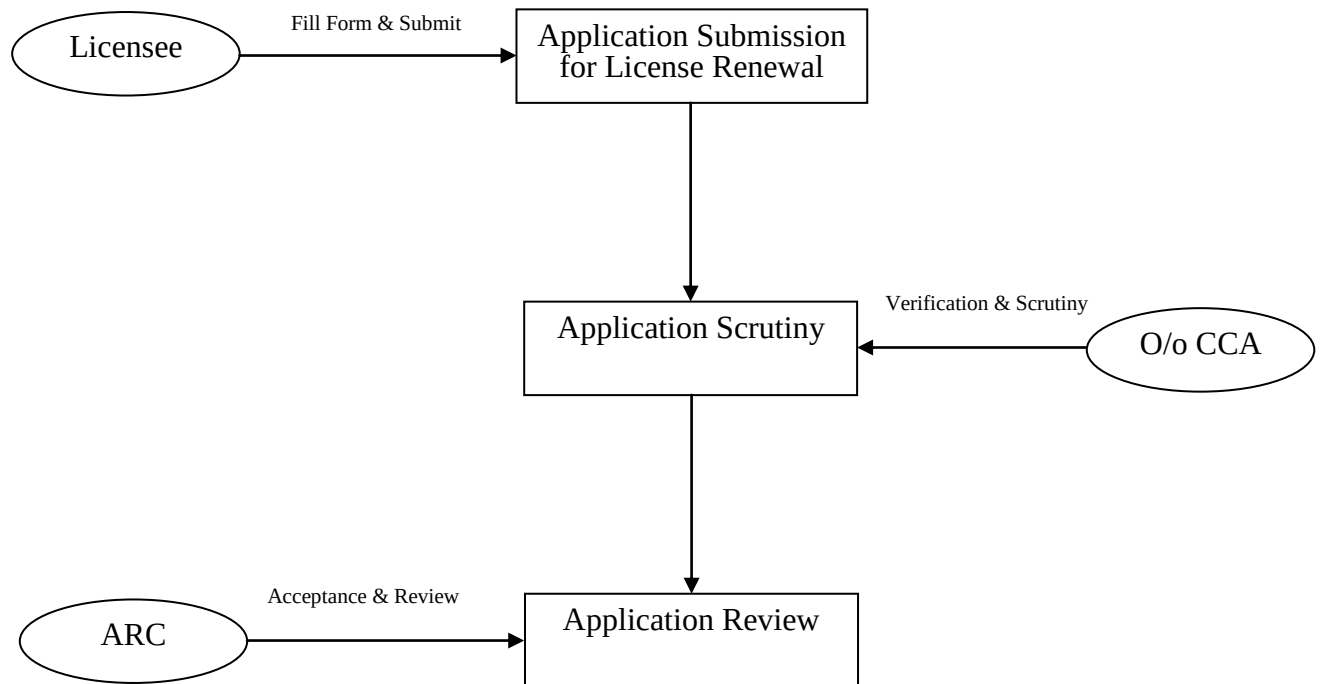
11. Level 2 DFD – Internal & Annual Audit



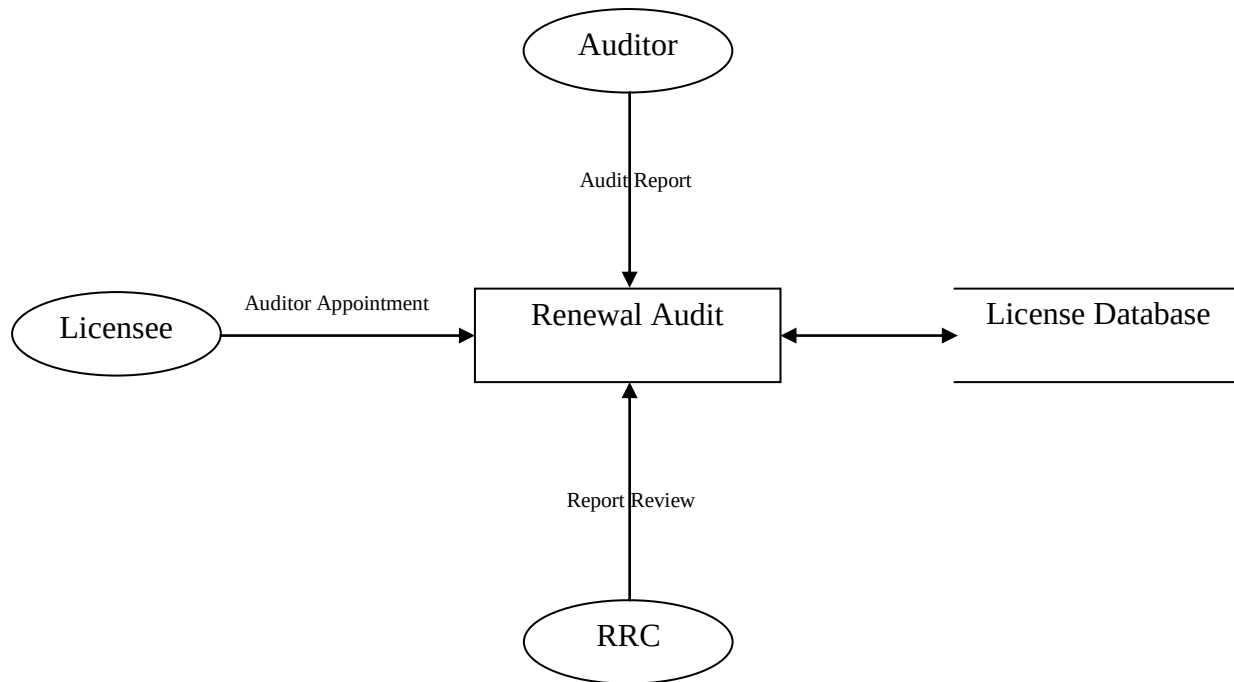
12. Level 1 DFD – Renewal Application



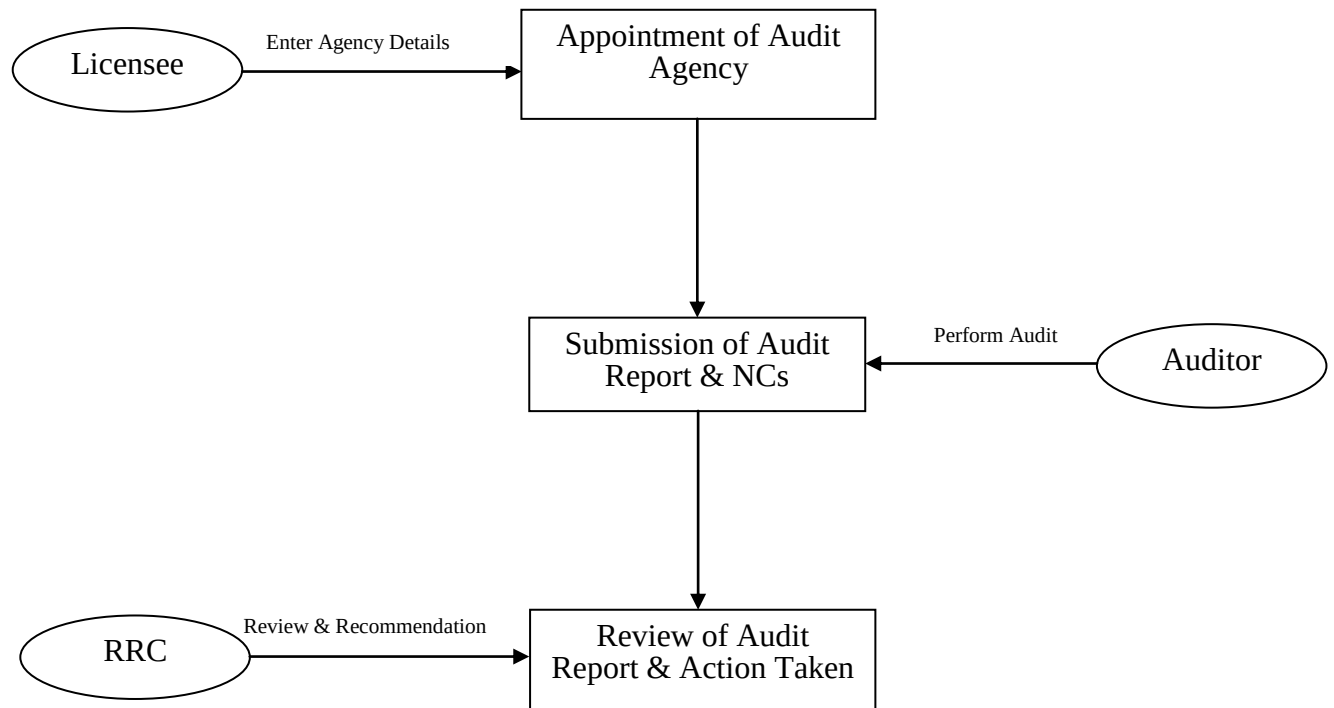
13. Level 2 DFD – Renewal Application



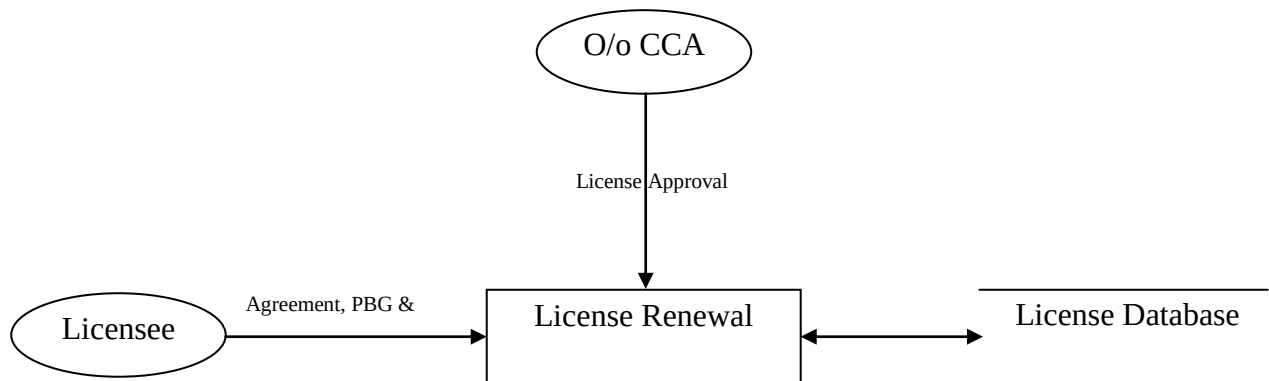
14. Level 1 DFD – Renewal Audit



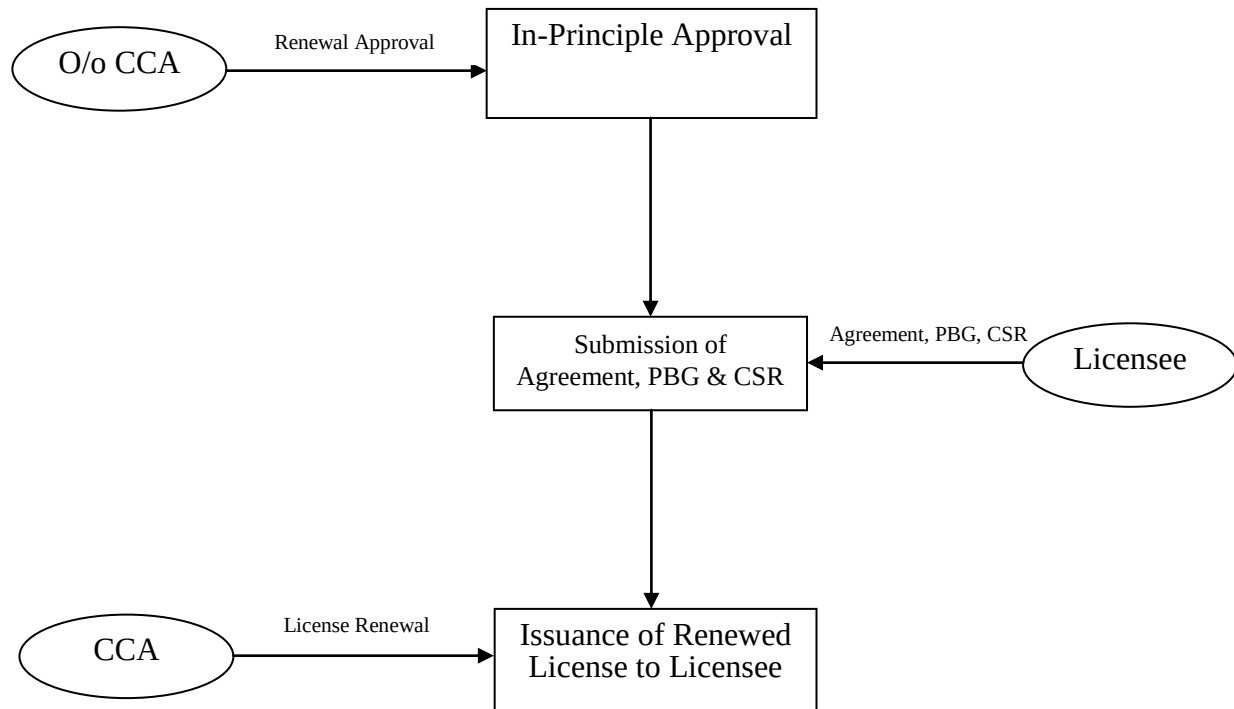
15. Level 2 DFD – Renewal Audit



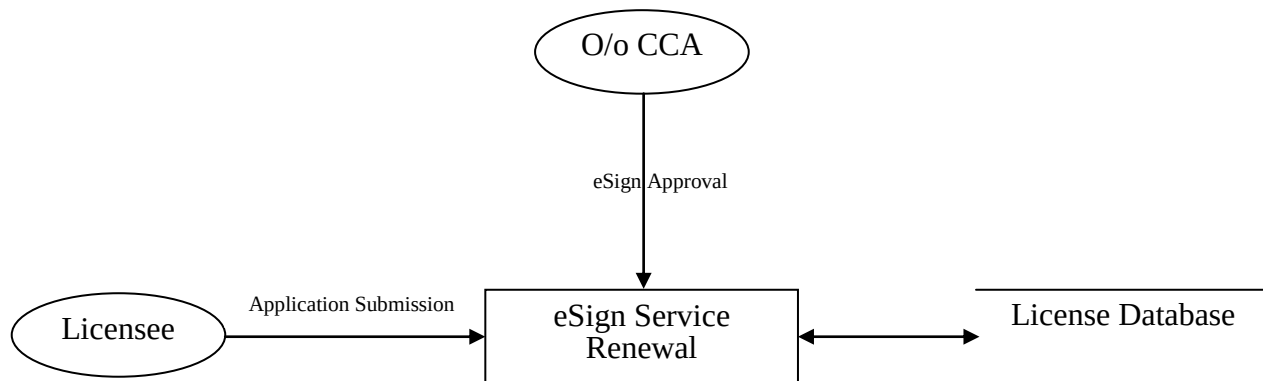
16. Level 1 DFD – License Renewal



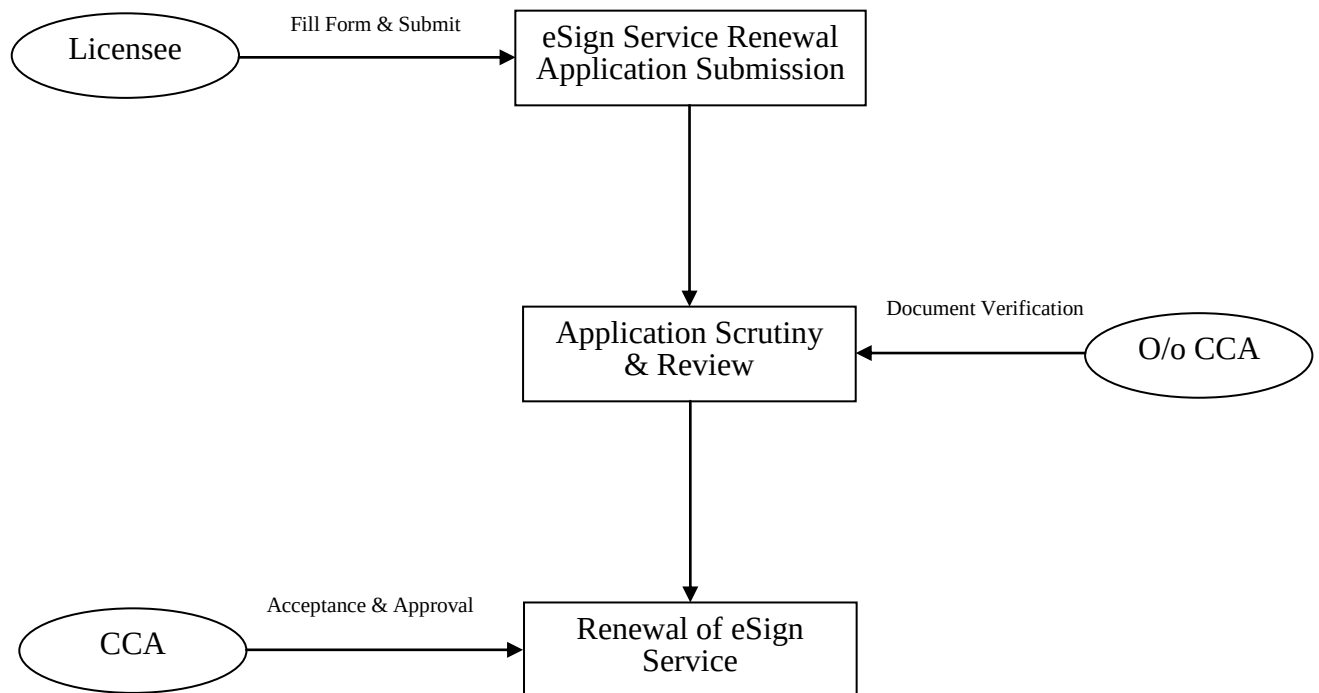
17. Level 2 DFD – License Renewal



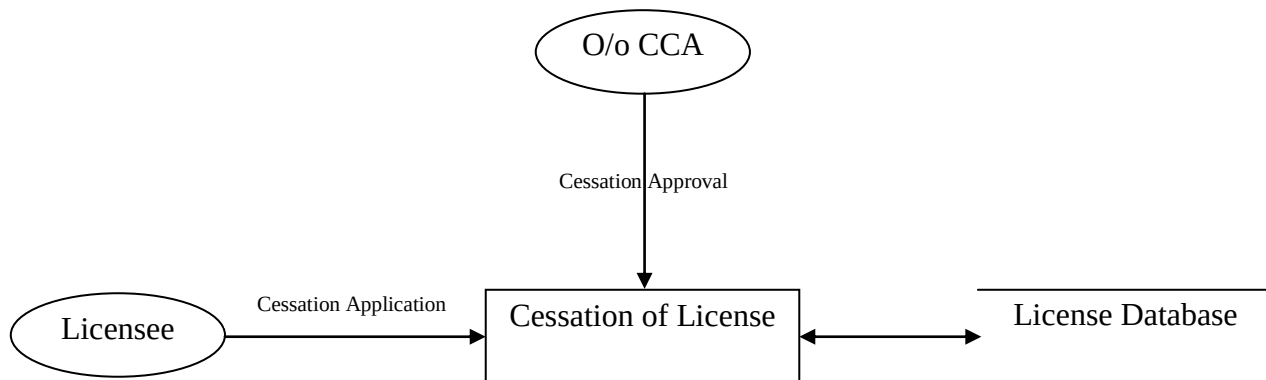
18. Level 1 DFD – eSign Service Renewal



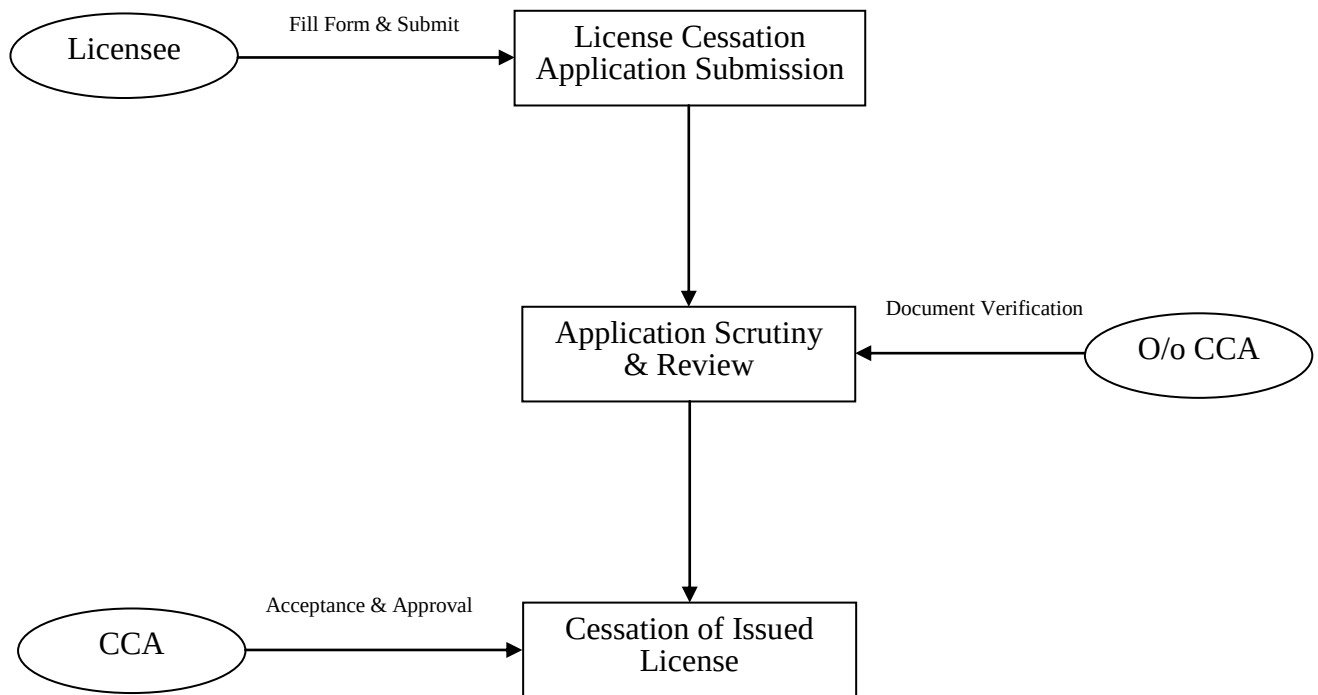
19. Level 2 DFD – eSign Service Renewal



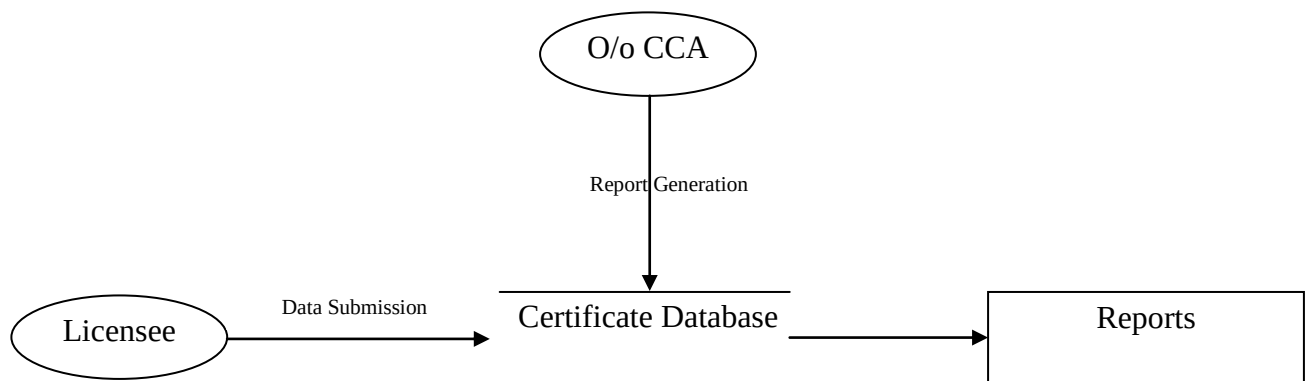
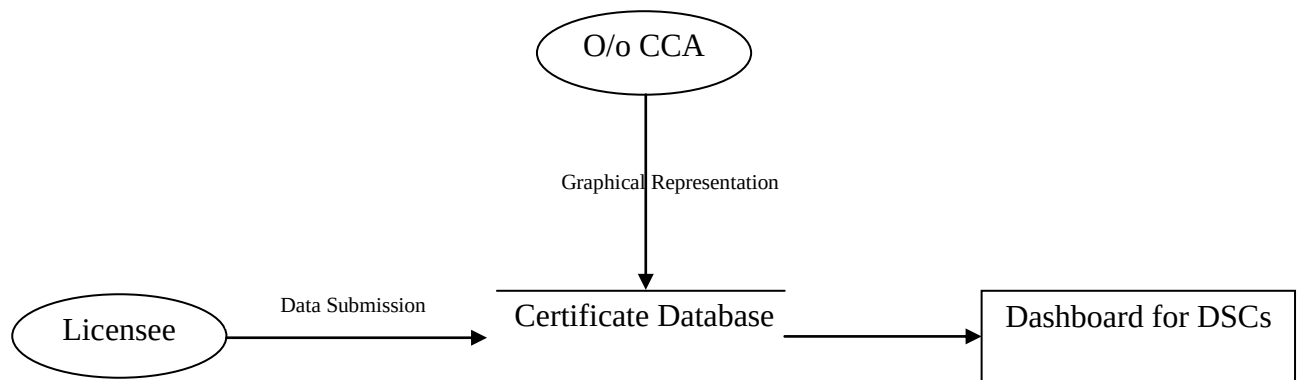
20. Level 1 DFD – Cessation of License



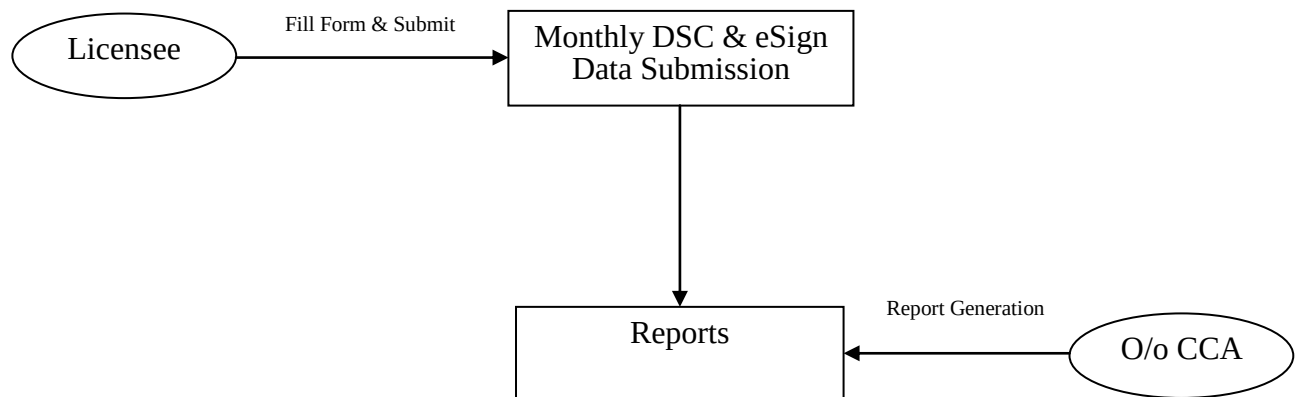
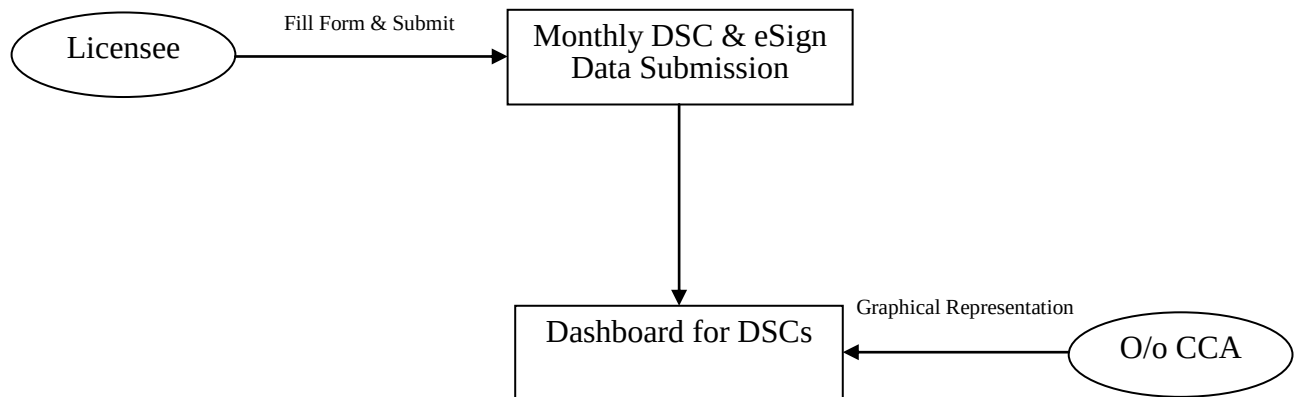
21. Level 2 DFD – Cessation of License



22. Level 1 DFD – Monitoring & Report

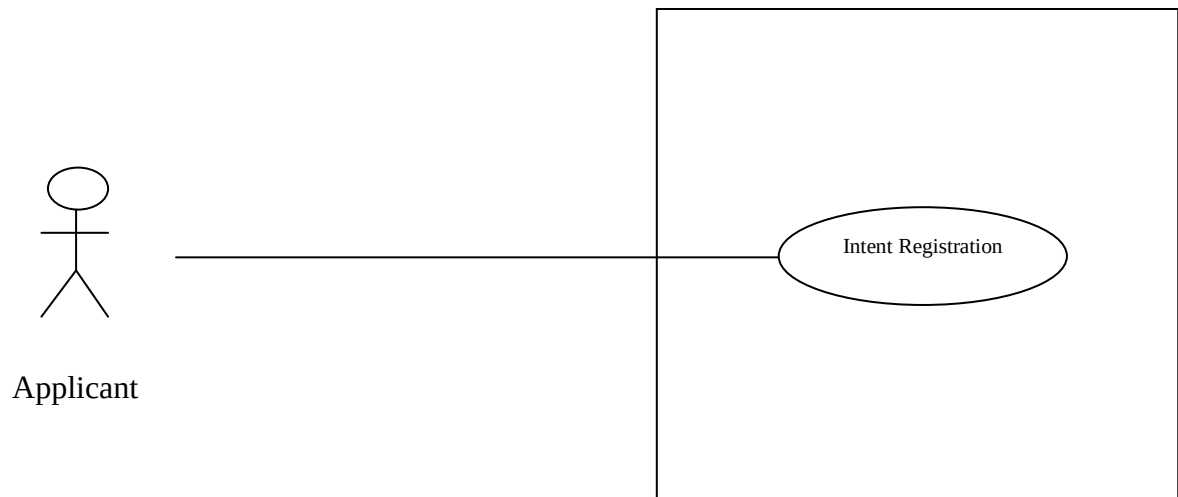
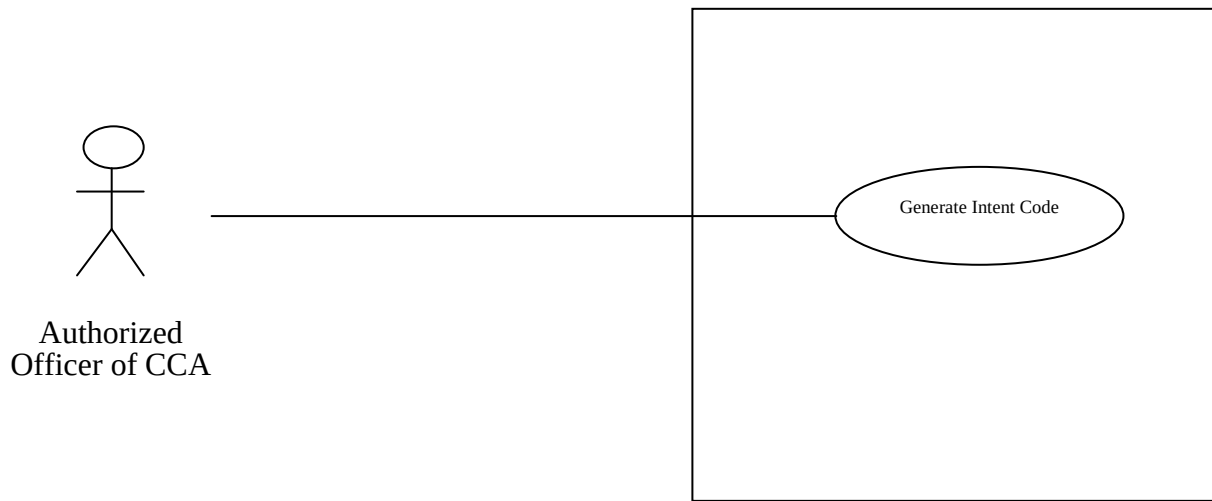


23. Level 2 DFD – Dashboard & Reports

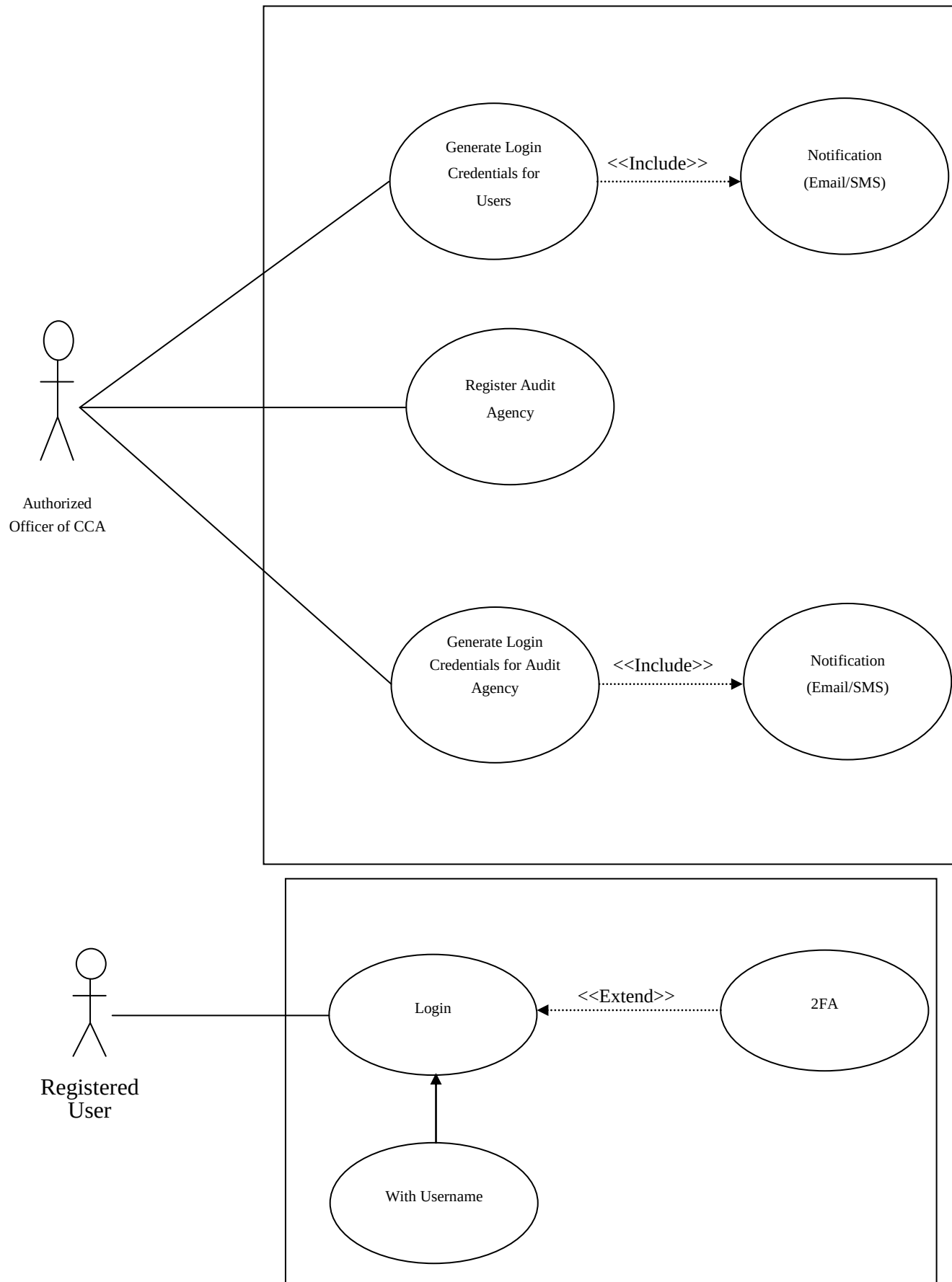


Appendix E: Use Case Diagram

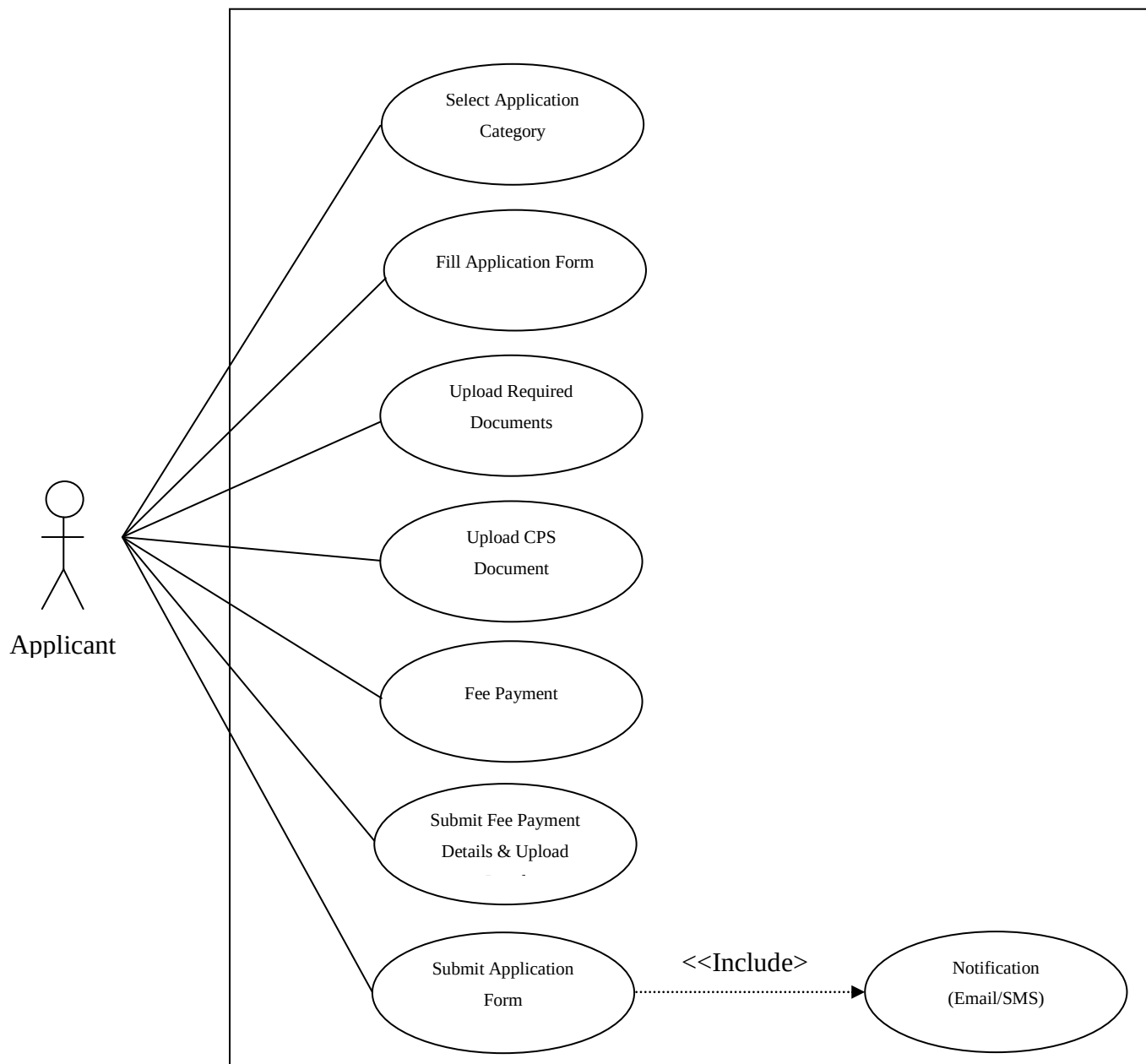
1. Intent for License

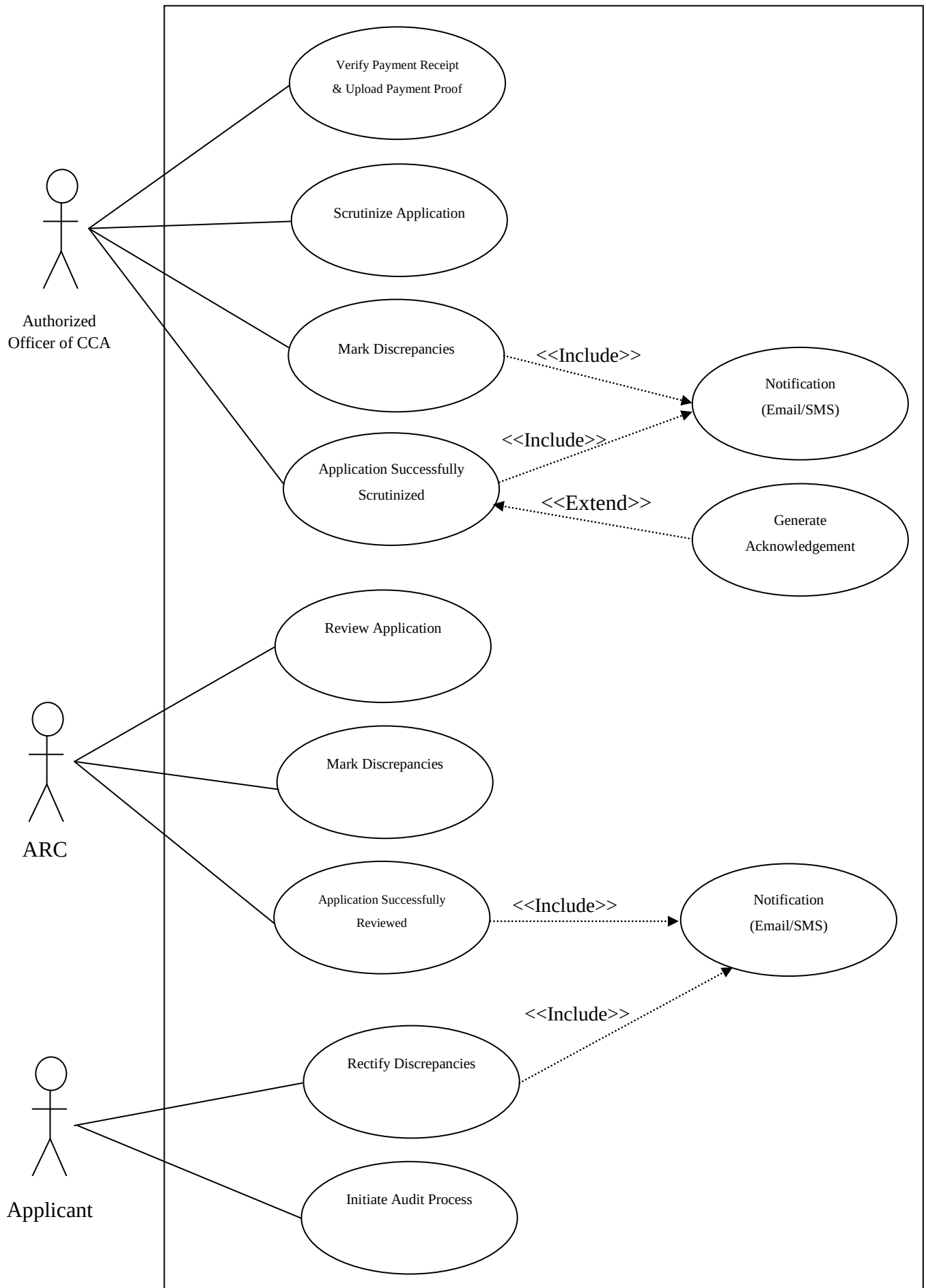


2. Verification & Login

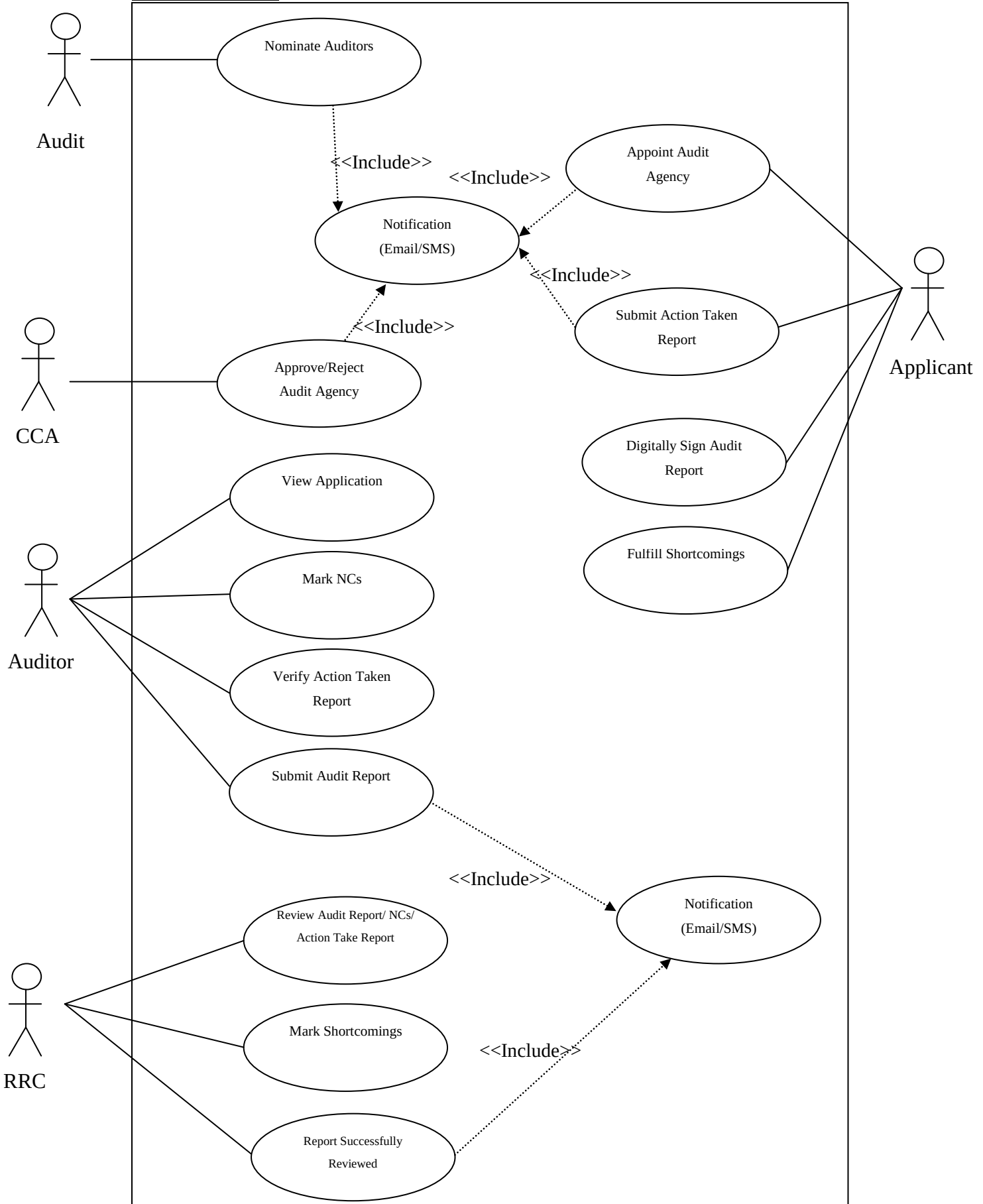


3. New License Application

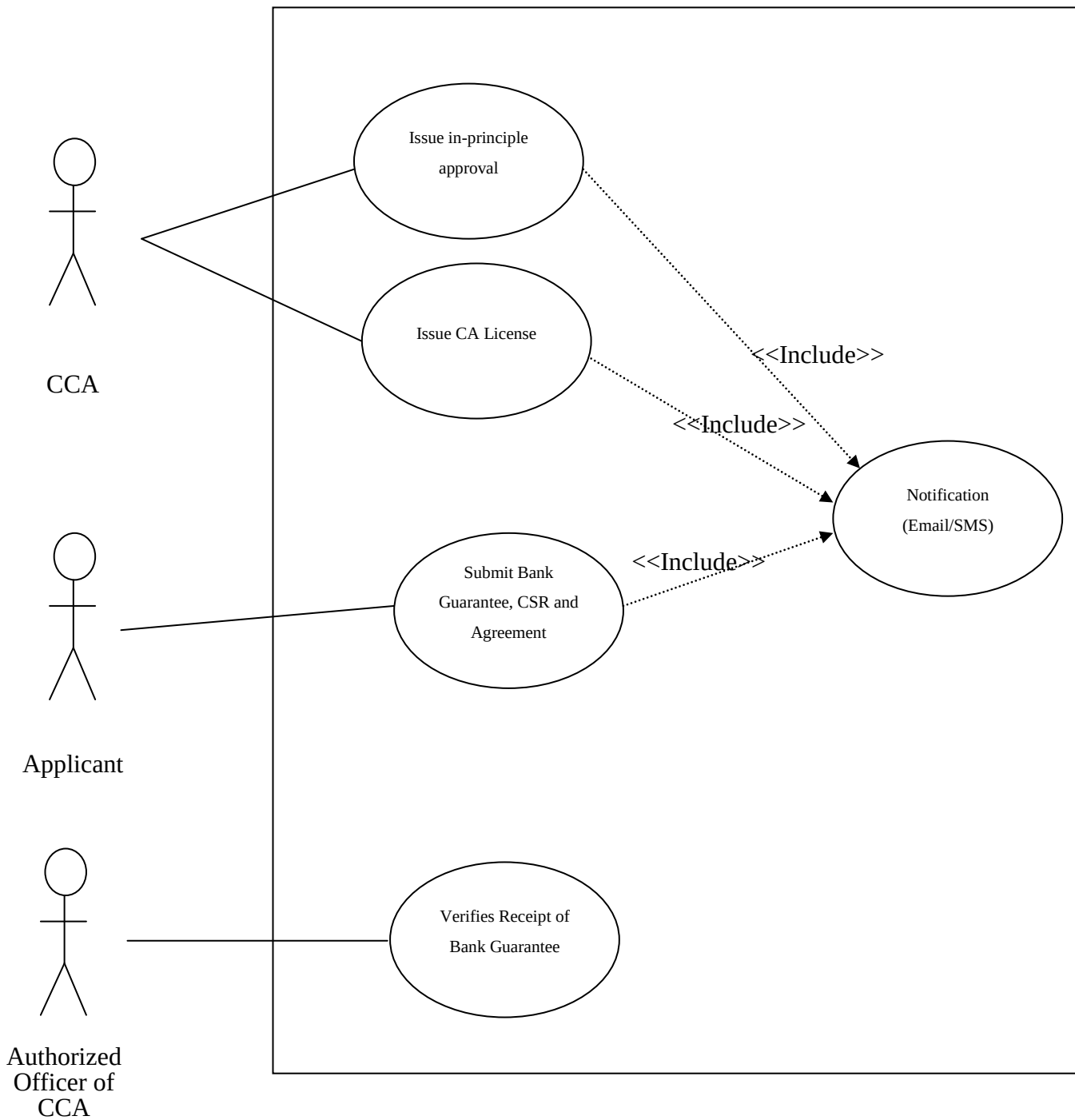




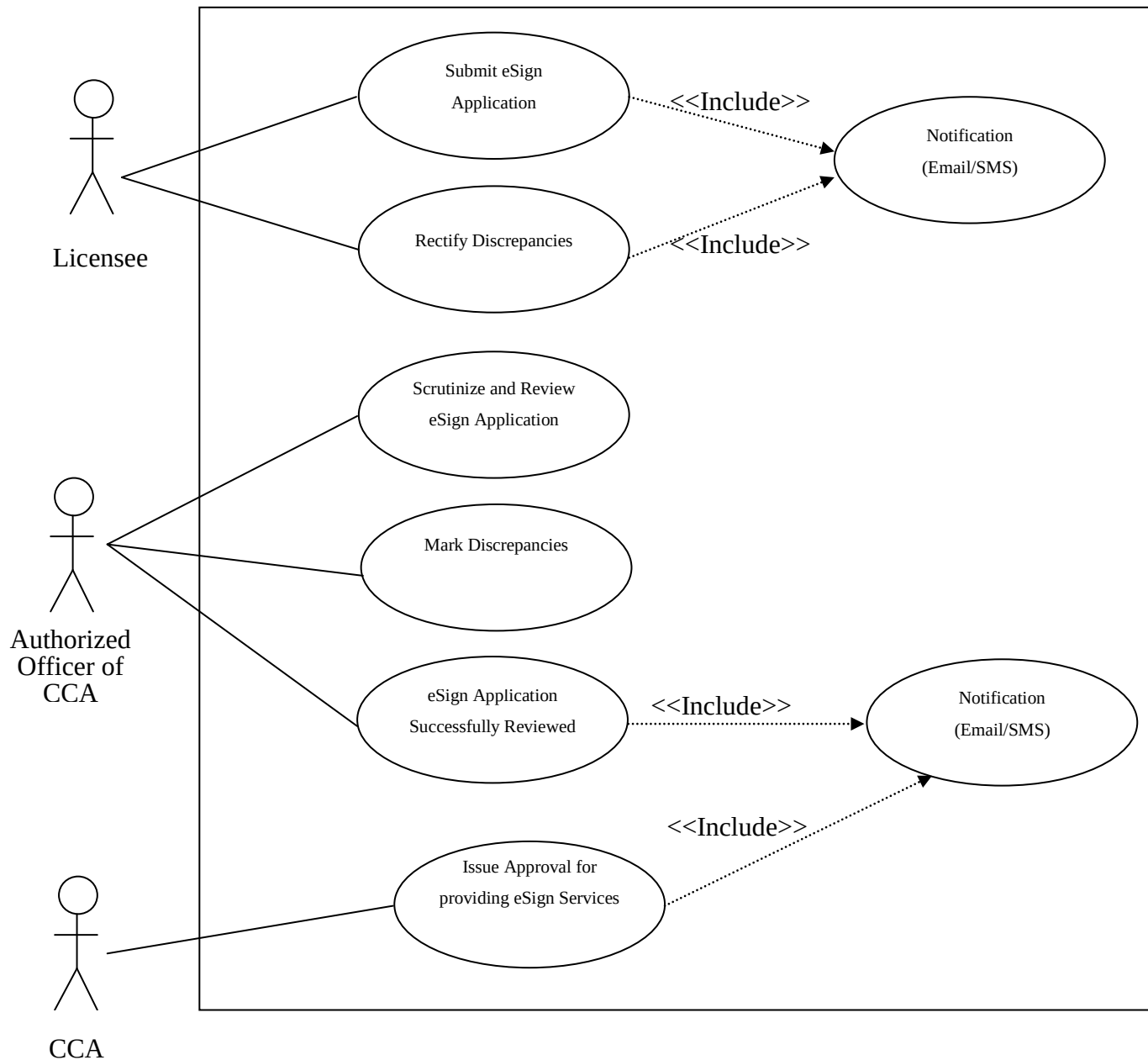
4. Applicant Audit



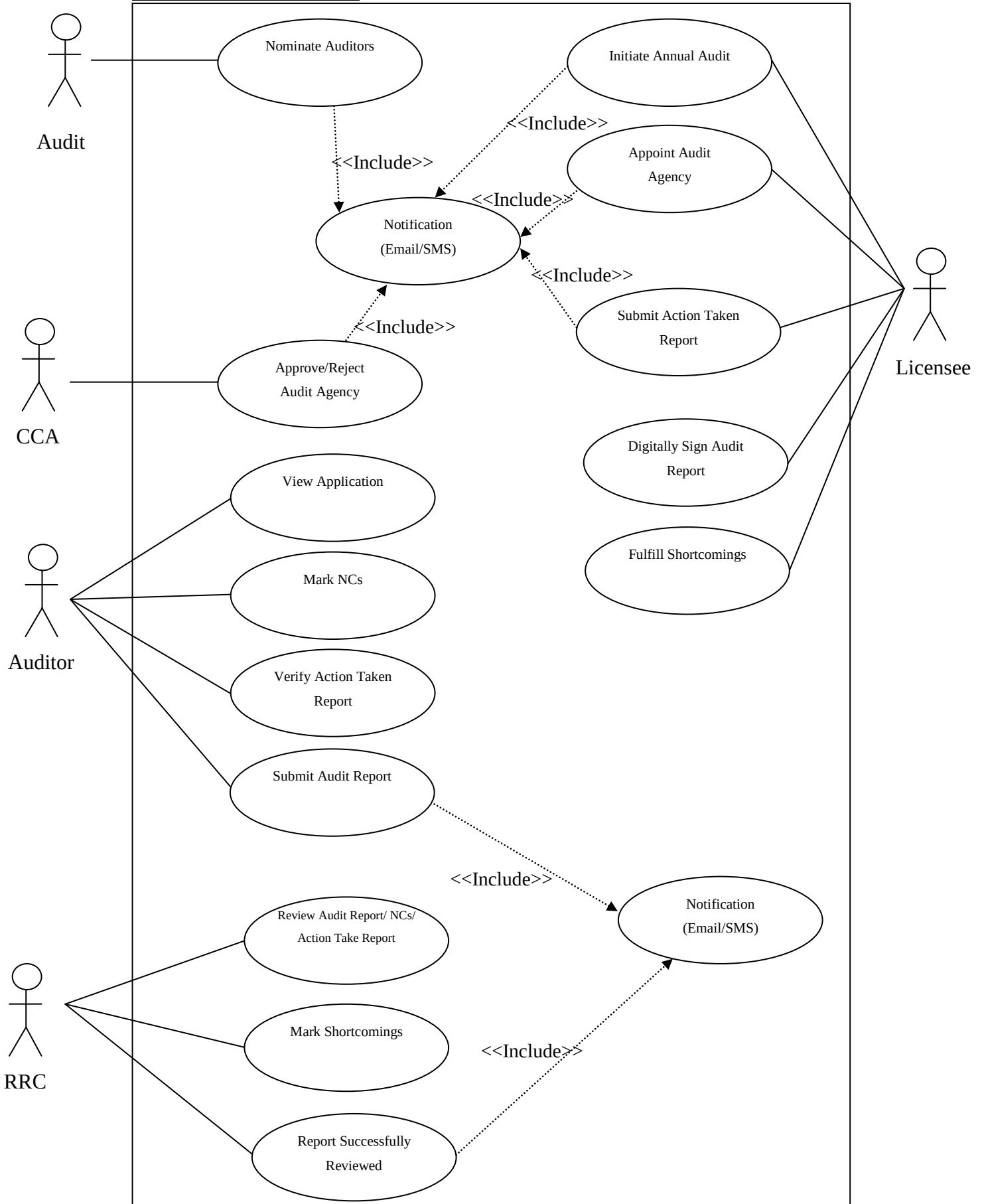
5. License Issuance



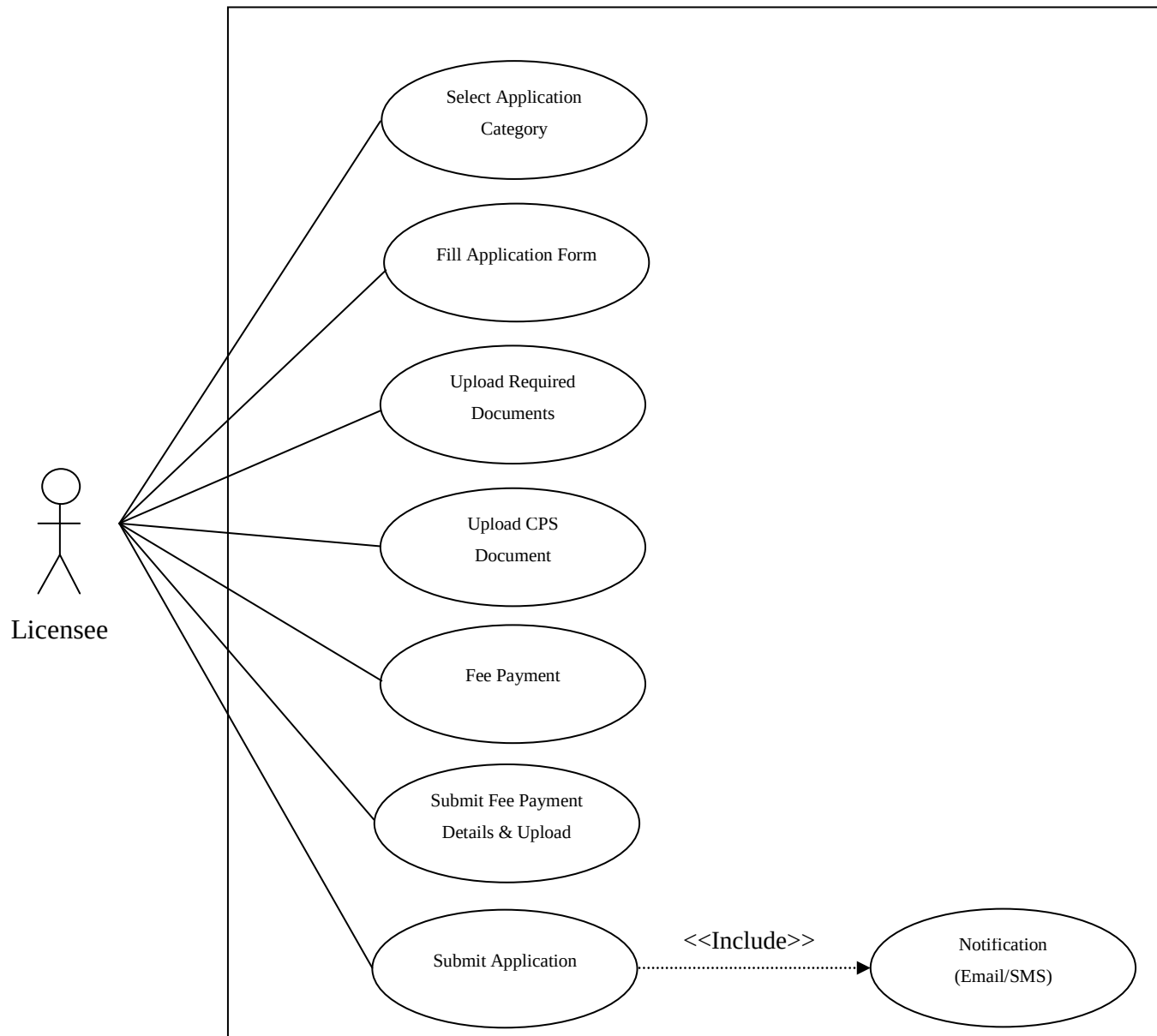
6. eSign Application

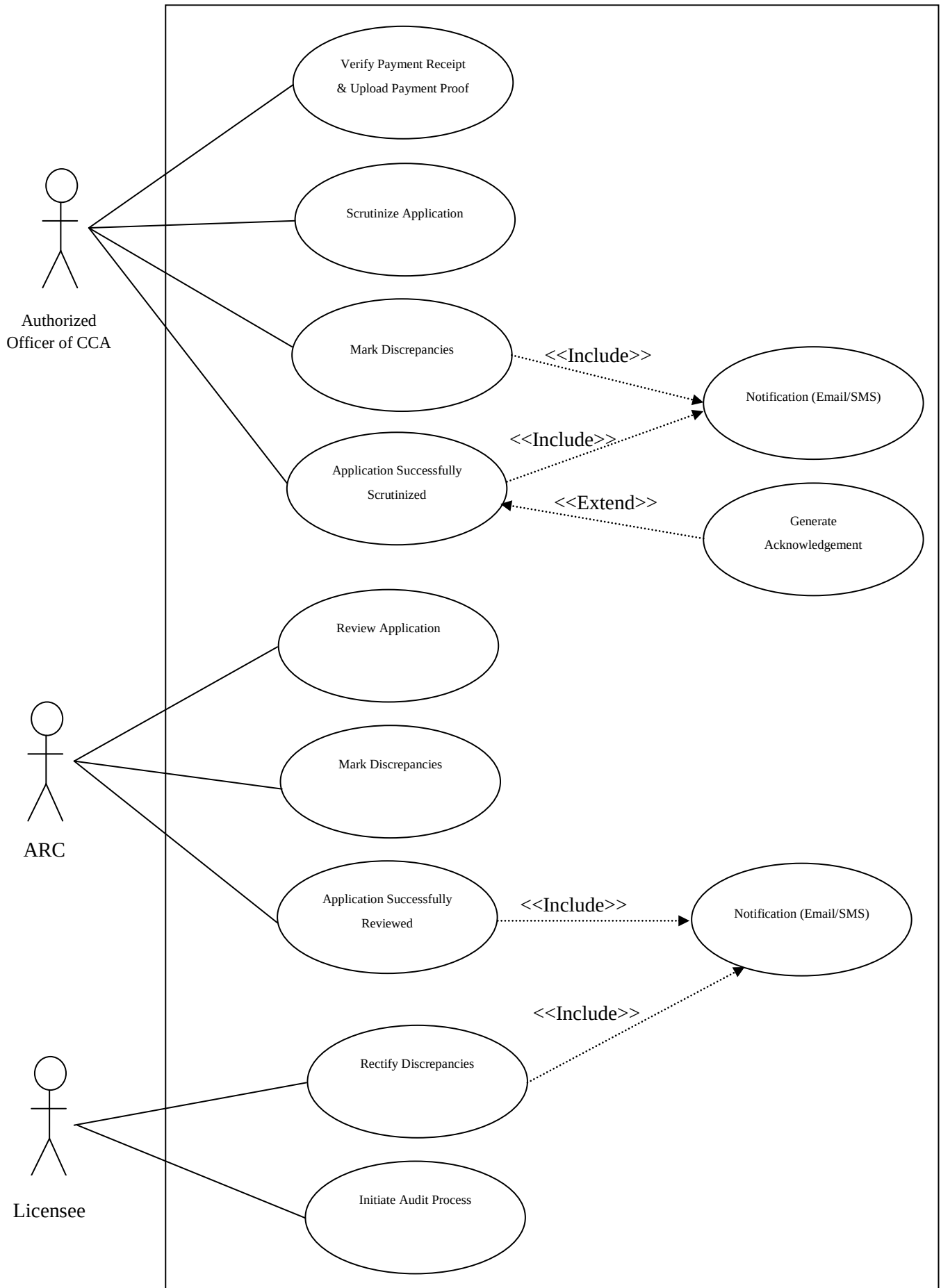


7. Internal & Annual Audit

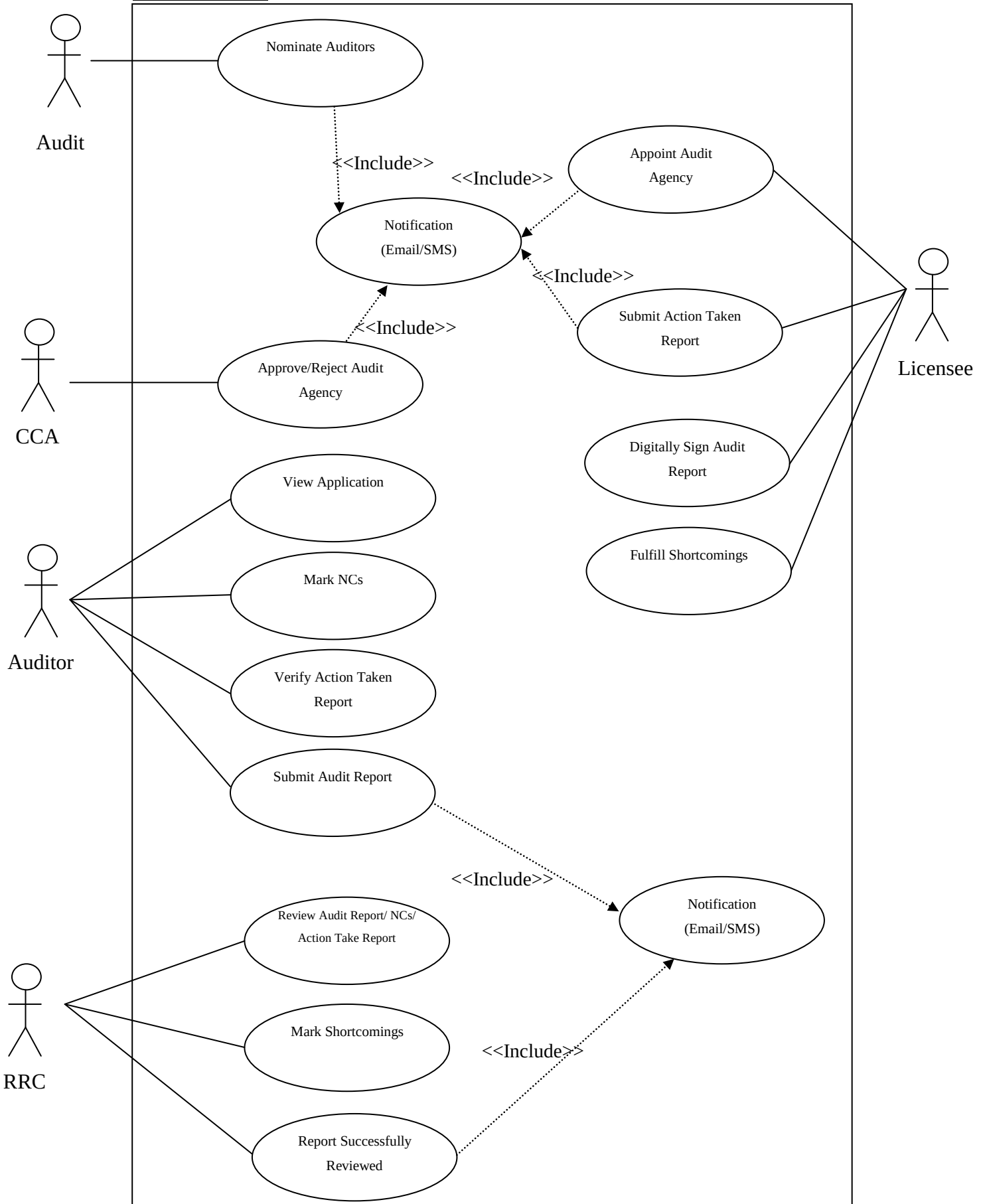


8. Renewal Application

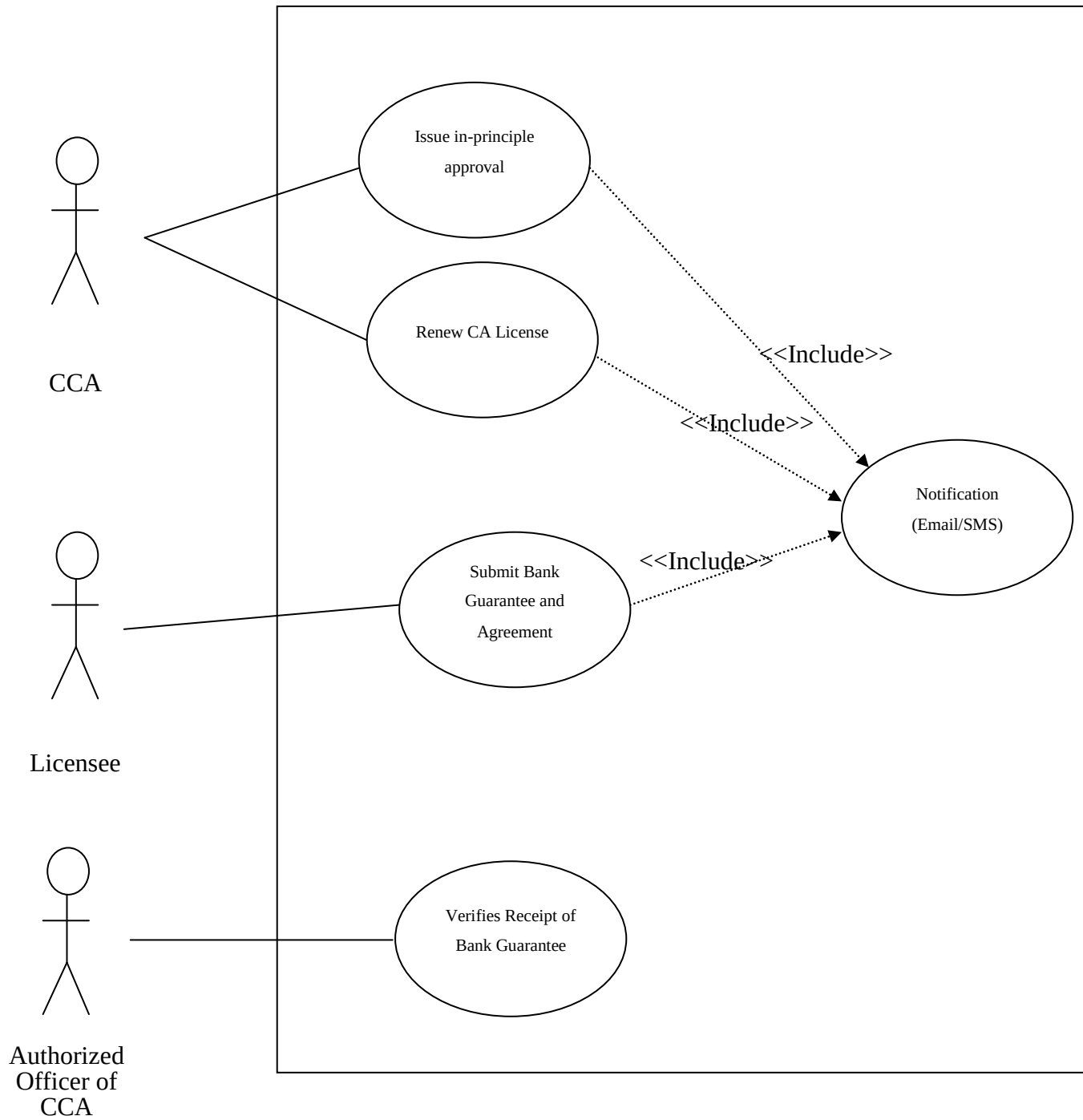




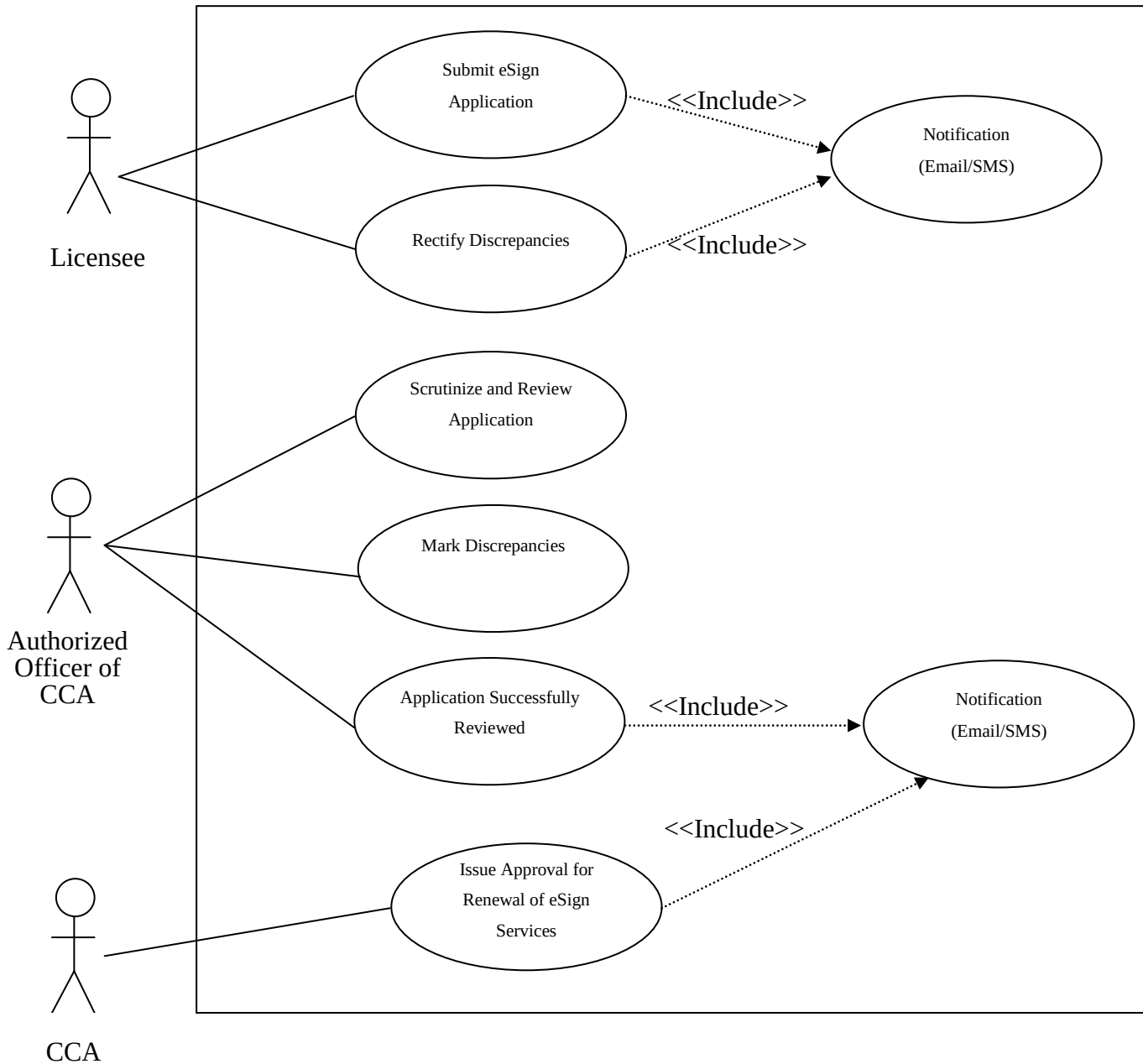
9. Renewal Audit



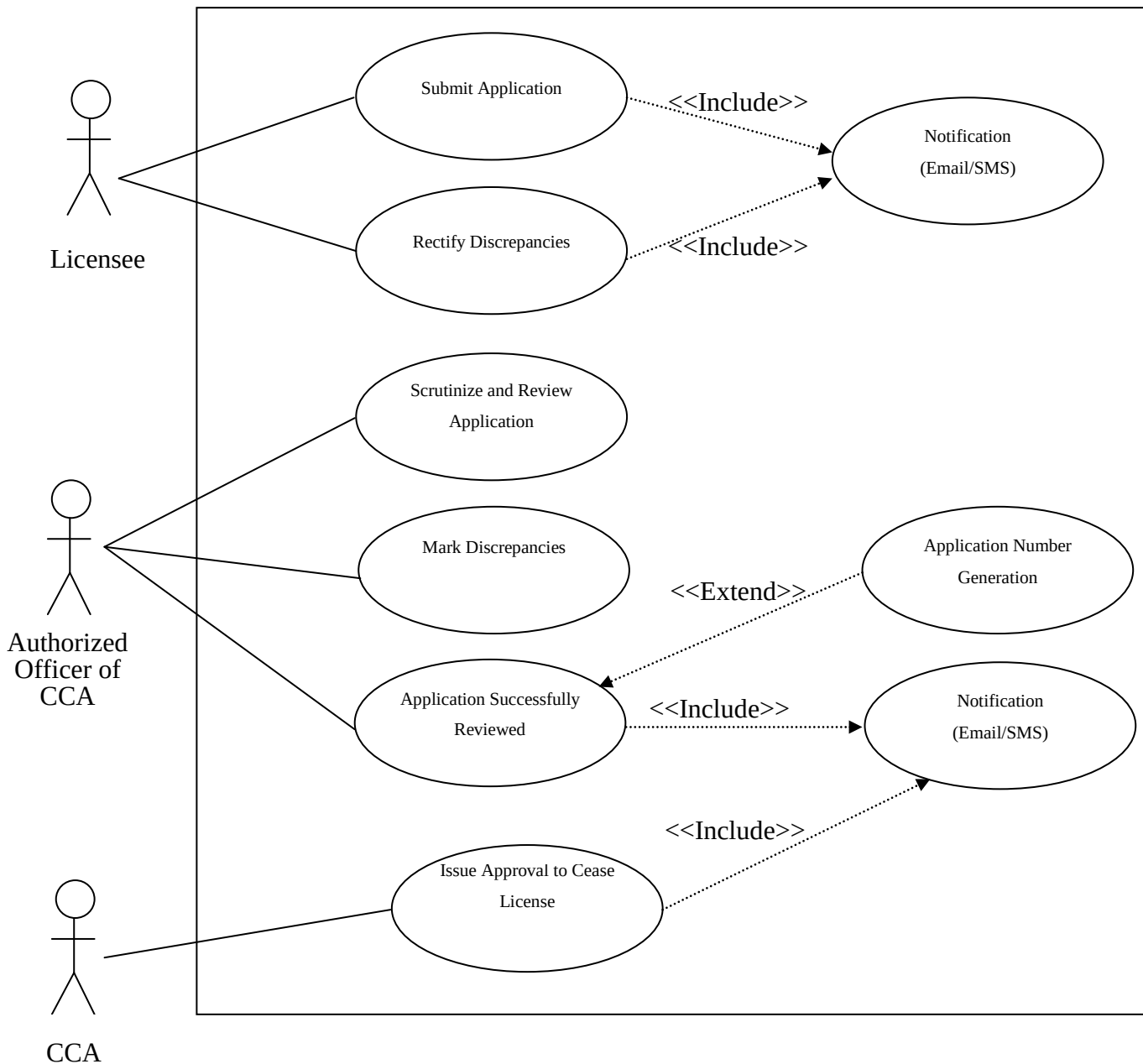
10. License Renewal



11. Renewal of eSign Service



12. Cessation of License



13. Dashboard & Reports

